

# WDW-6

## Quantum Constraint Survival and Renormalized Physical Realization of the Wheeler–DeWitt Family

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Complete proof-bearing pointed Branch-A capstone

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**Exact scope.** The exact unrenormalized WDW-5 lifts remain excluded. A distinct counterterm-enriched successor is constructed through finite-cap renormalization, a stable graph-topological carrier, constraint and closed anomaly compatibility, and a nonzero pointed physical realization. The endpoint theorem is Branch A and recordwise; stronger temporal, recovery, and Branch-B endpoint claims are not promoted.

**Integrated result.** This publication edition supplies the proof-bearing mathematics behind the WDW-6 endpoint. Parts I–IV prove the unrenormalized obstruction, the finite-cap counterterm calculus, the successor carrier, and the original zero-theta constraint/anomaly contract. Part V proves the terminal-map criterion and constructs the nonzero complete-constraint annihilator, positive quotient, represented scalar physical algebra, complete matching, and guarded continuation interfaces. Thus the exact lifts remain killed, the renormalized route remains a distinct successor, and the pointed Branch-A certified quotient contains exactly one equivalence class.

## Abstract

The accepted WDW-5 construction provides two algebraic Wheeler–DeWitt candidate families on a fixed compact, boundaryless regular canonical target. This manuscript gives the complete WDW-6 survival and realization argument. It first constructs a complete locally convex continuous-dual carrier and common operator domain, then proves that every exact unrenormalized lift fails continuum heat-regulator removal at a typed scalar witness. The negative theorem is not reversed. A separately typed, counterterm-enriched successor is then constructed using directed finite syntax caps, a global off-shell singular quotient, complete capwise heat asymptotics, source-attached counterterm columns, quantitative mixing inversion, and common-parameter remainder control. The inherited topology does not support the full endpoint; the paper therefore constructs a smooth trace source, the minimal composition-closed graph topology, its completion, and the corresponding continuous-dual successor carrier.

On that carrier the complete linear and Hamiltonian constraint families close at the declared baseline. The Standard Model anomaly coordinate is treated on the faithful  $G_6$  global form: the local polynomial, determinant/Pfaffian data, fifth spin-bordism obstruction, mapping-torus holonomy, regulator compatibility, and the declared trivial four-dimensional theta character are checked as one closed record. No BRST, BFM, master-constraint, or group-averaging replacement is invoked. Finally, a topological coinvariant criterion isolates the physical-state burden. A normalized joint solution yields a continuous nonzero terminal map, rank-one positive quotient, faithful scalar physical algebra, and the complete pointed Branch-A matching. The exact boundary is explicit: stronger temporal functorial trivializations, Branch-B physical realization, non-scalar observables, conditional dynamics, and semiclassical recovery are not claimed.

**Keywords:** Wheeler–DeWitt equation; canonical quantum gravity; heat-kernel renormalization; locally convex operator carrier; quantum constraints; anomaly line; spin bordism; physical state; AASC.

## Imported interfaces and proof responsibility

The dependency chain is one way: WDW-1 supplies the canonical projection, WDW-2 proper-redescription descent, WDW-3 classical refoliation, WDW-4 classical occupancy, and WDW-5 the typed candidate family [1, 2, 3, 4, 5, 6]. Those results fix the incoming objects and notation. They do not prove regulator removal, quantum commutator closure, anomaly triviality, or physical realization; those obligations are proved here. AASC supplies necessity constraints on admissible carriers and standing-bearing records, not a choice of topology, regulator, ordering, counterterms, anomaly character, state, form, or observable algebra.

The distinction between the two realization classes is immutable. The exact unrenormalized lifts fail at Q6R-07 and are never revived. The positive construction changes declared data—counterterm record, smooth trace source, graph topology, completion, and endpoint—and is therefore a distinct renormalized successor rather than a modification of an exact lift.

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## Part I

# Exact unrenormalized realization and obstruction

### 2.5 Exact handoff theorem

**Theorem W6.1 (Exact handoff and no-backflow theorem).**

The registered artifacts reconstruct the exact  $\mathfrak{H}_{5 \rightarrow 6}$  authorized by WDW-5 predecessor stage. Every WDW-6 construction in this manuscript acts downstream of the accepted WDW-5 graphs, changes no WDW-1–WDW-5 theorem or target coordinate, and introduces no Hilbert, anomaly, state, or observable conclusion by import.

**Proof.**

The incoming typed data agree across the accepted WDW-5 handoff and the source record. The branch labels, common modules, algebraic duals, complete constraint-label families, matching relation, and fourteen-row continuation register are taken from that object without replacement. The only new carrier is the continuous-dual completion constructed in Theorem W6.3, which is registered as WDW-6 data and restricts to each old continuous operator graph by the evaluation identity proved there. The only new negative conclusion is the regulator estimate in Theorem W6.6. Neither construction alters an upstream premise.  $\square$

### 2.6 Candidate quotient and status theorem

Let  $\mathcal{Q}_5$  be the disjoint union of the two accepted candidate families and let  $\mathcal{G}_5 \rightrightarrows \mathcal{Q}_5$  be the WDW-5 matching groupoid. A WDW-6 record lies over one object of  $\mathcal{Q}_5$ . A matching transports status only when it also transports the new carrier, domain, regulator endpoint, anomaly data, state, form, observables, and continuation interfaces.

**Theorem W6.2 (Candidate quotient and four-status theorem).**

KILLED, OPEN, CONDITIONAL, and CERTIFIED define a mutually exclusive partition of the current source-relative candidate records. WDW-5 equivalence alone does not transport a WDW-6 status. A complete WDW-6 matching does.

**Proof.**

Assign to each record the least certificate index at which it is obstructed or unoccupied. KILLED requires a proved contradiction at that least index; OPEN requires absence without such a contradiction; CONDITIONAL requires a proved implication whose hypotheses remain unoccupied; and CERTIFIED requires no least failure because all twenty clauses hold. These predicates are pairwise inconsistent and exhaustive. A WDW-5 matching does not contain the added WDW-6 coordinates, so it cannot preserve their witnesses. The complete matching tuple of Theorem W6.15 does contain them and preserves the least-failure record.  $\square$

## 3. Module A: complete carrier, common domain, and closed realization

### 3.1 Continuous-dual notation

Let  $V$  be a Hausdorff locally convex space with its declared topology  $\tau_V$ . Write  $V'$  for its continuous dual and reserve  $V^\times$  for the full algebraic dual. The two must not be identified: WDW-5 uses  $V^\times$  for its formal solution functor, whereas the topological completion below is built from  $V'$ .

The canonical evaluation map is

$$j_V : V \longrightarrow (V')^\times, \quad j_V(v)(\lambda) = \lambda(v).$$

Equip  $(V')^\times$  with

$$\tau_{w*} = \sigma((V')^\times, V'),$$

the locally convex topology of pointwise convergence on the continuous dual. Its defining seminorms are

$$p_\lambda(L) = |L(\lambda)|, \quad \lambda \in V'.$$

For a continuous linear map  $T : V \rightarrow V$ , its continuous transpose is

$$T' : V' \rightarrow V', \quad T'\lambda = \lambda \circ T.$$

Its algebraic transpose on the completion is

$$T'^\times : (V')^\times \rightarrow (V')^\times, \quad (T'^\times L)(\lambda) = L(T'\lambda).$$

No norm, measure, or choice of basis enters these definitions.

### 3.2 Complete weak-star continuous-dual lemma

**Lemma 3.1 (Dense complete continuous-dual envelope).** The locally convex duality facts used below are standard; see [7].

For every Hausdorff locally convex space  $V$ :

1.  $j_V : (V, \tau_V) \rightarrow ((V')^\times, \tau_{w*})$  is continuous, linear, and injective;
2.  $j_V(V)$  is dense in  $\mathcal{K}(V) = ((V')^\times, \tau_{w*})$ ;
3.  $\mathcal{K}(V)$  is Hausdorff and complete;
4. every continuous linear  $T : V \rightarrow V$  has the unique continuous extension  $T'^\times : \mathcal{K}(V) \rightarrow \mathcal{K}(V)$ ; and
5. this extension preserves identities and finite compositions.

**Proof.**

The continuous dual of a Hausdorff locally convex space separates points, so  $j_V$  is injective and the weak-star topology is Hausdorff. Moreover,

$$p_\lambda(j_V v) = |\lambda(v)|$$

is  $\tau_V$ -continuous for every  $\lambda \in V'$ ; hence  $j_V$  is continuous for the initial weak-star seminorms.

Embed  $(V')^\times$  into the product  $\mathbb{C}^{V'}$  by evaluation. It is the intersection of the closed linearity loci

$$L(a\lambda + b\mu) - aL(\lambda) - bL(\mu) = 0$$

for all  $a, b \in \mathbb{C}$  and  $\lambda, \mu \in V'$ . The product is complete, hence this closed subspace is complete.

To prove density, fix  $L \in (V')^\times$  and a basic weak-star neighborhood determined by  $\lambda_1, \dots, \lambda_n \in V'$ . The value vector  $(L(\lambda_1), \dots, L(\lambda_n))$  satisfies every linear relation satisfied by  $(\lambda_1, \dots, \lambda_n)$ . The evaluation map

$$V \longrightarrow \mathbb{C}^n, \quad v \longmapsto (\lambda_1(v), \dots, \lambda_n(v))$$

therefore contains that value vector in its image. Thus some  $v \in V$  agrees with  $L$  on all  $\lambda_i$ , which proves density.

For continuity of  $T'^\times$ ,

$$p_\lambda(T'^\times L) = p_{T'\lambda}(L).$$

The extension identity

$$T'^\times j_V(v) = j_V(Tv)$$

follows by evaluation. Density and Hausdorffness give uniqueness. The formulas

$$\text{id}'^\times = \text{id}, \quad (ST)'^\times = S'^\times T'^\times$$

follow directly.  $\square$

### 3.3 Branch-typed realization

For  $b \in \{A, B\}$ , set

$$V_b = (\Phi_{5b}, \tau_{5b}), \quad \mathcal{K}_{6b} = \mathcal{K}(V_b), \quad \iota_{6b} = j_{V_b}.$$

Here  $\tau_{5b}$  is the accepted WDW-5 locally convex topology: the branch- $A$  Bastiani test-function topology and, on branch  $B$ , the compact-open Bastiani topology in each completed exterior degree followed by the finite-particle locally convex direct sum. Denote the corresponding continuous dual by  $V_b'$ . The full algebraic dual  $V_b^\times$ , which contains the WDW-5 formal joint annihilators, remains a distinct object.

On branch  $B$ ,  $V_B$  is exactly the WDW-5 finite-particle, degree-wise completed exterior-section module. Passing to  $\mathcal{K}_{6B}$  completes the whole algebraic test module in the weak-star continuous-dual sense; it does not retroactively replace the completed exterior degrees, the finite-degree filtration, the strong-dual spinor carrier, or either CCR/CAR representation level on  $V_B$ .

The canonical pairing is

$$\langle L, \lambda \rangle_6 = L(\lambda), \quad L \in \mathcal{K}_{6b}, \quad \lambda \in V_b'.$$

No measure class is asserted. This pairing, rather than a silently chosen Gaussian measure, is the complete realization datum.

Every WDW-5 proper-redescription action  $\rho_b(g) : V_b \rightarrow V_b$  extends to

$$\rho_{6b}(g) = \rho_b(g)'^\times.$$

Because  $\rho_b(g^{-1})$  is its inverse on  $V_b$ , Lemma 3.1 makes  $\rho_{6b}(g)$  a bicontinuous automorphism. Proper and improper transformations retain the exact WDW-5 classification.

### 3.4 Kinematical carrier theorem

**Theorem W6.3 (Kinematical carrier and dense-core theorem).**

For every  $\alpha \in A_5$  and  $\beta \in B_5$ , the displayed  $\mathcal{K}_{6A}$  and  $\mathcal{K}_{6B}$  are complete Hausdorff locally convex carriers. The maps

$$\iota_{6b} : \Phi_{5b} \longrightarrow \mathcal{K}_{6b}$$

are continuous, injective, and dense. The full branch- $B$  fermionic filtration and both representation levels embed without identification, and every proper-redescription action extends bicontinuously.

**Proof.**

Completeness, Hausdorffness, injectivity, and density are Lemma 3.1 applied to each exact WDW-5 locally convex module. Continuity of  $\iota_{6b}$  follows because every defining target seminorm pulls back to the continuous functional  $F \mapsto |\lambda(F)|$ ,  $\lambda \in V'_b$ .

The WDW-5 branch- $B$  filtration consists of actual subspaces and continuous algebraic actions on  $\Phi_{5B}$ . Injectivity preserves their distinctions. Each redescription action and inverse extend by Lemma 3.1, so the extension is bicontinuous. No cross-branch map is introduced.  $\square$

**What W6.3 does not say**

The carrier is not a Hilbert space, no probability or kinematical measure is present, and no positivity or vacuum is inferred. The theorem is a genuine complete topological realization because the fixed standard permits a topological or Hilbert carrier. It is not yet a physical realization.

### 3.5 Complete operator monoid

Let  $\mathcal{P}_b$  be the complete inventory of WDW-5 primitives that are total continuous endomorphisms of  $V_b$ : proper redescrptions; the smooth-core CCR/CAR action; every spatial, internal-Gauss, six-direction Lorentz, and regulated Hamiltonian constraint; the Dirac/Yukawa  $d\Gamma$  actions; and each printed finite ordering component. WDW-5 proves totality and continuity on  $\Phi_{5b}$  for this inventory and for its admitted finite words. The full fixed-rung CCR/CAR seed remains, as in WDW-5, an action only on the initial seed algebra; it is not silently promoted to an endomorphism of  $V_b$  or of  $\mathcal{K}_{6b}$ .

Write  $\mathcal{W}_b$  for the monoid of finite words in the primitives whose WDW-5 domains compose. For  $w \in \mathcal{W}_b$ , let  $\tilde{w} = w'^\times$ . Lemma 3.1 gives

$$\widetilde{w_1 w_2} = \tilde{w}_1 \tilde{w}_2.$$

Define

$$\mathcal{D}_{6b} = \mathcal{K}_{6b}.$$

Every extended primitive and every word preserves  $\mathcal{D}_{6b}$ .

The joint graph topology is generated by

$$p_{\lambda, w}(L) = p_{\lambda}(\tilde{w}L) = |L(w'\lambda)|.$$

Since  $w'\lambda \in V'_b$ , every graph seminorm is already a weak-star seminorm. Conversely the empty word gives every  $p_{\lambda}$ . Therefore

$$\tau_{\text{graph}, b} = \tau_{w*, b}.$$

In particular, the joint graph domain is complete, Hausdorff, nonzero, and dense in itself, while  $\Phi_{5b}$  remains a dense invariant core.

### 3.6 Common graph-domain theorem

**Theorem W6.4 (Common invariant graph-domain theorem).**

For every record in  $\mathcal{R}_6^0$ , one nonzero complete domain  $\mathcal{D}_{6b} = \mathcal{K}_{6b}$  supports every extended constraint, declared transpose, finite ordered product, and finite commutator. All such maps are continuous, and their joint graph topology is exactly  $\tau_{w*, b}$ .

**Proof.**

Every word in  $\mathcal{W}_b$  is a continuous linear endomorphism of  $V_b$ . Lemma 3.1 extends it uniquely to a continuous endomorphism of  $\mathcal{K}_{6b}$ . The equality of the graph and weak-star seminorm families was proved above. The commutator

$$[\tilde{S}, \tilde{T}] = \tilde{S}\tilde{T} - \tilde{T}\tilde{S}$$

is consequently a continuous endomorphism of the same domain whenever the two WDW-5 products are admitted. No intersection of independently chosen unbounded domains occurs.  $\square$

### 3.7 Closedness and transpose data

A continuous linear map from a topological vector space to a Hausdorff topological vector space has closed graph. Hence every  $\tilde{T} = T'^{\times}$  is a closed operator with full domain  $\mathcal{K}_{6b}$  in the locally convex sense. Its continuous dual action on the defining dual pair is exactly

$$\tilde{T}'\lambda = T'\lambda, \quad \lambda \in V'_b,$$

because

$$\langle \tilde{T}L, \lambda \rangle_6 = \langle L, T'\lambda \rangle_6.$$

This is the complete adjoint-type datum available without a Hermitian kinematical form. Terms such as symmetric, self-adjoint, deficiency subspace, Friedrichs extension, and Krein extension are not typed on this carrier. They become relevant only after an additional Hilbert realization is supplied.

For completeness, if a future realization supplies a Hilbert carrier  $\mathcal{H}$ , a dense invariant core, and closed symmetric constraints, the imported extension theorem gives the following exhaustive classifier:

| $(n_+, n_-)$    | extension status                    |
|-----------------|-------------------------------------|
| $(0, 0)$        | unique self-adjoint closure         |
| $n_+ \neq n_-$  | no self-adjoint extension           |
| $n_+ = n_- > 0$ | unitary deficiency-space parameters |

Redescription-invariant parameters are unitary intertwiners of the two deficiency representations. Invariance alone leaves a scalar phase family when an intertwiner exists. Friedrichs or Krein selection requires the corresponding semibounded form-order endpoint hypothesis. None is presumed for the indefinite gravitational Hamiltonian.

### 3.8 Closed realization theorem

**Theorem W6.5 (Closed realization and extension-status theorem).**

Every WDW-5 primitive in the source-relative lift has a unique continuous closed realization on  $\mathcal{K}_{6b}$ , with full domain  $\mathcal{D}_{6b}$  and the exact transpose displayed above. There is no extension ambiguity inside this locally convex completion. Hilbert symmetry, self-adjointness, and deficiency indices are not asserted; any future Hilbert extension choice remains standing-bearing and is classified only under the explicit hypotheses of the imported self-adjoint-extension theorem.

**Proof.**

Existence, continuity, uniqueness, and preservation of finite composition follow from Lemma 3.1. Continuous maps into a Hausdorff carrier have closed graphs. The transpose identity follows from the evaluation pairing. Because no Hermitian form is part of the record, a Hilbert adjoint cannot be formed and no deficiency parameter is hidden. A later Hilbert carrier is new WDW-6 data; the imported classifier applies only after its hypotheses are proved.  $\square$

### 3.9 Module-A determination

Module A therefore establishes the carrier, dense embedding, complete common graph domain, continuous proper-redescription action, closed topological constraints, finite products and commutators, and transpose data. It deliberately does not call the carrier physical. The next question is whether the regulator-indexed Hamiltonian family converges on this carrier.

## 4. Module B: regulator endpoint, ordering, and quantum algebra

### 4.1 Two analytic estimates

Let  $\mathcal{L}_\phi^2$  be the infinite-dimensional scalar pivot space in either WDW-5 branch. For each  $\epsilon > 0$ , the scalar leg of the WDW-5 heat smoother is Hilbert–Schmidt. A continuum-removal claim requires its pivot realization to approach the identity strongly; otherwise the limiting principal functional differential is not the unregulated one.

**Lemma 4.1 (Hilbert–Schmidt approximate-identity divergence).**

Let  $\mathcal{H}$  be an infinite-dimensional Hilbert space and let  $S_\epsilon \in \mathcal{L}^2(\mathcal{H})$  be Hilbert–Schmidt for every  $\epsilon > 0$ . If

$$S_\epsilon x \longrightarrow x \quad \text{for every } x \in \mathcal{H},$$

then

$$\|S_\epsilon\|_2^2 \longrightarrow +\infty.$$

**Proof.**

Choose an orthonormal sequence  $(e_j)_{j \geq 1}$ . For every  $n$ ,

$$\|S_\epsilon\|_2^2 \geq \sum_{j=1}^n \|S_\epsilon e_j\|^2.$$

Strong convergence makes the right-hand side tend to  $n$ . Hence  $\liminf_{\epsilon \downarrow 0} \|S_\epsilon\|_2^2 \geq n$  for every  $n$ , which proves divergence.  $\square$

**Lemma 4.2 (Positive coefficient trace bound).**

If  $M$  is bounded and  $M \geq mI$  with  $m > 0$ , then for every Hilbert–Schmidt  $S$ ,

$$\text{Tr}(S^*MS) \geq m\|S\|_2^2.$$

**Proof.**

The positive trace-class operator  $S^*(M - mI)S$  has nonnegative trace.  $\square$

These lemmas remain valid auxiliary estimates. The source-certified witness below has a second-order scalar Hessian, rather than the unsupported pivot identity used in the first source, so the decisive estimate is the Laplace-type heat-trace calculation in Lemma 4.4.

## 4.2 The exact WDW-5 seed witness

Fix a branch  $b$ , its printed smooth chart center  $q_{b,*}$ , and a strictly positive smooth smearing  $W$ . WDW-5 Sections 6.1, 11.2, and 11.3 define  $\mathcal{L}_b$  to contain the inherited smeared scalar potential

$$P_{b,W}(q) = \frac{1}{2} \int_{\Sigma} W \sqrt{h_b(q)} (h_b(q)^{ij} D_i \phi(q) D_j \phi(q) + m^2 \phi(q)^2) d^3x, \quad m^2 \geq 0.$$

On branch  $B$ ,  $h_B = e^* \eta$ ; the clock scalar is a gauge singlet, so  $D_i \phi = \partial_i \phi$  in both branches. The exact source inclusion is

$$P_{b,W} \in \mathcal{L}_b \subset \mathcal{A}_{b,0}^{(0)} \subset \Phi_{5b}^{(0)} \subset \Phi_{5b}.$$

Put

$$\ell_{b,W,*} = dP_{b,W}(q_{b,*}) \in (\mathcal{H}_b^s)^*.$$

The WDW-5 CCR seed acts for every continuous  $\ell \in (\mathcal{H}_b^s)^*$ , and

$$\widehat{Q}_b[\ell] \mathbf{1}(q) = \ell(q - q_{b,*}) \in \mathcal{A}_{b,0}^{(0)}.$$

Therefore the centered functional

$$G_{b,W} = P_{b,W} - P_{b,W}(q_{b,*}) \mathbf{1} - \widehat{Q}_b[\ell_{b,W,*}] \mathbf{1}$$

is an exact WDW-5 seed element. More explicitly,

$$\begin{aligned} G_{A,W} &\in \mathcal{A}_A^{(0)} \subset \Phi_{5A}^{(0)} \subset \Phi_{5A}, \\ G_{B,W}^{(0)} &= G_{B,W} \otimes 1_{\Lambda^0} \in \mathcal{A}_{5B}^{(0)} \subset \Phi_{5B}^{(0)} \subset \Phi_{5B}. \end{aligned}$$

No assertion that  $\phi = 0$  lies in an arbitrarily chosen WDW-5 chart is needed. Direct differentiation at the actual chart center gives

$$G_{b,W}(q_{b,*}) = 0, \quad dG_{b,W}(q_{b,*}) = 0, \quad d^2 G_{b,W}(q_{b,*}) = d^2 P_{b,W}(q_{b,*}).$$

**Lemma 4.3 (Source-domain scalar witness).**

For every exact branch- $A$  or branch- $B$  WDW-5 candidate and every positive  $W$ , the displayed  $G_{b,W}$  belongs to the actual source module  $\Phi_{5b}$ ; on branch  $B$  it has exterior degree zero. Its value and full first differential vanish at  $q_{b,*}$ . Its scalar-scalar Hessian is the positive second-order form

$$d_{\phi\phi}^2 G_{b,W}(q_{b,*})[v, w] = \int_{\Sigma} W \sqrt{h_{b,*}} (h_{b,*}^{ij} D_i v D_j w + m^2 v w) d^3x.$$

The metric-metric or coframe-coframe Hessian block is order zero; the mixed metric-scalar or coframe-scalar block is order one.

**Proof.**

The membership chain is the WDW-5 definition of the inherited local potential algebra followed by Theorem W5.8 applied to  $\ell_{b,W,*}$ . Constants belong to the unital cylinder algebra. Subtraction therefore remains in the initial algebra, and tensoring by the degree-zero exterior unit gives the branch- $\mathcal{B}$  statement.

The affine correction has the same value and first differential as  $P_{b,W}$  at the chart center and zero second differential. Twice varying the scalar gives the displayed form. The dependence on  $h$ , or on  $e$  through  $h = e^*\eta$ , is algebraic in the metric coefficients and contains no derivative of a metric or coframe variation. A mixed variation can differentiate one scalar gradient. Those facts give the stated differential orders.  $\square$

### 4.3 The exact pivot trace

Choose a strictly positive Hamiltonian lapse  $N$ . Let

$$L_{\phi,*} = 1 + \nabla_{\phi,*}^* \nabla_{\phi,*}, \quad S_{\epsilon,\phi} = e^{-\epsilon^2 L_{\phi,*}/2}$$

be the scalar member of the exact WDW-5 reference elliptic family. At a finite controlling rung  $r$ , the scalar trace is the typed composition

$$\mathcal{L}_{\phi}^2 \xrightarrow{S_{\epsilon,\phi}^{(r)}} H^r \xrightarrow{B_{G_{b,W},r}^{\phi\phi}(q_{b,*})} H^{-r} \xrightarrow{M_{N\mathbb{K}_{\phi,b}(q_{b,*})}} H^{-r} \xrightarrow{S_{\epsilon,\phi}^{V,(r)}} \mathcal{L}_{\phi}^2.$$

Under the fixed WDW-5 reference-density and pivot identifications, cyclicity of the trace gives

$$\mathrm{Tr}_{\mathcal{L}_{\phi}^2} \left( S_{\epsilon,\phi}^{V,(r)} M_{N\mathbb{K}_{\phi,b}} B_{G_{b,W},r}^{\phi\phi} S_{\epsilon,\phi}^{(r)} \right) = \mathrm{Tr} \left( A_{\phi,b} e^{-\epsilon^2 L_{\phi,*}} \right),$$

where  $A_{\phi,b}$  is a second-order differential operator with scalar principal symbol

$$\sigma_2(A_{\phi,b})(x, \xi) = \frac{N(x)W(x)}{2} h_{b,*}^{ij}(x) \xi_i \xi_j.$$

Every density conversion in WDW-5 is a smooth positive order-zero identification, so it changes this symbol only by a smooth positive factor.

**Lemma 4.4 (Branch-uniform heat-trace dominance).**

For every exact WDW-5 reference elliptic family,

$$\mathrm{Tr} \left( A_{\phi,b} e^{-\epsilon^2 L_{\phi,*}} \right) = C_{\phi,b} \epsilon^{-5} + O(\epsilon^{-4}), \quad C_{\phi,b} > 0.$$

The complete metric-scalar or coframe-scalar trace of  $d^2 G_{b,W}(q_{b,*})$  has the same leading term:

$$\mathrm{Contr}_{\mathbb{K}_b, K_{\epsilon}} d^2 G_{b,W}(q_{b,*})[N] = C_{\phi,b} \epsilon^{-5} + O(\epsilon^{-4}).$$

**Proof.**

The standard heat-trace expansion for an order-two differential operator against a positive Laplace-type heat kernel on a three-manifold gives

$$C_{\phi,b} = (2\pi)^{-3} \int_{\Sigma} \int_{T_x^* \Sigma} \sigma_2(A_{\phi,b})(x, \xi) e^{-\sigma_2(L_{\phi,*})(x, \xi)} d\xi d\mu_*(x).$$

Positive  $N$ , positive  $W$ , positive  $h_{b,*}$ , and ellipticity of  $L_{\phi,*}$  make the integrand nonnegative and nonzero on a set of positive measure, hence  $C_{\phi,b} > 0$ .

The WDW-5 kinetic tensor and reference heat family are sector diagonal. Consequently the mixed Hessian blocks are off diagonal on the direct-sum pivot and have zero trace. Lemma 4.3 makes the metric-metric or coframe-coframe diagonal block order zero; its heat trace is  $O(\epsilon^{-3})$ , even though the DeWitt coefficient is indefinite. All scalar lower-order terms contribute at worst  $O(\epsilon^{-4})$ . No gauge, Higgs, or fermion Hessian block is present. The positive  $\epsilon^{-5}$  scalar term therefore controls the complete trace.  $\square$

#### 4.4 Ordering, fermionic, and potential isolation

For every admitted bosonic ordering, WDW-5 writes

$$\mathfrak{O}_o(\text{Contr } d^2)F = \text{Contr } d^2F + \mathcal{B}_o(dF).$$

Thus the first-order correction vanishes on  $G_{b,W}$  at  $q_{b,*}$ . The potential term vanishes because it multiplies  $G_{b,W}(q_{b,*}) = 0$ . On branch  $B$ ,  $d\Gamma_0(A'_N) = 0$ , while every explicitly recorded finite exterior ordering component acts linearly on the degree-zero value and therefore also vanishes at  $q_{b,*}$ . It follows from Lemma 4.4 that

$$\left(\widehat{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_b}^b[N]G_{b,W}\right)(q_{b,*}) = -\hbar^2 (C_{\phi,b}\epsilon^{-5} + O(\epsilon^{-4}))$$

for  $b = A, B$ . In particular, its absolute value tends to  $+\infty$ .

#### 4.5 Failure in the completed carrier

Define on the actual WDW-5 module

$$\lambda_{b,q_{b,*}}(F) = \text{pr}_0 F(q_{b,*}), \quad F \in \Phi_{5b},$$

where  $\text{pr}_0$  is the identity on branch  $A$  and the degree-zero projection on branch  $B$ . Point evaluation is continuous on the ambient compact-open Bastiani mapping space. Degree-zero projection is continuous for the branch- $B$  locally convex finite-degree direct sum, and restriction to the subspace topology on  $\Phi_{5b}$  remains continuous. Hence

$$\lambda_{b,q_{b,*}} \in (\Phi_{5b}, \tau_{5b})' = V_b' \subset V_b^\times.$$

Lemma 3.1 gives the exact extension identity

$$\widetilde{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_b}^b[N]\iota_{6b}(G_{b,W}) = \iota_{6b}(\widehat{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_b}^b[N]G_{b,W}).$$

Therefore

$$\begin{aligned} p_{\lambda_{b,q_{b,*}}}(\widetilde{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_b}^b[N]\iota_{6b}(G_{b,W})) &= |\lambda_{b,q_{b,*}}(\widehat{\mathcal{H}}_{\perp,\epsilon,\mathfrak{o}_b}^b[N]G_{b,W})| \\ &\longrightarrow +\infty. \end{aligned}$$

Weak-star convergence in  $\mathcal{K}_{6b}$  is therefore impossible. The obstruction is visible in the weakest topology used by the complete carrier; strengthening the topology cannot restore convergence.

#### 4.6 The regulator trilemma

The exact family proved in W5.7 uses the declared positive Laplace-type heat smoothers. For such a WDW-5 smoothing coordinate, exactly one of the following applies:

1. its heat scale tends to zero and it approaches the identity, in which case Lemma 4.4 and the typed detector prove divergence;
2. it does not approach the identity, in which case it is not removal of the smoothing from the principal functional carrier; or
3. it is retained at a nonzero physical scale, in which case the original continuum target has been changed unless physical inessentiality is separately proved.

A replacement profile not constructed by W5.7 is new candidate data and cannot be inserted into an exact lift by notation. The exact WDW-5 tuple contains no WDW-6 counterterm family, subtraction scheme, convergence topology for a renormalized family, or proof of regulator-independent states, forms, and observables. Such constructions may be legitimate, but they populate a new record in  $\mathcal{R}_6^{\text{ct}}$ .

#### 4.7 Regulator-endpoint theorem

**Theorem W6.6 (Unrenormalized regulator-obstruction theorem).**

No record in  $\mathcal{R}_6^0$  has a continuum regulator endpoint on the fixed target. For every branch, WDW-5 candidate index, admitted ordering, density convention, field-space connection, and declared Laplace-type heat smoother, either the heat scale fails to approach the identity, the source-certified centered scalar-potential witness diverges, or a retained regulator changes the target. Hence  $Q6R-07$  is obstructed for every exact unrenormalized lift.

**Proof.**

Lemma 4.3 places  $G_{b,W}$  in the actual source  $\Phi_{5b}$ . Lemma 4.4 proves the positive  $\epsilon^{-5}$  scalar leading term and shows that the lower-order indefinite metric or coframe contribution cannot cancel it. Section 4.4 eliminates every admitted first-order ordering, potential, and degree-zero fermionic correction at the chart center. Section

4.5 detects the resulting divergence by an element of the continuous dual defining the weak-star carrier. The three cases in Section 4.6 exhaust the exact W5.7 heat-smoothing endpoint. No counterterm datum occurs in  $\mathcal{R}_6^0$ .  $\square$

#### Scope of the obstruction

The theorem does not say that no renormalized Wheeler–DeWitt operator can exist. It says that the exact unrenormalized WDW-5 family does not itself possess the required continuum endpoint. A successful subtraction or a new regulator profile must be printed, typed, shown compatible with every constraint and anomaly, and re-verificationed as new WDW-6 data.

### 4.8 Ordering and smoothing comparison

The resolutioned witness has four stability properties:

1. its source membership is traced through  $\mathcal{L}_b \subset \mathcal{A}_b^{(0)} \subset \Phi_{5b}$ ;
2. its affine centering works at the actual center of every admitted chart, without assuming that a zero-scalar configuration lies in that chart;
3.  $G_{b,W} = dG_{b,W} = 0$  at that center, so potential and every admitted first-order ordering term vanish; and
4. the positive scalar principal heat coefficient has order  $\epsilon^{-5}$ , while the indefinite metric or coframe block is only  $O(\epsilon^{-3})$ .

Accordingly, no WDW-5 factor ordering, positive density convention, field-space connection correction, reference Laplace-type family, or declared heat scale removes the obstruction by presentation alone. A modification that cancels it is a counterterm or a change in principal operator data and must be registered accordingly.

### 4.9 Ordering and smoothing theorem

**Theorem W6.7 (Ordering/smoothing non-cancellation theorem).**

All exact WDW-5 orderings and declared identity-approaching Laplace-type heat profiles have the same first failure Q6R-07 on the source-certified witness. Equality of that failure does not make the orderings physically equivalent; no physical state, form, or observable intertwiner has been constructed.

**Proof.**

The ordering and field-connection corrections are linear in  $dG_{b,W}(q_{b,*})$ , the potential is multiplied by  $G_{b,W}(q_{b,*})$ , and the branch- $B$  finite-degree correction vanishes on the degree-zero value. Lemma 4.4 proves that every exact reference Laplace-type heat family retains a strictly positive scalar leading coefficient and that all other diagonal contributions have lower heat order. Physical equivalence requires the later data in Theorem W6.15 and therefore does not follow from a shared obstruction.  $\square$

### 4.10 Linear quantum constraints

The WDW-5 linear constraints are derivatives of the accepted proper groupoid actions:

$$\widehat{\mathcal{D}}[\xi] = -i\hbar\mathcal{L}_{X_\xi}, \quad \widehat{\mathcal{G}}[\lambda] = -i\hbar\mathcal{L}_{X_\lambda}, \quad \widehat{\mathcal{J}}[\vartheta] = -i\hbar\mathcal{L}_{X_\vartheta}.$$

They include every matter response and all six  $\mathfrak{spin}(1,3)$  directions. Differentiated group actions give

$$[\widehat{\mathcal{D}}[\xi], \widehat{\mathcal{D}}[\eta]] = -i\hbar\widehat{\mathcal{D}}[[\xi, \eta]]$$

and the corresponding gauge, Lorentz, and semidirect relations on  $\Phi_{5b}$ . These are identities of derivations, so no central term is present at this algebraic level. Lemma 3.1 transports every identity to  $\mathcal{D}_{6b}$ .

### 4.11 Linear quantum-constraint closure theorem

**Theorem W6.8 (Linear quantum-constraint closure theorem).**

The spatial-diffeomorphism constraints on both branches and the complete internal-Gauss and six-direction Lorentz constraints on branch  $B$  form the exact continuous transpose representation of their infinitesimal semidirect algebra on  $\mathcal{D}_{6b}$ . Every matter and fermionic response is retained, and there is no algebraic central term.

**Proof.**

WDW-5 constructs the operators as differentiated groupoid actions and proves their infinitesimal algebra on the common algebraic core. Continuous-dual extension preserves sums, scalar multiples, products, and hence commutators. Density of the core and continuity give the same identities on all of  $\mathcal{D}_{6b}$ .  $\square$

## 4.12 Hamiltonian commutators and structure functions

For each fixed  $\epsilon > 0$ , the continuous-dual construction makes

$$[\widetilde{\mathcal{H}}_\epsilon[N], \widetilde{\mathcal{D}}[\xi]], \quad [\widetilde{\mathcal{H}}_\epsilon[N], \widetilde{\mathcal{H}}_\epsilon[M]]$$

continuous endomorphisms of the declared common domain. Domain typing is therefore complete. WDW-5, however, supplies no equality between these regulated commutators and a structure-function combination, and Theorem W6.6 proves that the Hamiltonian family has no unrenormalized continuum endpoint. There is consequently no endpoint coefficient operator to which an exact, weak, quadratic-form, or cohomological closure claim could attach.

## 4.13 Hamiltonian and structure-function theorem

**Theorem W6.9 (Hamiltonian endpoint non-activation theorem).**

The regulated Hamiltonian products and commutators are well typed on  $\mathcal{D}_{6b}$ , but no record in  $\mathcal{R}_6^0$  reaches a continuum Hamiltonian commutator or operator-valued structure-function theorem. Clauses Q6R-10 and Q6R-11 are therefore not reached, and no classical hypersurface-deformation identity is promoted to replace them.

**Proof.**

The common-domain statement is Theorem W6.4. A continuum commutator requires a continuum Hamiltonian realization. Theorem W6.6 proves that no such realization exists in the declared unrenormalized class. The accepted classical bracket and the finite- $\epsilon$  existence of products do not supply the missing limit or equality.  $\square$

## 4.14 Module-B determination

The linear algebra survives on the complete common carrier. The Hamiltonian continuum endpoint does not. This first obstruction fixes the status of every exact lift before anomaly, cohomology, physical-state, inner-product, or observable gates are evaluated.

# 5. Module C: anomaly and cohomological status

## 5.1 Exact anomaly vocabulary

The fixed anomaly register distinguishes:

- local internal-gauge anomaly;
- local gravitational anomaly;
- mixed gauge-gravitational anomaly;
- global gauge and large-gauge obstruction;
- large-diffeomorphism phase;
- determinant-line holonomy;
- Pfaffian orientation;
- bordism anomaly;
- fermionic measure/polarization obstruction;
- regularization anomaly;
- boundary central extension; and
- total compatible anomaly class.

The statuses are VANISHES, TRIVIALIZED, CANCELLED, TARGET-UPDATE, OBSTRUCTED, and NOT-REACHED. Local cancellation in the three-generation Standard Model does not determine the global entries. Triviality of the principal bundle does not remove large gauge transformations. The empty boundary makes boundary inflow unavailable; it does not cancel a bulk obstruction.

## 5.2 Why the total anomaly class is not formed

A total anomaly certificate must belong to a specific endpoint quantum realization. It depends on:

1. the endpoint constraint representation;
2. the endpoint regulator and counterterm prescription;
3. the fermionic measure or determinant/Pfaffian line;
4. the action of large gauge transformations and mapping classes; and
5. compatibility of all local and global trivializations.

The exact unrenormalized lifts fail item 1 at the Hamiltonian endpoint and item 2 at the regulator gate. Attaching

a determinant line to the finite- $\epsilon$  algebraic family cannot resolution that failure. Conversely, a future counterterm prescription may itself carry a regularization anomaly and must be verificationed at its own endpoint.

### 5.3 Complete anomaly theorem

**Theorem W6.10 (Complete-anomaly non-promotion theorem).**

No record in  $\mathcal{R}_6^0$  has a total WDW-6 anomaly certificate. Every anomaly row is retained with status NOT-REACHED, except that boundary inflow is excluded by the fixed empty-boundary target. No local Standard Model cancellation, trivial-bundle assumption, classical closure identity, or finite- $\epsilon$  covariance is promoted to full anomaly freedom.

**Proof.**

A total anomaly class is an invariant of an endpoint quantum representation. Theorem W6.6 proves that the Hamiltonian family in every record lacks that endpoint. The determinant, Pfaffian, bordism, large-gauge, large-diffeomorphism, and regulator-trivialization data are not present in the accepted WDW-5 tuple. Therefore neither vanishing nor nonvanishing is licensed for those entries. The boundary entry is handled separately:  $\partial\Sigma = \emptyset$  gives no boundary charge or inflow term, and introducing one changes the target.  $\square$

### 5.4 Cohomological alternatives

No BRST charge, BFV complex, master constraint, group-averaging measure, or refined-algebraic-quantization rigging map is introduced in this manuscript. This is a substantive non-use decision:

- a BRST or BFV charge would require the endpoint constraint algebra and a typed common domain;
- a master constraint would require proof that its zero space, physical form, and observables agree with the original joint system; and
- group averaging would require an averaging object and convergence theorem for the actual proper-redescription groupoid.

None can be used to bypass Theorem W6.6.

### 5.5 Cohomological-equivalence theorem

**Theorem W6.11 (Cohomological non-use and non-necessity theorem).**

The source-relative classification uses no BRST/BFV, master-constraint, or group-averaging replacement. Because the exact records are already obstructed at *Q6R-07*, no cohomological construction is necessary to determine their status. Clause *Q6R-13* is satisfied only in the non-use sense and supplies no physical state or anomaly conclusion.

**Proof.**

The status proof uses the explicit scalar regulator witness and the four-valued classifier. It contains no cohomological replacement. Introducing one cannot remove the need for a continuum Hamiltonian realization or prove equality with the original joint system. Thus non-use is both explicit and sufficient for the bounded negative classification.  $\square$

## 6. Module D: state, physical form, and observables

### 6.1 The formal WDW-5 solution functor

For each candidate, WDW-5 defines

$$\text{Sol}_5(\mathbf{Q}) = \left\{ \Psi \in \Phi_5^\times : \widehat{C}_i^\times \Psi = 0 \text{ for every complete constraint label } i \right\}.$$

This object is typed and may be the zero subspace. Its construction does not prove:

- existence of a nonzero functional;
- continuity in a rigged topology;
- stability under regulator removal;
- positivity of a physical form; or
- separation by represented observables.

The continuous-dual carrier does not alter this conclusion. Its defining pairing uses  $V_b'$ , while the formal WDW-5 solution functor lives in the generally larger algebraic dual  $V_b^\times$ . Neither space comes with an exhibited nonzero joint annihilator, and the completion supplies no map that promotes a formal algebraic solution to a physical state.

## 6.2 State nontriviality after regulator failure

A physical state of the unrenormalized continuum theory would have to be annihilated by a continuum Hamiltonian constraint. Theorem W6.6 proves that there is no such operator in  $\mathcal{R}_6^0$ . It is therefore not meaningful to select a finite- $\epsilon$  annihilator and call it stable at the absent endpoint.

## 6.3 Nontrivial physical-state theorem

**Theorem W6.12 (Physical-state gate non-activation theorem).**

No nonzero continuum joint kernel or rigged solution carrier is constructed for a record in  $\mathcal{R}_6^0$ . The formal  $\text{Sol}_5$  functor is retained without nonemptiness promotion, and  $Q6R-14$  is not reached.

**Proof.**

The only inherited solution object is the algebraic joint annihilator, explicitly allowed to be zero. A WDW-6 state must solve the endpoint constraints and survive the endpoint regulator. Theorem W6.6 supplies no endpoint Hamiltonian, and no independent nonzero functional is exhibited. Therefore a state theorem cannot be formed.  $\square$

## 6.4 Physical form and null quotient

A physical form would require a specified solution carrier  $\mathcal{S}_6$  and a Hermitian sesquilinear form

$$\langle \cdot, \cdot \rangle_{\text{phys}} : \mathcal{S}_6 \times \mathcal{S}_6 \rightarrow \mathbb{C}$$

compatible with the complete constraints. Its radical

$$\mathcal{N}_{\text{phys}} = \{ \psi : \langle \psi, \varphi \rangle_{\text{phys}} = 0 \text{ for all } \varphi \}$$

must be quotiented, positivity proved on the quotient, and the resulting carrier completed.

The evaluation pairing of Module A is not such a form: it pairs the kinematical completion with the continuous test dual, is not a Hermitian form on a solution space, and has no positivity claim.

## 6.5 Physical inner-product theorem

**Theorem W6.13 (Physical-form gate non-activation theorem).**

No rigging map or physical sesquilinear form is constructed for  $\mathcal{R}_6^0$ ; hence no null radical, positive quotient, or physical completion is formed. Clauses  $Q6R-15$  and  $Q6R-16$  are not reached.

**Proof.**

Theorem W6.12 supplies no nonzero endpoint solution carrier on which a physical form could be defined. The Module-A evaluation pairing has a different type and cannot be renamed. The stated objects are therefore absent, and no positivity inference is made from the indefinite or non-Hilbert kinematical data.  $\square$

## 6.6 Physical observables and separation

A represented physical observable algebra must:

1. act on a dense physical domain;
2. preserve the solution space;
3. preserve the null radical;
4. descend to the positive quotient;
5. respect the physical star operation; and
6. separate physical rays or state functionals.

The formal commutant of finite- $\epsilon$  algebraic constraints satisfies none of those conditions merely by notation.

## 6.7 Physical observable theorem

**Theorem W6.14 (Physical-observable gate non-activation theorem).**

No represented physical star-algebra or separation theorem is constructed for  $\mathcal{R}_6^0$ . Clause  $Q6R-17$  is not reached, and no formal commutant is promoted to an observable algebra.

**Proof.**

Theorem W6.13 supplies neither a physical carrier nor a null quotient. Without those objects, descent and separation are not typed. The conclusion is therefore the exact gate status, not a claim that physical observables are impossible in every future realization.  $\square$

## 6.8 Modules C and D determination

The anomaly, state, form, and observable registers are completely adjudicated for the source-relative class: they are not reached after the regulator obstruction, and none is silently assigned a favorable value. This preserves the correct logical difference between a proved obstruction and an unattempted downstream positive construction.

## 7. Module E: physical equivalence, live residue, and handoffs

### 7.1 Complete WDW-6 matching record

Let  $\mathbf{R}_6(\mathbf{Q})$  and  $\mathbf{R}'_6(\mathbf{Q}')$  be fully populated WDW-6 realization records. A complete matching is the tuple

$$\mathfrak{M}_6 = (\mathfrak{M}_5, U_{\mathcal{X}}, U_{\mathcal{D}}, \mu_C, U_{\text{reg}}, U_{\text{anom}}, U_{\mathcal{S}}, U_{\text{phys}}, U_{\mathcal{O}}, U_{6 \rightarrow 7}, U_{6 \rightarrow 8}),$$

where:

1.  $\mathfrak{M}_5$  is an accepted complete WDW-5 matching;
2.  $U_{\mathcal{X}}$  is a bicontinuous carrier isomorphism satisfying  $U_{\mathcal{X}} \iota_6 = \iota'_6 T_5$ ;
3.  $U_{\mathcal{D}}$  maps the complete common graph domains bicontinuously;
4.  $\mu_C$  is a total bijection of complete constraint labels and

$$U_{\mathcal{X}} \overline{C}_i = \overline{C}'_{\mu_C(i)} U_{\mathcal{D}};$$

5.  $U_{\text{reg}}$  transports regulator nets, counterterms, convergence topology, and endpoint;
6.  $U_{\text{anom}}$  transports the compatible total anomaly trivialization;
7.  $U_{\mathcal{S}}$  transports the nonzero solution carrier;
8.  $U_{\text{phys}}$  is unitary or an explicitly typed rigged isometry of the physical form and null quotient;
9.  $U_{\mathcal{O}}$  is a represented star-algebra isomorphism compatible with the physical action; and
10.  $U_{6 \rightarrow 7}$  and  $U_{6 \rightarrow 8}$  transport the two continuation interfaces.

The relation  $\mathbf{R}_6 \sim_6 \mathbf{R}'_6$  holds exactly when such a tuple exists. No component may be replaced by the phrase “matched data.”

For the continuous-dual lift, a continuous WDW-5 map  $T_5 : V \rightarrow W$  induces

$$U_{\mathcal{X}}(L)(\mu) = L(\mu \circ T_5), \quad \mu \in W'.$$

When  $T_5$  is an isomorphism, this is a bicontinuous weak-star isomorphism with inverse induced by  $T_5^{-1}$ . Thus the Module-A carrier coordinate itself transports functorially.

### 7.2 Identity, inverse, and composition

The identity matching uses the identity in every component. The inverse of  $\mathfrak{M}_6$  is the componentwise inverse, with the label bijection  $\mu_C^{-1}$  and the transported regulator/anomaly data read in the opposite direction. If  $\mathfrak{M}_6^{12}$  and  $\mathfrak{M}_6^{23}$  are composable, define

$$\mathfrak{M}_6^{23} \circ \mathfrak{M}_6^{12}$$

componentwise and use  $\mu_C^{23} \circ \mu_C^{12}$  on constraint labels. Each intertwining square composes. The regulator, state, form, observable, and interface squares do likewise. Hence complete populated records and matchings form a groupoid.

### 7.3 Strengthened physical-equivalence theorem

**Theorem W6.15 (Strengthened physical-equivalence theorem).**

The complete matching tuple above defines an equivalence relation  $\sim_6$  on fully populated WDW-6 records. It preserves the WDW-5 candidate class, kinematical carrier, common domain, closed constraints, regulator endpoint, anomaly data, physical states and form, represented observables, and both continuation interfaces. The continuous-dual carrier transport is explicit and functorial.

**Proof.**

The identity tuple satisfies every square. Every component is invertible, so reversing the squares gives the inverse tuple. Componentwise composition and composition of  $\mu_C$  preserve each typed square. Associativity follows from associativity of map composition. The explicit continuous-dual formula is the transpose construction of Lemma 3.1 and therefore respects identities, inverses, and composition.  $\square$

## 7.4 Two-level status matrix

The exact classification distinguishes an unrenormalized realization record from its underlying WDW-5 candidate class.

| Object                                                                        | Branch | Status | First record | Reason                                                                       |
|-------------------------------------------------------------------------------|--------|--------|--------------|------------------------------------------------------------------------------|
| Canonical<br>source-relative lift<br>$\mathbf{R}_6^0(\mathbf{Q}_{5A,\alpha})$ | A      | KILLED | Q6R-07       | Source-certified<br>centered<br>scalar-potential heat<br>trace diverges      |
| Canonical<br>source-relative lift<br>$\mathbf{R}_6^0(\mathbf{Q}_{5B,\beta})$  | B      | KILLED | Q6R-07       | Same degree-zero<br>seed witness;<br>fermionic terms<br>vanish on it         |
| Underlying class<br>$[\mathbf{Q}_{5A,\alpha}]_5$                              | A      | OPEN   | Q6R-07       | No<br>counterterm-enriched<br>endpoint is<br>constructed or ruled<br>out     |
| Underlying class<br>$[\mathbf{Q}_{5B,\beta}]_5$                               | B      | OPEN   | Q6R-07       | Same; additional<br>fermionic/global<br>anomaly burdens<br>remain downstream |

These four rows are parametric: they apply to every  $\alpha \in A_5$  and  $\beta \in B_5$ . They do not collapse branch  $B$  into branch  $A$ .

Let

$$\text{Cert}_6 = \left\{ \mathbf{R}_6 : \bigwedge_{r=1}^{20} Q6R_r(\mathbf{R}_6) \right\}.$$

Theorem W6.6 gives

$$\text{Cert}_6 \cap \mathcal{R}_6^0 = \emptyset.$$

The counterterm-enriched class is outside the exact-lift quantifier at this stage, so it contributes no certified element to this sub-classification. Therefore

$$\text{Cert}_6 = \emptyset$$

within the declared construction class and

$$|\text{Cert}_6 / \sim_6| = 0.$$

## 7.5 Live-residue classification theorem

**Theorem W6.16 (Live-residue classification theorem).**

Every exact unrenormalized WDW-5 lift is killed first at  $Q6R-07$ . Every underlying branch- $A$  and branch- $B$  candidate class is open at the same clause because no theorem excludes all counterterm-enriched WDW-6 records. No record is conditional or certified. The certified physical-equivalence quotient has cardinality zero.

**Proof.**

Theorems W6.3–W6.5 occupy the carrier, dense-core, common-domain, and closed-topological-realization gates. Theorem W6.6 supplies a branch-uniform obstruction to the exact unrenormalized endpoint. It does not prove the nonexistence of every new subtraction scheme, so the parent WDW-5 classes are open rather than globally killed. The four-status partition of Theorem W6.2 then gives the displayed matrix. Since certification requires  $Q6R-07$ , the certified set is empty.  $\square$

## 7.6 Continuation interfaces

For a certified survivor  $\mathbf{R}_6$ , the planned interfaces would be:

$$\begin{aligned} \mathcal{I}_{6 \rightarrow 7}(\mathbf{R}_6) &= (\mathcal{U}_{\text{clk}}, \text{ClockFail}, \mathcal{S}_6, \langle \cdot, \cdot \rangle_{\text{phys}}, \mathcal{O}_{\text{phys}}), \\ \mathcal{I}_{6 \rightarrow 8}(\mathbf{R}_6) &= (\mathcal{H}_4^{\text{cov}}, \mathcal{S}_6, \mathcal{O}_{\text{phys}}, \mathcal{A}_{\text{approx}}, \mathcal{C}_{\text{compare}}, \mathcal{E}_{\text{error}}). \end{aligned}$$

The first contains no conditional evolution theorem; the second contains no semiclassical recovery theorem. Because the survivor domain is empty, there is a unique empty functor to each interface category. This proves

typing of the failure boundary but does not occupy the positive requirement that a surviving object carry the interface.

## 7.7 WDW-7 and WDW-8 interface theorem

### Theorem W6.17 (Empty-domain continuation-interface theorem).

The WDW-7 clock and WDW-8 recovery interface types are fixed as displayed, but no positive interface object is exported because  $\text{Cert}_6 = \emptyset$ . The only maps are the unique empty-domain functors. No clock dynamics or covariant/semiclassical recovery is claimed.

#### Proof.

The interface formulas use only coordinates prescribed by the fixed basis model. Theorem W6.16 makes their source category empty. A functor from the empty category is unique, but supplies no object in the target category.  $\square$

## 7.8 Certificate profile

For every source-relative lift, the twenty clauses have the profile:

| clauses     | status                                       |
|-------------|----------------------------------------------|
| $Q6R-01-06$ | SATISFIED                                    |
| $Q6R-07$    | OBSTRUCTED                                   |
| $Q6R-08$    | NOT REACHED as physical comparison           |
| $Q6R-09$    | SATISFIED                                    |
| $Q6R-10-17$ | NOT REACHED                                  |
| $Q6R-18$    | SATISFIED as classification/equivalence data |
| $Q6R-19-20$ | VACUOUS-NO SURVIVOR                          |

No conjunction  $\text{QSurv}_6$  is true.

## 7.9 Quantum-survival capstone

### Theorem W6.18 (Quantum-survival capstone).

Within the exact unrenormalized source-relative WDW-6 class considered in this part:

1. every WDW-5 candidate has a canonical complete continuous-dual carrier, dense embedding, common graph domain, and continuous closed constraint lift;
2. all linear quantum constraints close exactly on that domain;
3. every exact unrenormalized Hamiltonian lift fails continuum regulator removal at  $Q6R-07$ ;
4. counterterm-enriched realizations remain open and are not counted as survivors;
5. no anomaly, state, physical-form, observable, clock-dynamics, or recovery theorem is promoted across the failed gate; and
6. the certified quotient has exact cardinality

$$|\text{Cert}_6 / \sim_6| = 0.$$

No downstream physical-dynamics continuation is authorized from this exact-lift endpoint; any authorization must come from a separately typed successor.

#### Proof.

Items 1 and 2 are Theorems W6.3–W6.5 and W6.8. Item 3 is Theorem W6.6, stable under the admitted WDW-5 ordering and smoothing changes by Theorem W6.7. Theorem W6.9 prevents a Hamiltonian commutator promotion. Theorems W6.10–W6.14 give the exact downstream gate statuses. Theorems W6.15–W6.17 construct the equivalence classifier, compute the empty certified quotient, and type the empty continuation interfaces. Together they yield the six conclusions.  $\square$

## Part II

# Finite-cap counterterm construction

## 2. Exact source objects and notation

For  $b \in \{A, B\}$ , let

$$V_b = (\Phi_{5b}, \tau_{5b})$$

be the accepted WDW-5 test module with its compact-open Bastiani topology and, on branch  $B$ , its finite-particle locally convex direct-sum topology. Its initial algebra is

$$\mathcal{A}_b^{(0)} = \text{Cyl}_b^\infty \cdot \mathcal{L}_b$$

with the accepted branch-B exterior extension. WDW-5 defines

$$\Phi_{5b}^{(0)} = \text{Alg}_{\mathbb{C}}^{\text{gr}}(\mathcal{A}_b^{(0)})$$

and recursively closes under finite words in the total primitive family  $\mathcal{P}_b$ :

$$\Phi_{5b}^{(n+1)} = \text{Alg}_{\mathbb{C}}^{\text{gr}} \left( \Phi_{5b}^{(n)} \cup \{P(F) : F \in \Phi_{5b}^{(n)}, P \in \mathcal{P}_b\} \right), \quad \Phi_{5b} = \bigcup_{n \geq 0} \Phi_{5b}^{(n)}.$$

The primitive label includes its smearing, regulator, ordering, density, branch, and exterior-sector data. In particular, two Hamiltonian actions with different positive regulators are different primitive labels even when their displayed differential form is otherwise identical.

The exact W5.7 smoothing family is

$$S_{\epsilon,r} = e^{-\epsilon^2 L_{r,*}/2}, \quad K_{\epsilon,r} = e^{-\epsilon^2 L_{r,*}}, \quad \epsilon > 0,$$

on the compact boundaryless three-manifold  $\Sigma$ . Its decisive pivot composition is

$$\mathcal{L}_b^2 \xrightarrow{S_{\epsilon,b}^{(r)}} H^r \xrightarrow{B_{F,r}(q)} H^{-r} \xrightarrow{M_{N\mathbb{K}_b(q)}} H^{-r} \xrightarrow{S_{\epsilon,b}^{V,(r)}} \mathcal{L}_b^2.$$

The accepted WDW-6 scalar witness satisfies

$$G_{b,W} \in \Phi_{5b}, \quad p_{\lambda_{b,q_{b,*}}} \left( \widetilde{\mathcal{H}}_{\epsilon,b}^0 \iota_{6b}(G_{b,W}) \right) = C_b \epsilon^{-5} + O(\epsilon^{-4}), \quad C_b > 0.$$

This is a necessary Gate-II diagnostic. It is not used to define any Gate-I basis or coefficient.

**Theorem W6R.1 (Fixed handoff and no-amendment theorem).**

WDW-6R constructs only records in  $\mathcal{R}_6^{\text{ct}}$ . It neither changes nor reclassifies any record in  $\mathcal{R}_6^0$ . In particular,

$$\text{Cert}_6 \cap \mathcal{R}_6^0 = \emptyset$$

remains the accepted WDW-6 result, both exact unrenormalized branch lifts remain KILLED at Q6R-07, and no WDW-1–WDW-6 theorem is reopened by a later counterterm construction.

**Proof.**

The source component of a WDW-6R record is an exact WDW-5 candidate, but the record also contains a cap system, singular quotient, mixing data, counterterm data, and a renormalized endpoint. None of those additional fields occurs in the exact-lift record type exhausted by WDW-6. Hence the inclusion of record types is not  $\mathcal{R}_6^{\text{ct}} \subset \mathcal{R}_6^0$ , and a future member of the former cannot be a revived member of the latter.

The registered WDW-6 theorem quantifies over the exact unrenormalized lifts and proves their first failure at Q6R-07. WDW-6 explicitly retains counterterm-enriched constructions as open rather than placing them inside that quantifier domain. The accepted source record fixes the theorem text and candidate registers. WDW-6R imports them without alteration. Therefore the old empty intersection and both first-failure assignments remain true regardless of the status of any new record.  $\square$

## 4. Yang-Mills transfer firewall

### 4.1 Import classes

Each use of the Yang-Mills project receives exactly one label:

| Label                | Meaning in WDW-6R                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Abstract import      | A finite-dimensional or algebraic lemma is used only after its hypotheses are proved on the WDW cap |
| Proof architecture   | Only the order of construction is reused                                                            |
| WDW transfer theorem | The statement is proved anew on the exact WDW-5/6 carrier                                           |

The only Lane-A abstract import used at Gate I is the elementary finite block-lower-triangular inverse assembly. Directed-union uniqueness is reserved for Gate II. The finite-cap ordering of quotient, coefficient extraction, inverse control, and remainder compatibility is proof architecture.

None of the following is imported:

- the Yang-Mills engineering-dimension quotient;
- small-flow substitution;
- asymptotic freedom or weak-coupling control;
- flowed Euclidean states;
- Wilson/RG identification;
- insertion-RG or polymer bounds;
- reflection positivity or Osterwalder-Schrader reconstruction;
- bounded-base cyclicity;
- any sharp-local net or mass-gap conclusion.

The WDW heat expansion in Section 9 is a new theorem on the W5.7 spatial heat family. Its small parameter is the Wheeler-DeWitt regulator  $\epsilon$ , not Yang-Mills flow time, and its proof uses the WDW Hessian decomposition established in Section 8.

## 4.2 Counterterm-enriched comparison record at Gate I

For an exact candidate label  $(b, \eta)$ , the Gate-I partial record is

$$\mathbf{R}_{6R}^I(b, \eta) = \left( \mathbf{Q}_{5b, \eta}, \mathfrak{C}_b, \{\Phi_{5b}^\epsilon\}_c, \{\Phi_{5b}^{\epsilon, +}\}_c, \{\mathcal{J}_{b, \eta; \epsilon}^{\text{sing}}\}_c, \{\mathcal{C}_{c, g, \alpha}\}_{c, g, \alpha}, \{\text{Ins}_{\epsilon, c, g}\}_{c, g}, \{\mathcal{Y}_{\tau, c}\}_{\tau, c}, \widetilde{\mathbf{M}}, \mathbf{R} \right).$$

It is not yet a member of the constructed comparison class  $\mathcal{R}_6^{\text{ct}}$ , because the complete subtraction and renormalized operator fields required at Gate II are absent. The partial record is construction data, not a survivor.

### Theorem W6R.2 (Transfer-firewall and comparison-class theorem).

The Gate-I partial record is well typed over the exact WDW-5 candidate and contains no Yang-Mills physical or analytic conclusion. Every result used below is either an exact WDW-5/6 import, an elementary finite-dimensional abstract lemma with WDW hypotheses independently verified, or a new WDW theorem. In particular, no Yang-Mills grading, running coupling, state, positivity, reconstruction, or endpoint enters the WDW comparison class.

#### Proof.

The record components displayed above are defined solely from  $\mathbf{Q}_{5b, \eta}$ , the finite syntax of its WDW-5 primitive family, the W5.7 heat operator, and exact WDW quotient relations. Sections 6–14 construct every component. The sole imported inverse lemma says that a finite block-lower-triangular matrix is invertible when its diagonal blocks are invertible and supplies the usual finite recursion for its inverse. That statement contains no Yang-Mills datum.

The Yang-Mills archive is used only to verification the proof order that motivated the construction. No symbol  $u(\mu)$ , Yang-Mills flow, Euclidean state, Wilson observable, OS form, or local-net object occurs in the WDW record. The exact WDW imports are separately registered by hash. This exhausts all dependencies and proves the firewall.  $\square$

## 5. The finite-cap operator topology

Let  $W_\epsilon$  and  $W_\epsilon^+$  be the finite syntax sets constructed in Section 6. Their semantic spans are finite-dimensional subspaces of the accepted ambient section carrier:

$$\Phi_{5b}^\epsilon = \text{span}_{\mathbb{C}} \text{ev}(W_\epsilon), \quad \Phi_{5b}^{\epsilon, +} = \text{span}_{\mathbb{C}} \text{ev}(W_\epsilon^+).$$

W6R.4 freezes an ordered quotient basis and an ordered raw representative for each counterterm block before coefficient extraction. Those choices are registered normalization coordinates for the mixing matrix. Changing them conjugates the fixed-block principal identity and changes only the recorded lower-block coordinates; existence and nondegeneracy are basis-independent, while later physical equivalence of two normalization records remains W6R.14. Equip the finite spaces with the subspace topology inherited from the exact compact-open Bastiani/direct-sum topology. In finite dimension, that topology is the unique Hausdorff finite-dimensional locally convex topology.

For a finite collection of continuous seminorms  $\mathcal{P}_c^+$  that separates  $\Phi_{5b}^{c,+}$ , define

$$\|T\|_c = \max_{p \in \mathcal{P}_c^+} \sup_{\|F\|_c \leq 1} p(TF).$$

Any two such cap norms are equivalent. Every convergence or inverse bound at Gate I is in this declared finite-cap operator topology, with constants allowed to depend on  $c$ . No uniform estimate on the complete directed union is claimed.

The finite mixing spaces carry one more specific norm. After W6R.4 has fixed the ordered block bases, give each coordinate block its Euclidean norm and set

$$\left\| \bigoplus_g z_g \right\|_{\text{blk},c} = \max_g \|z_g\|_2.$$

All coefficient-space matrix norms in W6R.8–W6R.10 are the induced block-max operator norm. This norm is fixed before the mixing matrix is computed and generates the same finite-dimensional topology. In particular, for

$$\mathbf{D}_c(\epsilon)^{-1} = \text{diag}_g(\epsilon^{\rho_g} I_{d_g}), \quad 0 < \epsilon \leq 1,$$

one has the exact bound

$$\|\mathbf{D}_c(\epsilon)^{-1}\|_{\text{blk},c} = \max_g \epsilon^{\rho_g} \leq 1.$$

The cap operator seminorms continue to control the underlying WDW task maps; finite-dimensional coefficient extraction is continuous from that topology into this declared block norm.

## 5.1 Complete task-output carrier

The inverse-remainder statement uses a larger carrier than the local nonpositive coefficient block. This carrier is fixed with the finite task syntax, before a heat coefficient or remainder is evaluated. For every  $\tau \in \mathfrak{T}_\Gamma$ , let  $\mathcal{Z}_{\tau,c}^{\text{reg}}$  be the finite tagged packet of the predeclared regular output slots of that task:

$$\mathcal{Z}_{\tau,c}^{\text{reg}} = \mathcal{Z}_{\tau,c}^{\text{loc},+} \oplus \mathcal{Z}_{\tau,c}^{\text{fr}} \oplus \mathcal{Z}_{\tau,c}^{\text{sm}}.$$

The first summand contains terminal positive-order local outputs, the second the separated finite-rank outputs, and the third the smoothing outputs. Every summand remains decomposed by the branch, exterior degree, constraint role, density, tensor, and target labels already carried by  $\Gamma$ . In particular, a branch-B completed exterior target remains present in its complete affected degree; it is not replaced by a scalar coordinate or suppressed as a fourth untyped channel.

The displayed sum is the external direct sum of syntax-tagged slots. Finite-rank and smoothing operators can have overlapping semantic images, so their tags are not erased before the estimate. The exact finite task tree supplies one tagged tuple, and a continuous semantic evaluation map sends that tuple to the actual cap operator. Proving convergence in the tagged carrier is therefore stronger than proving convergence only after possible cancellations among semantic channel images.

After W6R.4 freezes the coefficient source and target spaces  $\mathcal{S}_c$  and  $\mathcal{T}_c$ , define

$$\tilde{\mathcal{S}}_{\tau,c} = \mathcal{S}_c \oplus \mathcal{Z}_{\tau,c}^{\text{reg}}, \quad \mathcal{Y}_{\tau,c} = \mathcal{T}_c \oplus \mathcal{Z}_{\tau,c}^{\text{reg}}.$$

These are syntax-tagged direct sums, not complements selected after a response map is known. Write their canonical injections and projections as

$$\iota_T, \iota_{\text{reg}}, \quad \pi_T, \pi_{\text{reg}}.$$

They obey

$$\pi_T \iota_T = I, \quad \pi_{\text{reg}} \iota_{\text{reg}} = I, \quad \pi_T \iota_{\text{reg}} = 0, \quad \pi_{\text{reg}} \iota_T = 0,$$

and the literal reconstruction identity

$$y = \iota_T \pi_T y + \iota_{\text{reg}} \pi_{\text{reg}} y \quad (y \in \mathcal{Y}_{\tau,c}).$$

Give every finite tag-coordinate span a separating norm, use their finite maximum as  $\|\cdot\|_{\text{reg},\tau,\mathfrak{c}}$ , and set

$$\|y\|_{\text{full},\tau,\mathfrak{c}} = \max \{ \|\pi_T y\|_{\text{blk},\mathfrak{c}}, \|\pi_{\text{reg}} y\|_{\text{reg},\tau,\mathfrak{c}} \}.$$

The four canonical maps then have norm at most one. The tagged semantic evaluation of the regular summand into  $\mathcal{L}(\Phi_{5b}^\mathfrak{c}, \Phi_{5b}^{\mathfrak{c},+})$  is continuous, so convergence in this full norm implies convergence in every cap operator seminorm used downstream. The task tags and coefficient slots are functorial under cap enlargement: on a shared task packet the injections and projections commute with the literal inclusions. This is a carrier-typing statement only; it does not assert the Gate-II overlap equality of renormalized endpoints.

## 6. Directed WDW syntax caps

### 6.1 Exact finite label packet

A cap is

$$\mathfrak{c} = (E, N, r, j, \Delta, \Gamma).$$

The six entries have the following exact meanings.

1.  $E$  is a finite set of elements of the initial algebra  $\mathcal{A}_b^{(0)}$ , each accompanied by one exact WDW-5 syntax witness.
2.  $N$  bounds algebraic word length and finite product length.
3.  $r$  bounds functional-differential order appearing in a registered word or task.
4.  $j$  bounds spatial jet order of its local coefficient.
5.  $\Delta$  bounds the predeclared renormalization weight.
6.  $\Gamma$  is a finite typed packet. It contains:
  - a finite subset of exact WDW-5 primitive labels, including every selected smearing, positive internal regulator, ordering, and density coordinate;
  - a finite subset of branch, exterior-degree, ghost, symmetry, and constraint-role labels;
  - a finite list  $\mathfrak{T}_\Gamma$  of verification tasks, such as one outer regulated Hamiltonian action, a selected linear-constraint action, one product, or one ordered commutator.

Thus “all linear constraints on a cap” means all finitely many fully specified linear-constraint labels in  $\Gamma$ , not the infinite smearing space. Every individual WDW-5 action is eventually present, because the directed system ranges over all finite label packets.

### 6.2 Raw finite word set

Let  $W_0(E) = E \cup \{1\}$ . Given  $W_k(E, \Gamma)$ , form  $W_{k+1}(E, \Gamma)$  by adjoining:

- every product of already formed words whose total length is at most  $N$ ;
- $P_\gamma(w)$  for every primitive label  $\gamma \in \Gamma$ , whenever the exact WDW type rules admit the expression and its bounds do not exceed  $(r, j, \Delta)$ ;
- finite sums only at the semantic-span stage, rather than as new syntax.

Stop at  $k = N$ , and write the resulting finite set as  $W_\mathfrak{c}$ . The evaluation map

$$\text{ev}_{b,\eta} : W_\mathfrak{c} \longrightarrow \mathcal{E}_{5b}^{\text{sec}}$$

is the WDW-5 semantic evaluation. Totality of every primitive on the accepted ambient section algebra makes every word defined. Set

$$\Phi_{5b}^\mathfrak{c} = \text{span}_\mathbb{C} \text{ev}_{b,\eta}(W_\mathfrak{c}).$$

There are finitely many words because the starting and operator alphabets are finite, the word depth is bounded, and every formation rule has finite arity. Semantic identities may reduce the dimension, never increase it:

$$\dim_\mathbb{C} \Phi_{5b}^\mathfrak{c} \leq |W_\mathfrak{c}| < \infty.$$

This is the finiteness proof. No engineering-dimension statement is used.

### 6.3 Finite task saturation

For each  $\tau \in \mathfrak{T}_\Gamma$ , let  $\tau(w_1, \dots, w_{a_\tau})$  be the fully typed finite syntax tree of that task, with  $a_\tau < \infty$ . Define

$$W_\mathfrak{c}^{\text{pre},+} = W_\mathfrak{c} \cup \left\{ \tau(w_1, \dots, w_{a_\tau}) : \tau \in \mathfrak{T}_\Gamma, \quad w_i \in W_\mathfrak{c} \right\}.$$

At Gate I, coefficient-extraction tasks add only the finitely many formal coefficient slots constructed in Section 7. They do not presume that a counterterm endpoint exists. The final Gate-I task set  $W_c^+$  is the union of the pre-saturation and those finite slots. Once the quotient basis in Section 7.4 is fixed, the slots are realized by the finitely many concrete basis-operator applications  $\mathcal{C}_{c,g,\alpha}(w)$ , with  $w \in W_c$ . This realizes the declared counterterm actions without selecting a subtraction coefficient or assuming a renormalized endpoint. Its semantic span is

$$\Phi_{5b}^{c,+} = \text{span}_{\mathbb{C}} \text{ev}_{b,\eta}^+(W_c^+).$$

This is deliberately not the smallest invariant subspace under an infinite family of actions. It is the output space required by a finite registered calculation. Its construction terminates after the printed finite task list, so

$$\dim \Phi_{5b}^{c,+} < \infty.$$

## 6.4 Directed order and exhaustion

Write  $c \preceq c'$  when

$$E \subseteq E', \quad N \leq N', \quad r \leq r', \quad j \leq j', \quad \Delta \leq \Delta', \quad \Gamma \subseteq \Gamma'.$$

Any two caps have an upper bound obtained by taking unions in the finite entries and maxima in the numerical entries. Hence the family is directed.

Every  $F \in \Phi_{5b}$  is, by W5.11, a finite complex linear combination of semantic values of finite primitive words. Only finitely many initial generators, primitive labels, smearings, positive internal regulators, orderings, and branch/exterior labels occur in that expression. Their union is one finite  $E, \Gamma$ , and the expression has finite bounds  $(N, r, j, \Delta)$ . Therefore  $F \in \Phi_{5b}^c$  for some  $c$ . Conversely every cap word is an admitted WDW-5 word. Thus

$$\Phi_{5b} = \bigcup_{c \in \mathcal{C}_b} \Phi_{5b}^c.$$

### Theorem W6R.3 (Directed finite-cap exhaustion theorem).

For each exact WDW-5 branch and candidate label, the construction above defines a directed family of finite-dimensional semantic caps and finite-dimensional task-saturated enlargements. The caps exhaust the exact WDW-5 module, and task saturation terminates without assuming any renormalized endpoint.

#### Proof.

Finite-word induction gives the cardinality bound in Section 6.2. The finite task list applied to a finite Cartesian power of the finite word set gives the saturation bound. Union and maximum give a common upper cap. The two inclusions in the exhaustion identity are proved in Section 6.4: one uses the finite-word definition of  $\Phi_{5b}$ , and the other uses the fact that every cap label belongs to the exact primitive family. No closure under an unbounded label family is taken.  $\square$

## 7. Global off-shell quotient and finite counterterm blocks

### 7.1 Predeclared structural filtration

Before any heat coefficient is computed, fix the lexicographic renormalization label

$$\text{deg}_{\text{ren}} = (\rho, r_{\text{fun}}, j_{\text{sp}}, d_{\text{field}}, c_{\text{role}}, s_{\text{sym}}).$$

Here  $\rho$  is the predicted heat order; the next three entries are the functional, spatial-jet, and field orders;  $c_{\text{role}}$  records Hamiltonian, spatial, Gauss, Lorentz, or ordering role; and  $s_{\text{sym}}$  records the branch, density, tensor, exterior, chirality, and internal symmetry type. Exact integration-by-parts relations can redistribute spatial jets, so this is used as a finite filtration. Its associated blocks are formed only after the exact off-shell quotient below. Heat coefficients play no role in choosing the order or the blocks.

### 7.2 Directed raw symbol system

Each word in  $W_c^+$  contains finitely many local coefficient expressions and has bounded functional and jet order. Let  $J_{b,\eta;c}^{\text{raw}}$  contain every typed local monomial obtainable from:

- the jets, through order  $j$ , of the fields and source coefficients in the word packet;
- the principal and lower symbols of the registered  $L_{r,\star}$ ;
- the kinetic tensor, lapse, density, ordering, and field-space connection coefficients in  $\Gamma$ ;
- all finite contractions permitted by the branch, exterior, and constraint-role types; and

- every coefficient order capable of contributing a nonpositive power of  $\epsilon$  in spatial dimension three.

No arbitrary smooth function or target vector is inserted. A cylinder function in  $E$  is one symbol together with the finitely many derivatives required by the cap. A branch-B strong-dual coefficient is carried by its exact source label rather than replaced by a basis of its infinite ambient carrier.

Put

$$\mathbf{V}_\epsilon = \text{span}_{\mathbb{C}} \mathbf{J}_{b,\eta;\epsilon}^{\text{raw}}.$$

The raw packet is finite. It is chosen functorially so that  $\epsilon \preceq \epsilon'$  gives  $\mathbf{V}_\epsilon \subseteq \mathbf{V}_{\epsilon'}$ . The algebraic directed union

$$\mathbf{V}_\infty = \bigcup_{\epsilon} \mathbf{V}_\epsilon$$

is the source-wide raw local-symbol space.

### 7.3 One global off-shell relation space

Let  $\mathbf{R}_\infty^{\text{off}} \subset \mathbf{V}_\infty$  be the single linear span of all instances of the exact relation schemes in the quotient registry:

- total spatial divergences on the closed target;
- tensor and graded algebraic symmetries;
- geometric and gauge Bianchi identities;
- integration by parts in the registered smooth coefficient classes; and
- exact Clifford, exterior, representation, and density identities.

Every relation is an identity of typed local operator maps on the entire smooth WDW source. Classical field equations, quantum constraints, physical-state equations, regulator limits, and equations valid only after renormalization are excluded. Define once and for all

$$q : \mathbf{V}_\infty \longrightarrow \mathcal{J}_\infty := \mathbf{V}_\infty / \mathbf{R}_\infty^{\text{off}}.$$

The cap object is the image subspace

$$\mathcal{J}_{b,\eta;\epsilon}^{\text{sing}} = \mathcal{J}_\epsilon := q(\mathbf{V}_\epsilon) \subset \mathcal{J}_\infty.$$

Equivalently,

$$\mathcal{J}_\epsilon \cong \mathbf{V}_\epsilon / (\mathbf{R}_\infty^{\text{off}} \cap \mathbf{V}_\epsilon).$$

Because  $\mathbf{V}_\epsilon$  is finite dimensional, the intersection has a finite basis  $(r_{\epsilon,1}, \dots, r_{\epsilon,k_\epsilon})$ . Each basis vector is, by the algebraic-span definition of  $\mathbf{R}_\infty^{\text{off}}$ , a finite linear combination of printed exact relation instances. Record those finite decompositions as the cap relation certificate and row-reduce them against the fixed raw packet. This produces an ordered quotient basis without inspecting a heat coefficient. The certificate may depend on the cap; the quotient map and the represented classes do not.

This is not a family of unrelated cap quotients. If  $\epsilon \preceq \epsilon'$ , both are literal subspaces of the same global quotient and the refinement map is the inclusion

$$\iota_{\epsilon\epsilon'} : \mathcal{J}_\epsilon \hookrightarrow \mathcal{J}_{\epsilon'}.$$

In particular, a larger cap cannot create a retrospective relation that collapses an earlier class: every global off-shell consequence was already used by  $q$ .

### 7.4 Fixed finite blocks and completeness

The filtration of Section 7.1 descends to  $\mathcal{J}_\infty$ , because the exact relation schemes preserve heat weight, functional order, role, and symmetry and are filtered in spatial-jet order. For a finite cap, take the associated blocks

$$\mathcal{S}_{\epsilon,g} = \text{Gr}_g \mathcal{J}_\epsilon, \quad \mathcal{T}_{\epsilon,g} = \text{a typed target copy of } \mathcal{S}_{\epsilon,g}.$$

The source and target blocks, their dimensions, their exact types, and ordered quotient bases are fixed before any coefficient extraction. For each basis class, the row-reduction certificate also freezes its first ordered raw representative. This representative section is normalization data. It is not changed after a heat matrix is seen. No kernel of a heat-coefficient map, image restriction, pivot row, or post-calculation dimension choice is allowed to change them.

The raw packet consists of typed local operator densities, not unattached scalar labels. Write the first ordered representative of the  $\alpha$ -th class in block  $g$  as

$$\hat{\theta}_{\epsilon,g,\alpha}[F](q, x).$$

It includes the fixed lapse, density conversion, branch, tensor, exterior, constraint-role, functional-differential, and spatial-jet data. Before any response coefficient is evaluated, define the concrete cap counterterm operator

$$(\mathcal{C}_{\mathfrak{c},g,\alpha} F)(q) = \int_{\Sigma} \hat{\theta}_{\mathfrak{c},g,\alpha}[F](q, x) d\mu_{\star}(x).$$

For a branch-B target-valued density, the integral is taken in the complete affected exterior-degree target and is characterized by all its continuous seminorms. Exact off-shell relations are identities of these operator maps, so changing a representative within its quotient class does not change the represented class. The fixed first representative fixes the finite- $\epsilon$  normalization used below.

Each  $\mathcal{C}_{\mathfrak{c},g,\alpha}$  is a total map

$$\Phi_{5b}^{\epsilon} \longrightarrow \Phi_{5b}^{\epsilon,+}.$$

This is part of the finite task saturation in Section 6.3. It neither chooses its eventual coefficient nor adds it to the Hamiltonian. The ordered operator family

$$\mathcal{B}_{\mathfrak{c},g}^{\text{ct}} = (\mathcal{C}_{\mathfrak{c},g,1}, \dots, \mathcal{C}_{\mathfrak{c},g,d_g})$$

is therefore fixed before heat evaluation, determinant calculation, or inverse construction.

Completeness has the exact cap-relative meaning:

Every local coefficient in the heat expansion of a registered cap task represents a class in  $\mathcal{J}_{\mathfrak{c}}$ .

It does not mean that the quotient contains every counterterm proposed in every quantum-gravity theory. Lemma 8.1 confines cap Hessians to finite-order diagonal differential operators plus finite-rank or smoothing parts. Symbol composition uses only the jets and contractions above, and W6R.5 shows that only finitely many can occur at nonpositive heat order.

**Theorem W6R.4 (Hereditary global-quotient and fixed-block theorem).**

For every exact branch, candidate, and cap,  $\mathcal{J}_{\mathfrak{c}} = q(\mathcal{V}_{\mathfrak{c}})$  is a finite-dimensional subspace of the single global off-shell quotient. Cap refinement is injective, and

$$\mathcal{R}_{\infty}^{\text{off}} \cap \mathcal{V}_{\mathfrak{c}} = \ker(q|_{\mathcal{V}_{\mathfrak{c}}}).$$

The predeclared associated blocks are complete for every local nonpositive-order coefficient of the registered Gate-I tasks. Their ordered quotient bases have concrete total operator realizations  $\mathcal{C}_{\mathfrak{c},g,\alpha}$  on the cap, fixed before any heat response is evaluated. No on-shell relation or heat-response kernel is used to define the quotient, its blocks, their dimensions, or those operators.

**Proof.**

The equality with the restricted kernel is the definition of a quotient map. If  $\mathfrak{c} \preceq \mathfrak{c}'$ , then  $\mathcal{V}_{\mathfrak{c}} \subseteq \mathcal{V}_{\mathfrak{c}'}$ , so their images under the same map  $q$  are nested subspaces. This proves the refinement injection without a cap-dependent relation identity. Finite-dimensionality follows from the finite raw packet. The structural Hessian and heat-expansion theorems show that coefficient extraction stays inside the packet. A finite basis of the global-relation intersection exists because it is a subspace of the finite cap space, and every member has a finite relation-instance decomposition by definition of algebraic span. This gives the printed finite row-reduction certificate. Finally, the associated blocks are formed from the globally quotiented, predeclared filtration before any heat matrix is evaluated. Every fixed representative is a typed local operator density from the raw packet, so integration gives the displayed cap action. The finite application slots in Section 6.3 make its output part of  $\Phi_{5b}^{\epsilon,+}$ ; W6R.6 below proves its locality, equivariance, and continuity.  $\square$

## 8. Structural Hessian lemma

The analytic step requires more than ambient Bastiani smoothness. Define  $\mathfrak{CL}_{b,\Gamma}$  to be the finite-word class whose scalar coefficients are finite sums of products of:

1. finite-probe cylinder functions;
2. finitely many integrals of local densities depending on finite spatial jets of the exact fields and registered source coefficients;
3. fixed positive-internal-regulator heat-kernel diagonals and their finite source derivatives;
4. on branch  $B$ , degreewise smooth exterior sections obtained by the registered finite  $d\Gamma$ , exterior multiplication, contraction, Dirac, Yukawa, and ordering maps.

Products of local integrals are allowed. Their cross variations are finite sums of separated kernels and belong to the finite-rank channel.

**Lemma 8.1 (Cap structural Hessian decomposition).**

For each finite cap there is an integer  $m_c < \infty$  such that every scalar or degree-wise target-valued word  $F \in \Phi_{5b}^{c,+}$  has, locally on the smooth configuration core, a Hessian decomposition

$$B_{F,r}(q) = A_F(q) + K_F(q),$$

where:

- $A_F(q)$  is a finite-order differential operator, supported on the spatial diagonal, of order at most  $m_c$ ;
- $K_F(q)$  is a finite sum of finite-rank or smoothing kernels;
- all coefficient jets belong to the raw packet of Section 7;
- the decomposition and every configuration derivative required by the cap is uniform on compact derivative-test sets;
- in branch  $B$ , the assertions hold after every continuous seminorm on the finitely many affected completed exterior degrees.

**Proof.**

Initial cylinder Hessians have finite rank. Initial smeared local functionals have diagonal finite-order differential Hessians by W5.5 and the definition of  $\mathcal{L}_b$ . The product rule preserves the stated sum: diagonal-by-coefficient terms remain diagonal, while cross products of first variations are separated finite sums.

Constant configuration differentiation, spatial/gauge/Lorentz Lie derivatives, and multiplication by registered local coefficients preserve finite differential order. For a fixed positive internal regulator  $\epsilon_0$ , the W5.7 trace of a diagonal differential Hessian is an integral of its local coefficients against the smooth fixed diagonal of  $e^{-\epsilon_0^2 L}$ ; it therefore remains in the local-density class. Its action on a finite-rank Hessian remains a finite-probe smooth coefficient. Repeating a finite number of primitive actions increases the orders only finitely.

On branch  $B$ , W5.10–W5.11 prove that the one-particle transposes,  $d\Gamma_n$  maps, finite ordering maps, and target-valued regulated trace are continuous in every completed degree and Bastiani smooth in  $q$ . The projective-tensor Leibniz rule gives the same finite local versus finite-rank decomposition after each of the finitely many cap operations. Compact derivative-test sets and the finite cap allow the finitely many local estimates to be replaced by their maximum.  $\square$

## 9. Capwise WDW heat expansion

### 9.1 Closed-manifold heat lemma

The parameter-dependent expansion used here is the classical closed-manifold heat calculus [8, 9].

Let  $L$  be one of the positive Laplace-type operators  $L_{r,\star}$  on a smooth finite-rank bundle over the compact three-manifold  $\Sigma$ , with  $\partial\Sigma = \emptyset$ . If  $A$  is a differential operator of order  $m$  with smooth coefficients, the parameter-dependent heat calculus gives

$$\mathrm{Tr}(Ae^{-\epsilon^2 L}) \sim \sum_{k \geq 0} a_k(A, L) \epsilon^{k-3-m}.$$

For each integer  $K$ , subtraction through  $k = K$  leaves a remainder bounded by the next positive power in every fixed finite family of coefficient seminorms. Each  $a_k(A, L)$  is the integral of a universal local contraction of the symbols and their finite jets.

There are no boundary coefficients because  $\partial\Sigma = \emptyset$ . There are no logarithmic or mixed power-logarithmic coefficients for the classical differential-operator/Laplace-type heat trace used here. Those rows are part of the complete register and have coefficient zero. This is not a general no-log theorem for arbitrary pseudodifferential or boundary problems.

If  $K$  is finite rank or smoothing, then

$$\mathrm{Tr}(Ke^{-\epsilon^2 L}) \longrightarrow \mathrm{Tr}(K)$$

as  $\epsilon \downarrow 0$ . For one fixed  $K$ , this is trace-norm continuity. For the finite-dimensional span of  $K$ 's appearing in a cap, it is uniform on bounded sets.

### 9.2 Reduction of the W5.7 pivot

For the diagonal component in Lemma 8.1, set

$$A_{F,b,N}(q) = M_{N\mathbb{K}_b(q)} A_F(q)$$

with the exact reference-density and bundle identifications of W5.7. Ideal cyclicity, first on the smooth core and then by the W5.7 trace-class extension, gives sectorwise

$$\begin{aligned} \text{Tr}_{\mathcal{L}_b^2} \left( S_{\epsilon,b}^{\vee,(r)} M_{N\mathbb{K}_b(q)} A_F(q) S_{\epsilon,b}^{(r)} \right) \\ = \text{Tr} \left( A_{F,b,N}(q) e^{-\epsilon^2 L_{b,*}} \right). \end{aligned}$$

The finite-rank/smoothing component is treated by the second part of the heat lemma. This reduction retains the indefinite DeWitt block. No positive-definite replacement and no inverse of the kinetic tensor is used.

The gravitational/coframe, scalar, gauge, and Higgs sectors are a finite direct sum at a fixed branch and cap. Branch-B exterior degree is a target degree, not an additional trace direction. Hence the scalar heat calculation applies after each continuous target seminorm in the degree-wise complete carrier, exactly as in W5.7.

### 9.3 Complete elementary capwise expansion

Let  $P_\epsilon \subset (0, \infty)$  be the finite set of positive numbers  $\rho = 3 + m - k$  that occur before the exponent  $k - 3 - m$  becomes nonnegative, across the elementary action leaves of the finite cap task packet. A product or commutator in which several outer regulators share the same variable is not obtained by iterating this single-leaf statement; its joint expansion is proved separately in W6R.10. For  $F \in \Phi_{5b}^\epsilon$ , define the coefficient maps by the exact heat-symbol extraction:

$$\mathcal{C}_{\rho,b,\eta;\epsilon}[N]F = \sum_{\substack{\text{cap sectors} \\ k-3-m=-\rho}} a_k(A_{F,b,N}, L_{b,*}).$$

The finite coefficient is the sum of:

- the  $\epsilon^0$  differential heat coefficient, when present;
- the strong finite-rank/smoothing limit;
- the exact potential, first-order ordering, Dirac, and Yukawa terms that are independent of the outer regulator;
- every lower-order heat coefficient whose exponent is zero.

Call the resulting finite map  $\mathcal{F}_{b,\eta;\epsilon}[N]$ . Then

$$\begin{aligned} \widehat{\mathcal{H}}_{\epsilon,b,\eta;\epsilon}^0[N] &= \sum_{\rho \in P_\epsilon} \epsilon^{-\rho} \mathcal{C}_{\rho,b,\eta;\epsilon}[N] \\ &\quad + \sum_{m=1}^{m_\epsilon} (\log(\epsilon\mu))^m \underbrace{\mathcal{L}_{m,b,\eta;\epsilon}[N]}_{=0} \\ &\quad + \mathcal{F}_{b,\eta;\epsilon}[N] + \mathcal{R}_{\epsilon,b,\eta;\epsilon}[N], \end{aligned}$$

where mixed power-log rows also have zero coefficient for this exact heat class.

Choose the heat expansion order separately in each of the finitely many local channels until the remainder has a positive exponent. Combine it with the strong  $o(1)$  finite-rank remainder. In the cap operator norm,

$$\|\mathcal{R}_{\epsilon,b,\eta;\epsilon}[N]\|_\epsilon \longrightarrow 0.$$

### 9.4 Uniformity

The local coefficient estimates are uniform on compact derivative-test sets because:

1. Lemma 8.1 supplies only finitely many coefficient maps;
2. their required derivatives are continuous on the accepted smooth configuration core;
3. continuous functions and derivatives are bounded on each registered compact test set;
4. the heat remainder uses only finitely many such seminorms at a fixed truncation order.

Uniformity on bounded subsets of  $\Phi_{5b}^\epsilon$  then follows from finite dimensionality. For a basis  $F_1, \dots, F_d$ , a bounded set has uniformly bounded coordinates, and the maximum of the  $d$  basis remainders controls the set.

No operator-norm convergence of the heat semigroup to the identity on the full infinite-dimensional Sobolev space is asserted. The finite-rank cylinder channel needs only strong convergence for finitely many fixed probe kernels, which becomes uniform after restriction to the finite cap.

#### Theorem W6R.5 (Elementary capwise WDW heat-expansion theorem).

For every exact branch, candidate label, cap, and elementary registered Gate-I action, the regulated Hamiltonian action has the complete expansion displayed above in the declared cap operator topology. The same statement

holds for a task leaf whose internal regulators are fixed positive labels. The singular set is finite; all local coefficients lie in the global quotient block of W6R.4; the finite-rank channel has no negative-power coefficient; all allowed boundary, logarithmic, and mixed power-log rows are either computed or proved zero; and the remainder tends to zero uniformly on cap-bounded sets and compact derivative-test sets. No common-variable product or commutator expansion is inferred here.

### Proof.

Lemma 8.1 splits every cap Hessian into its diagonal differential and finite-rank/smoothing channels. Section 9.2 reduces the former to the closed-manifold heat trace and the latter to trace-norm continuity. Only finitely many differential orders occur, so the negative exponent set is finite. The local heat coefficients are exactly the symbol contractions enumerated in W6R.4. The uniformity argument in Section 9.4 passes the scalar estimates to the finite cap operator norm and, in branch  $B$ , to each of the finitely many affected target seminorms. Adding the outer-regulator-independent potential and fermionic terms changes only the finite coefficient. The proof deliberately stops at one outer regulator; W6R.10 forms and estimates a full common-parameter task tree before expansion.  $\square$

## 9.5 Recovery of the accepted obstruction order

For the exact centered scalar witness  $G_{b,W}$ , the local scalar-scalar Hessian has spatial differential order  $m = 2$ . Hence the leading exponent is

$$0 - 3 - 2 = -5.$$

The principal scalar coefficient is the positive heat-symbol contraction already computed in W6.6:

$$\mathcal{C}_{5,b}[N]G_{b,W} = C_b > 0$$

under the typed detector. The Gate-I expansion therefore reproduces

$$C_b \epsilon^{-5} + O(\epsilon^{-4})$$

without using that witness to define the general coefficient family.

## 10. Locality, covariance, and source attachment

### 10.1 Locality

Each singular coefficient is obtained from a diagonal differential Hessian and a local heat coefficient. It is therefore a finite integral of a density constructed from:

- the exact dynamical field jets present in the cap word;
- the lapse and inherited source coefficients;
- the kinetic tensor and ordering data;
- the symbol and finite jets of the registered  $L_{r,*}$ ;
- exact branch-specific tensor, gauge, exterior, and density contractions.

Finite-rank cylinder and multilocal cross channels contribute only to the finite part and remainder; they are not relabeled as local singular counterterms.

### 10.2 Source attachment

The reference metric, connections, bundle metrics, heat family, pivot, and adjoint conventions are candidate coordinates already registered in W5.7. Their appearance in a coefficient is therefore source attached to the exact candidate record. A new reference operator, boundary condition, bundle, field, coupling, or nonlocal kernel would be new standing-bearing data and is not part of the present coefficient family.

At Gate I, source attachment means that the coefficient defines a continuous typed map

$$\Phi_{5b}^c \longrightarrow \Phi_{5b}^{c,+} \subset \mathcal{E}_{5b}^{\text{sec}}$$

on the registered finite calculation. It does not yet prove that the assembled finite-part or subtraction map is a continuous endomorphism of the full  $(\Phi_{5b}, \tau_{5b})$ . That is the separate Gate-II burden W6R.13–W6R.15.

### 10.3 Proper-redescription covariance

The WDW-5 proper matching transports the configuration point, fields, smearings, regulator background, bundle connections, density maps, and ordering coordinates together. Let  $T$  be such a complete matching. Naturality

of the differential, symbol composition, heat coefficients, and fiber trace gives

$$T \mathcal{C}_{\rho,b,\eta;\epsilon}[N] = \mathcal{C}_{\rho,b',\eta';T\epsilon}[TN] T.$$

This is equivariance over the complete candidate record. It is not the false assertion that an active diffeomorphism leaves a fixed untransported reference metric invariant.

## 10.4 Continuity

On a finite cap, every local coefficient is a finite sum of the same compact-open derivative seminorms used to define the accepted topology. The coefficient maps are therefore continuous. On branch  $B$ , the target estimate is checked degree-wise and then placed in the finite-degree direct sum. Only finitely many degree shifts occur in a fixed ordering packet.

### Theorem W6R.6 (Locality, covariance, and source-attachment theorem).

Every singular coefficient in W6R.5 is a local, source-attached, proper-redescription-equivariant, continuous cap map with its exact branch, density, functional-order, jet-order, exterior-degree, and constraint-role type. The finite-rank and multilocal channels are retained separately and are not used as local counterterms. No full-source endomorphism or counterterm endpoint is claimed.

#### Proof.

Locality follows from diagonal heat-symbol composition. Source attachment follows from the exhaustive list of coefficient data in Section 10.2. Naturality proves the displayed transport identity when the complete candidate data are transported. Continuity follows from finite sums of the accepted compact derivative-test seminorms and, in branch  $B$ , from the degree-wise projective exterior estimates already used in W5.7 and W5.11.  $\square$

## 11. Full heat jet and the independently fixed WDW lift

### 11.1 Recursive response filtration

Fix an exact branch, candidate label, and cap  $\mathfrak{c}$ . Write

$$\mathcal{J}_{\mathfrak{c}} = \mathcal{J}_{b,\eta;\mathfrak{c}}^{\text{sing}} \subset \mathcal{J}_{\infty}$$

for the finite subspace of the global off-shell quotient constructed in W6R.4. Let

$$\rho_0 > \rho_1 > \cdots > \rho_q \geq 0$$

be the complete ordered list of nonpositive heat orders in the local differential response of W6R.5. All structural labels at one heat order are kept together.

Let  $\mathcal{Y}_s$  be the typed coefficient-output space at order  $\rho_s$ . The actual WDW coefficient extraction defines

$$\mathbf{A}_{s,\mathfrak{c}} : \mathcal{J}_{\mathfrak{c}} \longrightarrow \mathcal{Y}_s.$$

This is the full map at that heat order: it contains principal and lower differential symbols and all admitted tensor, jet, field, and constraint-role entries. Define

$$\mathcal{K}_{\mathfrak{c}}^0 = \mathcal{J}_{\mathfrak{c}}, \quad \mathcal{K}_{\mathfrak{c}}^{s+1} = \mathcal{K}_{\mathfrak{c}}^s \cap \ker \mathbf{A}_{s,\mathfrak{c}}.$$

Thus a direction with zero principal response is tested at every later nonpositive coefficient. Its detected response layer is

$$\text{Gr}_H^s \mathcal{J}_{\mathfrak{c}} = \mathcal{K}_{\mathfrak{c}}^s / \mathcal{K}_{\mathfrak{c}}^{s+1},$$

and the terminal heat-invisible sector is

$$\mathcal{K}_{\mathfrak{c}}^{\text{reg}} = \bigcap_{s=0}^q \ker \mathbf{A}_{s,\mathfrak{c}}.$$

The terminal sector remains part of the WDW source. It is not added to  $\mathbf{R}_{\infty}^{\text{off}}$ , declared physically zero, or used to alter a counterterm block.

For example, if

$$A_1 = -\Delta + V_1, \quad A_2 = -\Delta + V_2,$$

then  $A_1 - A_2$  has zero order-two principal symbol but

$$\text{Tr}((A_1 - A_2)e^{-\epsilon^2 L}) = (4\pi)^{-3/2} \epsilon^{-3} \int_{\Sigma} \text{tr}(V_1 - V_2) d\mu_{\star} + O(\epsilon^{-1}).$$

The difference therefore descends through the principal kernel and is retained at its first nonzero lower coefficient.

## 11.2 Hereditary restriction

For  $\mathfrak{c} \preceq \mathfrak{c}'$ , W6R.4 places both cap spaces in the same  $\mathcal{J}_\infty$ . Source attachment gives

$$A_{s,\mathfrak{c}'}|_{\mathcal{J}_\mathfrak{c}} = \iota_s A_{s,\mathfrak{c}},$$

where  $\iota_s$  is the typed coefficient-space inclusion. Induction on  $s$  now gives the well-defined intersection formula

$$\mathcal{K}_\mathfrak{c}^s = \mathcal{K}_{\mathfrak{c}'}^s \cap \mathcal{J}_\mathfrak{c}.$$

Unlike the predecessor stage source, this argument does not assume an embedding between two cap-dependent quotients; the embedding was already proved by the single global quotient.

## 11.3 Predeclared counterterm operators and blocks

The response filtration records which actual WDW coefficients occur. It does not define the inverse-bearing operator. For that operator, use the blocks and the concrete counterterm operators already fixed in W6R.4:

$$\begin{aligned} \mathcal{S}_{\mathfrak{c},g} &= \text{Gr}_g \mathcal{J}_\mathfrak{c}, & \mathcal{T}_{\mathfrak{c},g} &\cong \mathcal{S}_{\mathfrak{c},g}, \\ \mathcal{B}_{\mathfrak{c},g}^{\text{ct}} &= (\mathcal{C}_{\mathfrak{c},g,1}, \dots, \mathcal{C}_{\mathfrak{c},g,d_g}). \end{aligned}$$

The corresponding source and target coordinate bases are

$$\mathcal{B}_{\mathfrak{c},g}^S = (\Theta_{g,1}, \dots, \Theta_{g,d_g}), \quad \mathcal{B}_{\mathfrak{c},g}^T = (\Theta_{g,1}^T, \dots, \Theta_{g,d_g}^T).$$

All dimensions, types, representatives, and operator actions were fixed before the calculation below. The target basis is a typed copy of the same global quotient classes, not an image basis of a response map.

## 11.4 Actual W5.7-regulated counterterm insertion

Let  $L_{0,\star}$  be the rank-one scalar reference Laplace-type operator already registered in W5.7 and put

$$S_{\epsilon,0} = e^{-\epsilon^2 L_{0,\star}/2}, \quad K_{\epsilon,0} = S_{\epsilon,0} S_{\epsilon,0}^* = e^{-\epsilon^2 L_{0,\star}}.$$

Under the fixed scalar  $L^2$ -Sobolev pivot of W5.7,  $S_{\epsilon,0}^\vee$  is the same half-heat map as  $S_{\epsilon,0}^*$  after the declared density identifications. Write  $k_{\epsilon,\star}(x, y)$  for the kernel of  $K_{\epsilon,0}$ . If  $\mathcal{C}_{\mathfrak{c},g,\alpha}$  has heat weight  $\rho_g$ , its fixed local operator density is  $\hat{\theta}_{\mathfrak{c},g,\alpha}[F](q, x)$ . Define the regulated insertion of this concrete operator by

$$\begin{aligned} &\text{Ins}_{\epsilon,\mathfrak{c},g}(\mathcal{C}_{\mathfrak{c},g,\alpha}) F \\ &:= \epsilon^{3-\rho_g} \text{Tr}_{L^2}^V \left( S_{\epsilon,0}^\vee M_{\hat{\theta}_{\mathfrak{c},g,\alpha}[F]} S_{\epsilon,0} \right). \end{aligned}$$

Here  $V = \mathbb{C}$  in a scalar channel and is the appropriate complete target in a branch-B exterior-valued channel. In the latter case,  $\text{Tr}^V$  is the complete-target extension from W5.7: for an orthonormal basis  $(e_n)$ , its defining series is evaluated in every continuous seminorm of  $V$ . The W5.7 Hilbert–Schmidt estimate gives, on each registered compact configuration set,

$$p\left(\text{Tr}^V \left( S_{\epsilon,0}^\vee M_{\hat{\theta}_{\mathfrak{c},g,\alpha}[F]} S_{\epsilon,0} \right)\right) \leq \|S_{\epsilon,0}\|_2^2 \sup_{x \in \Sigma} p\left(\hat{\theta}[F](q, x)\right).$$

This proves absolute seminorm convergence and basis independence.

This is the registered WDW-6R rule for inserting the predeclared counterterm operator into the finite- $\epsilon$  Hamiltonian. It is not an auxiliary coordinate lift. The factor  $\epsilon^{3-\rho_g}$  is the dimensional scaling attached to the predeclared heat weight before any matrix is computed. No numerical normalization by the leading heat coefficient is made.

Ideal cyclicity in the scalar channel, followed by the complete-target extension, proves the operator identity

$$\begin{aligned} &\text{Ins}_{\epsilon,\mathfrak{c},g}(\mathcal{C}_{\mathfrak{c},g,\alpha}) F \\ &= \epsilon^{3-\rho_g} \int_{\Sigma} k_{\epsilon,\star}(x, x) \hat{\theta}_{\mathfrak{c},g,\alpha}[F](q, x) d\mu_\star(x). \end{aligned}$$

Thus the matrix below is extracted from the actual regulated action of the already fixed operators. The identity precedes every determinant, inverse, or use of the regulated Hamiltonian response vector.

The insertion is total on the cap because multiplication by the smooth diagonal kernel preserves the finite local task class, and its output is one of the saturation slots in  $\Phi_{5b}^{\mathfrak{c},+}$ . The complete-target W5.7 estimate proves

continuity. Transporting the operator density, reference metric, heat operator, density, and branch data together proves proper-redescription equivariance. The scalar kernel cannot contract or erase tensor, constraint-role, or exterior indices.

Multiplication by nonconstant lower diagonal coefficients need not descend independently of the fixed representative. The complete finite- $\epsilon$  insertion therefore uses that representative. A change of representative is later scheme data for W6R.14. Its leading constant heat coefficient is representative independent.

With  $\gamma_3 = (4\pi)^{-3/2}$ , the diagonal heat expansion is

$$k_{\epsilon,\star}(x, x) = \gamma_3 \epsilon^{-3} + \epsilon^{-1} u_1(x) + \dots + \epsilon^{2N-3} u_N(x) + O(\epsilon^{2N-1})$$

in every finite family of cap seminorms. Substitution into the actual insertion identity gives

$$\begin{aligned} \text{Ins}_{\epsilon,\mathbf{c},g}(\mathcal{C}_{\mathbf{c},g,\alpha}) \\ = \gamma_3 \epsilon^{-\rho_g} \mathcal{C}_{\mathbf{c},g,\alpha} + \sum_{n=1}^N \epsilon^{2n-\rho_g} \mathcal{C}_{\mathbf{c},g,\alpha}^{(n)} + O(\epsilon^{2N+2-\rho_g}). \end{aligned}$$

Each  $\mathcal{C}^{(n)}$  is the source-attached local operator obtained by multiplying the fixed density by  $u_n$ . Task saturation places every nonpositive output in the corresponding lower predeclared block.

The associated-graded operator of the actual insertion is therefore

$$\text{Gr}_g(\epsilon^{\rho_g} \text{Ins}_{\epsilon,\mathbf{c},g}) = \gamma_3 I_{\mathcal{S}_{\mathbf{c},g}}.$$

In the fixed Euclidean block coordinates,

$$\mathbf{L}_{\mathbf{c},g} = \gamma_3 I_{d_g}, \quad \det \mathbf{L}_{\mathbf{c},g} = \gamma_3^{d_g} \neq 0, \quad \sigma_{\min}(\mathbf{L}_{\mathbf{c},g}) = \gamma_3.$$

The nonunit constant is retained so the diagonal certificate displays the computed heat coefficient. No heat-response quotient, image restriction, dimension adjustment, or pivot selection occurs.

**Theorem W6R.7 (Global-quotient filtration and actual-insertion principal theorem).**

The complete nonpositive response jet defines the hereditary recursive filtration of Section 11.1 and retains every lower-order divergence. Independently, each predeclared W6R.4 basis class has a concrete total counterterm operator, and its actual W5.7-regulated insertion satisfies the displayed trace identity. The normalized associated-graded diagonal of that action is  $\gamma_3 I$ , with determinant  $\gamma_3^{d_g}$  and smallest singular value  $\gamma_3$ . The response filtration is not used to manufacture this nondegeneracy.

**Proof.**

W6R.5 supplies the finite coefficient family and W6R.6 supplies source attachment. Successive kernel intersections prove the response filtration; the global quotient of W6R.4 makes restriction under cap inclusion literal. W6R.4 separately freezes each concrete operator  $\mathcal{C}_{\mathbf{c},g,\alpha}$ . The W5.7 half-heat maps make the displayed insertion trace class, and ideal cyclicity gives its diagonal kernel integral. Substitution of  $k_{\epsilon,\star}(x, x) = \gamma_3 \epsilon^{-3} + O(\epsilon^{-1})$  then gives  $\gamma_3 \epsilon^{-\rho_g} \mathcal{C}_{\mathbf{c},g,\alpha}$  plus strict lower heat weights. Coefficient extraction in the bases fixed before this computation proves the determinant and singular-value statements on the required operator.  $\square$

## 12. The actual WDW counterterm insertion matrix

### 12.1 Definition from concrete insertion columns

Order the fixed blocks by decreasing heat weight and then by the predeclared structural order. Put

$$\mathcal{S}_{\mathbf{c}} = \bigoplus_g \mathcal{S}_{\mathbf{c},g}, \quad \mathcal{T}_{\mathbf{c}} = \bigoplus_g \mathcal{T}_{\mathbf{c},g}.$$

Let  $\text{Coeff}_{\mathbf{c}}^{\leq 0}$  be the coordinate extraction of all nonpositive local operator coefficients in the fixed target basis. Define

$$\mathbf{M}_{\mathbf{c}}(\epsilon) e_{g,\alpha} = \text{Coeff}_{\mathbf{c}}^{\leq 0} [\text{Ins}_{\epsilon,\mathbf{c},g}(\mathcal{C}_{\mathbf{c},g,\alpha})].$$

Thus every column of  $\mathbf{M}_{\mathbf{c}}$  is the computed coefficient vector of an actual regulated counterterm action. This definition contains no regulated Hamiltonian response vector and no inverse.

For a source coefficient column  $\mathbf{b} = (b_{g,\alpha})$ , the actual finite-cap counterterm insertion is

$$\Delta \widehat{\mathcal{H}}_{\epsilon,\mathbf{c}}^{\text{ct}}[\mathbf{b}] = \sum_{g,\alpha} b_{g,\alpha} \text{Ins}_{\epsilon,\mathbf{c},g}(\mathcal{C}_{\mathbf{c},g,\alpha}).$$

Linearity and the preceding column definition give the operator-level coefficient identity

$$\text{Coeff}_{\epsilon}^{\leq 0} [\Delta \widehat{\mathcal{H}}_{\epsilon, \mathbf{c}}^{\text{ct}} [\mathbf{b}]] = \mathbf{M}_{\epsilon}(\epsilon) \mathbf{b}.$$

This is the missing insertion bridge: it identifies the matrix with the counterterm action before an inverse is considered.

Let

$$\mathbf{D}_{\epsilon}(\epsilon) = \text{diag}_g (\epsilon^{-\rho_g} I_{d_g}), \quad \mathbf{L}_{\epsilon} = \text{diag}_g (\gamma_3 I_{d_g}).$$

Matrices act on source columns, so normalization is on the right:

$$\mathbf{M}_{\epsilon}(\epsilon) \mathbf{D}_{\epsilon}(\epsilon)^{-1} = \mathbf{L}_{\epsilon} + \mathbf{E}_{\epsilon}(\epsilon).$$

## 12.2 Exhaustive block calculation

The actual insertion formula gives every block:

| Target block $g$ versus source block $h$ | Unnormalized entry                                           | Normalized disposition                                                    |
|------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------|
| $g = h$                                  | $\gamma_3 \epsilon^{-\rho_h} I_{d_h}$ plus lower heat orders | computed principal block $\gamma_3 I_{d_h}$                               |
| $\rho_g > \rho_h$                        | 0                                                            | no heat coefficient can raise the source weight                           |
| $\rho_g = \rho_h$ , different exact type | 0                                                            | forbidden by branch, density, tensor, exterior, and representation typing |
| $\rho_g < \rho_h$                        | $\epsilon^{-\rho_g} \mathbf{B}_{gh}$                         | $\epsilon^{\rho_h - \rho_g} \mathbf{B}_{gh}$ , hence decaying             |
| terminal positive-order output           | retained in the remainder                                    | outside the nonpositive square block                                      |

All same-weight field, jet, tensor, and constraint-role coordinates were already grouped in the fixed block. Its complete leading matrix is  $\gamma_3 I$  on that whole block. No equal-order term is placed in  $\mathbf{E}_{\epsilon}$ .

There are finitely many cap blocks. In the declared block-max norm, there are constants

$$C_{\epsilon} < \infty, \quad \delta_{\epsilon} > 0, \quad p_{\epsilon} \in \mathbb{N}$$

such that

$$\|\mathbf{E}_{\epsilon}(\epsilon; \mu)\|_{\text{blk}, \epsilon} \leq C_{\epsilon} \epsilon^{\delta_{\epsilon}} (1 + |\log(\epsilon \mu)|^{p_{\epsilon}}).$$

For the exact closed differential heat class,  $p_{\epsilon} = 0$ . One may take  $\delta_{\epsilon}$  to be the least positive gap among the finitely many admitted source and target heat weights.

## 12.3 Matching the regulated WDW coefficient vector

Let  $\mathbf{h}_{\epsilon, \mathbf{c}} \in \mathcal{T}_{\epsilon}$  be the complete local nonpositive coefficient vector of the actual regulated Hamiltonian operator, computed independently by W6R.5 before evaluation on any  $F$ . Its entries are coordinates of typed local operator maps, not numbers fitted to a witness. Evaluating those maps on  $F$  gives  $\mathbf{h}_{\epsilon, \mathbf{c}}[F]$ . After W6R.9 proves invertibility, define

$$\mathbf{b}_{\epsilon, \mathbf{c}}^{\text{match}} = \mathbf{M}_{\epsilon}(\epsilon)^{-1} \mathbf{h}_{\epsilon, \mathbf{c}}.$$

The already proved insertion identity, rather than a coordinate relabeling of  $\mathbf{h}$ , then gives

$$\text{Coeff}_{\epsilon}^{\leq 0} [\Delta \widehat{\mathcal{H}}_{\epsilon, \mathbf{c}}^{\text{ct}} [\mathbf{b}_{\epsilon, \mathbf{c}}^{\text{match}}]] = \mathbf{h}_{\epsilon, \mathbf{c}}.$$

The equality is an operator-coefficient identity and hence holds after application to every  $F \in \Phi_{5b}^{\epsilon}$ ; no counterterm coefficient depends on the tested vector. Gate II may later use the negative of this coefficient column to assemble a subtraction. Gate I neither chooses that sign nor asserts that the capwise choices glue.

### Theorem W6R.8 (Actual WDW counterterm-insertion mixing theorem).

For every finite cap, coefficient extraction from the concrete regulated counterterm actions defines a square operator on the predeclared W6R.4 blocks and satisfies

$$\mathbf{M}_{\epsilon}(\epsilon) \mathbf{D}_{\epsilon}(\epsilon)^{-1} = \mathbf{L}_{\epsilon} + \mathbf{E}_{\epsilon}(\epsilon), \quad \mathbf{L}_{\epsilon, g} = \gamma_3 I_{d_g}.$$

Every upper or cross-type block in the table is exactly zero and every strictly lower block decays after right source-column normalization. For every coefficient column  $\mathbf{b}$ , the actual inserted operator has coefficient vector  $\mathbf{M}_\epsilon \mathbf{b}$ . Consequently, once inverted, this same operator—not an auxiliary lift—matches the complete W6R.5 coefficient vector without discarding any response-kernel direction.

**Proof.**

W6R.7 proves the trace identity for every concrete basis operator and computes its full same-block leading coefficient. Collecting those actual columns is precisely the definition of  $\mathbf{M}_\epsilon$ . Every later diagonal heat coefficient lowers heat weight, while source attachment preserves exact type; this proves the zero and lower-block statements. Finiteness gives the positive-gap estimate. Linearity proves  $\text{Coeff}^{\leq 0}[\Delta \widehat{\mathcal{H}}^{\text{ct}}[\mathbf{b}]] = \mathbf{M}\mathbf{b}$  before any inverse is used. Applying the inverse afterward proves the matching statement.  $\square$

### 13. Quantitative inverse control

Define the cap threshold

$$\epsilon_\epsilon(\mu) = \sup \left\{ \epsilon_0 > 0 : \gamma_3^{-1} C_\epsilon \epsilon^{\delta_\epsilon} (1 + |\log(\epsilon\mu)|^{p_\epsilon}) \leq \frac{1}{2} \text{ for } 0 < \epsilon \leq \epsilon_0 \right\}.$$

It is positive because every logarithm is dominated by the printed positive power. Below the threshold,

$$\|\mathbf{L}_\epsilon^{-1} \mathbf{E}_\epsilon(\epsilon; \mu)\|_{\text{blk}, \epsilon} \leq \frac{1}{2}.$$

Therefore

$$\begin{aligned} (\mathbf{M}_\epsilon \mathbf{D}_\epsilon^{-1})^{-1} &= (I + \mathbf{L}_\epsilon^{-1} \mathbf{E}_\epsilon)^{-1} \mathbf{L}_\epsilon^{-1}, \\ \|(\mathbf{M}_\epsilon \mathbf{D}_\epsilon^{-1})^{-1}\|_{\text{blk}, \epsilon} &\leq 2\|\mathbf{L}_\epsilon^{-1}\|_{\text{blk}, \epsilon} = 2\gamma_3^{-1}, \\ \mathbf{M}_\epsilon^{-1} &= \mathbf{D}_\epsilon^{-1} (I + \mathbf{L}_\epsilon^{-1} \mathbf{E}_\epsilon)^{-1} \mathbf{L}_\epsilon^{-1}. \end{aligned}$$

The block-max norm fixed in Section 5 gives, for  $0 < \epsilon \leq 1$ ,

$$\|\mathbf{D}_\epsilon^{-1}\|_{\text{blk}, \epsilon} = \max_g \epsilon^{\rho_g} \leq 1.$$

Hence

$$\|\mathbf{M}_\epsilon(\epsilon)^{-1}\|_{\text{blk}, \epsilon} \leq 2\|\mathbf{D}_\epsilon^{-1}\|_{\text{blk}, \epsilon} \|\mathbf{L}_\epsilon^{-1}\|_{\text{blk}, \epsilon} \leq 2\gamma_3^{-1}.$$

More sharply, a source coordinate in block  $g$  receives its expected factor  $\epsilon^{\rho_g}$ .

**Theorem W6R.9 (WDW-native quantitative inverse-control theorem).**

Every finite-cap actual counterterm-insertion matrix is invertible for  $0 < \epsilon \leq \min(1, \epsilon_\epsilon(\mu))$ . Its normalized inverse has norm at most  $2\|\mathbf{L}_\epsilon^{-1}\| = 2\gamma_3^{-1}$ , and its unnormalized inverse is the displayed right-sided source-column formula. The proof uses the calculated principal coefficient of the actual insertion and the declared block-max norm. It uses no image quotient, pivot selection, Yang-Mills running coupling, or asymptotic-freedom estimate.

**Proof.**

W6R.8 gives  $\mathbf{L}_\epsilon + \mathbf{E}_\epsilon$  with  $\|\mathbf{L}_\epsilon^{-1} \mathbf{E}_\epsilon\| \leq 1/2$ . The Neumann series proves the normalized inverse formula and its bound. The exact block-max estimate for  $\mathbf{D}_\epsilon^{-1}$  then proves the unnormalized formula and bound.  $\square$

### 14. Complete-carrier common-parameter remainders

#### 14.1 Full-cap mixing extension fixed before the remainder

Use the complete task-output carrier of Section 5.1. For every registered task  $\tau$ , define before expanding that task

$$\begin{aligned} \widetilde{\mathbf{M}}_{\tau, \epsilon}(\epsilon) &= \mathbf{M}_\epsilon(\epsilon) \oplus I_{\mathcal{Z}_{\tau, \epsilon}^{\text{reg}}}, \\ \widetilde{\mathbf{D}}_{\tau, \epsilon}(\epsilon) &= \mathbf{D}_\epsilon(\epsilon) \oplus I_{\mathcal{Z}_{\tau, \epsilon}^{\text{reg}}}, \\ \widetilde{\mathbf{L}}_{\tau, \epsilon} &= \mathbf{L}_\epsilon \oplus I_{\mathcal{Z}_{\tau, \epsilon}^{\text{reg}}}, \\ \widetilde{\mathbf{E}}_{\tau, \epsilon}(\epsilon) &= \mathbf{E}_\epsilon(\epsilon) \oplus 0. \end{aligned}$$

Thus

$$\widetilde{\mathbf{M}}_{\tau, \epsilon} \widetilde{\mathbf{D}}_{\tau, \epsilon}^{-1} = \widetilde{\mathbf{L}}_{\tau, \epsilon} + \widetilde{\mathbf{E}}_{\tau, \epsilon}.$$

The extension is not selected from the remainder. It is part of the Gate-I normalization record: singular local coefficient slots carry the actual counterterm insertion matrix proved in W6R.8, while every regular slot is

carried unchanged. Identity on the regular summand preserves its branch, exterior degree, constraint role, tensor, density, and target type. It also commutes with restriction to every shared tagged task slot. For products and commutators, the entire task tree is formed before this extension is applied; no factorwise projection is used. More generally, if a registered downstream map  $Q$  is to act on a task output, the finite composite syntax tree  $Q \circ \tau$  is itself the task expanded in Section 14.3. W6R.10 applies the task-specific full inverse after that complete expansion. It never exchanges  $Q$  with  $\tilde{\mathbf{M}}_{\tau, \epsilon}$ , so no unproved commutation between counterterm mixing and an intermediate product, commutator, evaluation, or type projection is used.

W6R.9 and the full max norm give

$$\begin{aligned}\tilde{\mathbf{M}}_{\tau, \epsilon}^{-1} &= \mathbf{M}_{\epsilon}^{-1} \oplus I_{\mathcal{Z}_{\tau, \epsilon}^{\text{reg}}}, \\ (\tilde{\mathbf{M}}_{\tau, \epsilon} \tilde{\mathbf{D}}_{\tau, \epsilon}^{-1})^{-1} &= (\mathbf{M}_{\epsilon} \mathbf{D}_{\epsilon}^{-1})^{-1} \oplus I_{\mathcal{Z}_{\tau, \epsilon}^{\text{reg}}},\end{aligned}$$

and hence, below the W6R.9 threshold,

$$\|\tilde{\mathbf{M}}_{\tau, \epsilon}^{-1}\|_{\text{full}, \tau, \epsilon}, \quad \left\| (\tilde{\mathbf{M}}_{\tau, \epsilon} \tilde{\mathbf{D}}_{\tau, \epsilon}^{-1})^{-1} \right\|_{\text{full}, \tau, \epsilon} \leq B_{\epsilon}, \quad B_{\epsilon} := \max\{2\gamma_3^{-1}, 1\}.$$

This is an inverse on the complete task-output carrier, not an inverse on the coefficient projection alone.

## 14.2 Strong-dual Taylor control

Let  $u \in (\mathcal{S}_{\infty})'_b$  be any one of the finitely many cap vectors and let  $B \subset \mathcal{S}_{\infty}$  be bounded. Spectral calculus and duality give, for every  $K \geq 0$ ,

$$e^{-\epsilon^2 L} u = \sum_{n=0}^K \frac{(-\epsilon^2)^n}{n!} L^n u + \mathcal{Q}_{K, \epsilon} u,$$

with

$$\sup_{\varphi \in B} |\langle \mathcal{Q}_{K, \epsilon} u, \varphi \rangle| \leq \frac{\epsilon^{2K+2}}{(K+1)!} \sup_{\psi \in B_{K+1}} |\langle u, \psi \rangle|.$$

Here

$$B_{K+1} = \{L^{K+1} e^{-tL} \varphi : \varphi \in B, 0 \leq t \leq 1\}$$

is bounded in  $\mathcal{S}_{\infty}$ . The bound follows because  $L^{K+1} e^{-tL}$  is an equicontinuous family on the smooth inverse-limit core for  $0 \leq t \leq 1$ . Thus every fixed strong-dual cap vector has a Taylor remainder of arbitrarily high even order in each registered strong seminorm. This closes the rate gap left by mere strong convergence.

## 14.3 Finite common-parameter task trees

Let  $\tau \in \mathfrak{T}_{\Gamma}$  be a product or ordered commutator whose outer regulators are all the same  $\epsilon$ . Expand the finite syntax tree before taking any asymptotic. Repeated use of Lemma 8.1 writes each term as a finite sum of:

1. local multi-heat kernels

$$\int_{\Sigma^{v_{\tau}}} a_{\tau}(q; \mathbf{x}) \prod_{e \in E_{\tau}} \nabla^{\beta_e} K_{c_e \epsilon}(x_{s(e)}, x_{t(e)}) d\mathbf{x},$$

where  $c_e > 0$ , the derivative orders are finite, and the amplitudes have uniformly bounded jets on the registered compact test sets; and

2. finite-rank or smoothing factors to which Section 14.2 applies.

This is a direct common-parameter representation, not an iteration of a fixed-internal-regulator asymptotic.

For a local multi-heat term, cover the compact target by finitely many normal-coordinate charts and split away from the partial diagonals. The off-diagonal part is  $O(\epsilon^N)$  for every  $N$  by the Gaussian heat-kernel bound. Near a partial diagonal, set  $x_i = x_0 + \epsilon z_i$ . Taylor expansion of the amplitude and metric Jacobian through order  $K$ , followed by the finite Gaussian moment integrals in the  $z_i$ , gives all joint coefficients through that order. Taylor's integral remainder and the Gaussian moment bound give

$$\|\mathcal{R}_{\tau, \epsilon}^{(K)}\|_{\epsilon} \leq C_{\tau, \epsilon} \epsilon^{K+1-P_{\tau}},$$

where  $P_{\tau} < \infty$  is the largest common-parameter pole order of the finite tree. Configuration derivatives only differentiate the finite amplitude and preserve the estimate. For every strong-dual factor, choose the Taylor order in Section 14.2 above  $K + P_{\tau}$ . Hence the same joint bound holds for the complete local, finite-rank, smoothing, and section-valued task.

Choosing  $K > P_{\tau}$  gives a complete common-parameter expansion with a strictly positive remainder exponent after all nonpositive coefficients are removed. Antisymmetrizing the two ordered products gives the commutator expansion with the same bound. No product of a divergent factor with an unrated remainder occurs.

## 14.4 Bounded projections, reconstruction, and inverse compatibility

For every finite task tree, choose  $K$  so that  $\sigma_{\tau,\epsilon} = K + 1 - P_\tau > 0$ . Regard its actual complete operator remainder as the semantic evaluation of the full-carrier element

$$\mathbf{R}_{\tau,\epsilon}^{\text{full},(K)} \in \mathcal{Y}_{\tau,\epsilon}.$$

The element is the tagged tuple produced by the already formed finite task tree; it is not chosen by decomposing the evaluated operator afterward. Define its two actual components, without discarding either one, by

$$\mathbf{R}_{\tau,\epsilon}^{T,(K)} = \pi_T \mathbf{R}_{\tau,\epsilon}^{\text{full},(K)}, \quad \mathbf{R}_{\tau,\epsilon}^{\text{reg},(K)} = \pi_{\text{reg}} \mathbf{R}_{\tau,\epsilon}^{\text{full},(K)}.$$

The reconstruction identity from Section 5.1 is literal:

$$\mathbf{R}_{\tau,\epsilon}^{\text{full},(K)} = \iota_T \mathbf{R}_{\tau,\epsilon}^{T,(K)} + \iota_{\text{reg}} \mathbf{R}_{\tau,\epsilon}^{\text{reg},(K)}.$$

The multi-heat and strong-dual estimates of Sections 14.2–14.3 were proved in the complete tagged output norm. Since both projections have norm at most one,

$$\left\| \mathbf{R}_{\tau,\epsilon}^{T,(K)} \right\|_{\text{blk},\epsilon}, \quad \left\| \mathbf{R}_{\tau,\epsilon}^{\text{reg},(K)} \right\|_{\text{reg},\tau,\epsilon} \leq C_{\tau,\epsilon,K} \epsilon^{\sigma_{\tau,\epsilon}}.$$

After every nonpositive joint coefficient has been removed,  $\mathbf{R}_{\tau,\epsilon}^{T,(K)} = 0$ ; the displayed two-component estimate is retained because it also controls intermediate truncations. The regular component contains every terminal positive-order local, finite-rank, smoothing, and complete branch-B target-valued remainder.

Applying the already fixed full operator gives

$$\begin{aligned} \left\| \left( \widetilde{\mathbf{M}}_{\tau,\epsilon} \widetilde{\mathbf{D}}_{\tau,\epsilon}^{-1} \right)^{-1} \mathbf{R}_{\tau,\epsilon}^{\text{full},(K)} \right\|_{\text{full},\tau,\epsilon} &\leq B_\epsilon C_{\tau,\epsilon,K} \epsilon^{\sigma_{\tau,\epsilon}} \longrightarrow 0, \\ \left\| \widetilde{\mathbf{M}}_{\tau,\epsilon}^{-1} \mathbf{R}_{\tau,\epsilon}^{\text{full},(K)} \right\|_{\text{full},\tau,\epsilon} &\leq B_\epsilon C_{\tau,\epsilon,K} \epsilon^{\sigma_{\tau,\epsilon}} \longrightarrow 0. \end{aligned}$$

In particular, the predecessor stage direct-sum test  $\mathbf{R}_\epsilon = \epsilon z$  with  $0 \neq z \in \mathcal{Z}_{\tau,\epsilon}^{\text{reg}}$  is inside the domain and satisfies

$$\widetilde{\mathbf{M}}_{\tau,\epsilon}^{-1} \mathbf{R}_\epsilon = \epsilon z \longrightarrow 0.$$

Continuous semantic evaluation into the finite cap operator space, every registered commutator map formed before expansion, and every continuous evaluation functional preserve the proved limit. This constructs neither a state nor a physical pairing.

### Theorem W6R.10 (Joint common-regulator inverse-remainder theorem).

Every finite registered Gate-I action, product, and ordered commutator has a direct joint expansion along its common outer regulator. Local multi-heat terms have the uniform scaled-diagonal remainder above, and finite-rank, smoothing, and branch-B strong-dual terms have arbitrary finite Taylor order in the declared cap seminorms. After all nonpositive joint coefficients are removed,

$$\left\| \left( \widetilde{\mathbf{M}}_{\tau,\epsilon} \widetilde{\mathbf{D}}_{\tau,\epsilon}^{-1} \right)^{-1} \mathbf{R}_{\tau,\epsilon}^{\text{full}} \right\|_{\text{full},\tau,\epsilon} \longrightarrow 0, \quad \left\| \widetilde{\mathbf{M}}_{\tau,\epsilon}^{-1} \mathbf{R}_{\tau,\epsilon}^{\text{full}} \right\|_{\text{full},\tau,\epsilon} \longrightarrow 0.$$

Here the full remainder is reconstructed from bounded coefficient and regular projections. The regular projection retains every terminal positive-order, finite-rank, smoothing, and complete branch-B target component; none is erased by coefficient extraction.

#### Proof.

Sections 14.2 and 14.3 prove the common-parameter expansion and a positive remainder exponent in every local and strong-dual channel. Section 5.1 freezes the complete direct-sum carrier, bounded projections, and reconstruction before the remainder is evaluated. Section 14.1 extends the W6R.9 inverse by the identity on every typed regular summand and proves the full-cap bound  $B_\epsilon$ . Section 14.4 applies that bound to both reconstructed components. This proves the two limits on the complete carrier and hence in every continuous downstream cap channel.  $\square$

## 15. Gate-I capstone

Combining W6R.1–W6R.10 yields

$$\bigwedge_{k=1}^{15} \text{Ren6R}_k.$$

The construction has reached the last analytic prerequisite before a complete singular subtraction may be assembled.

## Exact Gate-I status

W6R.1–W6R.10: **PROVED**,  
 Ren6R-01–Ren6R-15: **SATISFIED**,  
 W6R.11–W6R.24: **NOT-REACHED**,  
 $\mathcal{R}_6^{\text{ct}}$  : **OPEN-UNPOPULATED**,  
 Gate II: **PENDING BOUNDED HOSTILE RE-REVIEW**,  
 WDW-7: **UNAUTHORIZED**.

## Part III

## Renormalization-stable successor carrier

## 2.1 Trace-admissible probe split and smooth reserve

For each even sector with Sobolev model  $H^s(\Sigma, E_r)$ , WDW-5 permits continuous probes in

$$H^{-s}(\Sigma, E_r^\vee \otimes \text{Dens}).$$

The fixed reference density and bundle metrics give a continuous inclusion

$$L^2(\Sigma, E_r^\vee; d\mu_\star) \hookrightarrow H^{-s}(\Sigma, E_r^\vee \otimes \text{Dens}).$$

A probe is  $L^2$ -**trace-admissible** when its covector belongs to this image. This condition is sufficient for one ordinary finite-rank trace; it is not by itself stable under differential constraint actions.

For the successor source, define in every even typed sector

$$\mathcal{S}_{\infty, r}^\vee = \bigcap_{k \geq 0} H^{s+k}(\Sigma, E_r^\vee \otimes \text{Dens}), \quad p_{r, k}(a) = \|a\|_{H^{s+k}}.$$

On the compact smooth target this is the Fréchet space of smooth covector-densities with the indicated tensor type. It is contained in  $L^2$ , is dense in  $H^{-s}$ , and separates every declared configuration direction. The branch- $B$  odd test space is the already imported WDW-5 smooth inverse-limit space  $\mathcal{S}_{\text{SM}, \infty}$ , with its strong-dual exterior carriers. Thus every successor seed has an infinite regularity reserve; no claim is made that an arbitrary  $L^2$  covector has this property.

If

$$F(q) = f(\ell_{a_1}(q), \dots, \ell_{a_m}(q)), \quad a_j \in L^2,$$

then its finite-rank Hessian is a finite sum of  $L^2$  rank-one operators. For a bounded kinetic multiplier  $A(q, N)$ ,

$$\text{Tr}(S_\epsilon^\vee A K_F(q) S_\epsilon) \longrightarrow \text{Tr}(A K_F(q))$$

because  $S_\epsilon \rightarrow I$  strongly on  $L^2$  and convergence is trace-norm on each fixed finite-rank operator. This proves the ordinary finite-rank endpoint on each fixed  $L^2$ -trace-admissible cylinder. For a general  $H^{-s}$  probe, the regulated expression at  $\epsilon > 0$  remains defined by W5.7, but no ordinary  $\epsilon = 0$  trace is asserted.

## 2.2 Explicit smooth local-separated grammar

Define  $\Phi_{5b}^{\text{tr}}$  before any successor word family. It is the complex graded span of exact WDW-5 semantic expressions generated by the following finite grammar.

1. A cylinder coefficient has the form

$$f(\ell_{a_1}(q), \dots, \ell_{a_m}(q)), \quad a_j \in \mathcal{S}_{\infty, r_j}^\vee, \quad f \in C^\infty(\mathbb{R}^m).$$

2. A local coefficient is a finite integral of a smooth density depending on finitely many spatial jets of the exact fields and smooth registered source coefficients.
3. A separated coefficient is a finite sum of tensor products whose covector factors are smooth maps into the corresponding  $\mathcal{S}_{\infty, r}^\vee$ . A regular-kernel coefficient is one of the inherited smoothing kernels, with all configuration derivatives smoothing.

4. Finite sums, products, constant smooth-core Fréchet differentials, multiplication by registered smooth local coefficients, and the fixed smoothing and target constructors are admitted exactly when their displayed local/separated/smoothing decomposition has smooth factors. Linear constraints, proper redesciptions, and the renormalized Hamiltonian are not inserted as closure clauses.
5. On branch  $B$ , coefficients take values in one of the accepted completed exterior degrees, have finite exterior support, and are obtained by the WDW-5 smooth strong transposes, finite  $d\Gamma$ -lifts, exterior multiplications, contractions, and printed finite ordering maps.

Equivalently, for every configuration derivative order, an expression in  $\Phi_{5b}^{\text{tr}}$  has a finite local-diagonal part with smooth finite jets, a finite separated part with every covector factor in  $\mathcal{S}_{\infty}^{\vee}$ , and a smoothing part. This is an analytic membership condition on the evaluated expression, not a label attached after applying a formal word.

For every descendant-closed cap put

$$\Phi_{5b}^{\text{c},\text{tr}} = \Phi_{5b}^{\text{c}} \cap \Phi_{5b}^{\text{tr}}, \quad \Phi_{5b}^{\text{c},+,\text{tr}} = \Phi_{5b}^{\text{c},+} \cap \Phi_{5b}^{\text{tr}}.$$

Only finite syntax and finitely many derivative orders occur in one cap, but membership requires smooth factors at every Sobolev rung.

### 2.3 Non-Hamiltonian stability and density

Let  $P$  be any registered non-Hamiltonian primitive. In a typed even probe sector it induces a finite-order differential operator  $P^{\vee}(q)$  with smooth coefficients. For every  $k$  and every configuration derivative order  $\ell$ , there are finite  $d_P$ ,  $r_{k,\ell}$ , and a locally bounded constant such that

$$p_{r,k}(d^{\ell} P^{\vee}(q)[h_1, \dots, h_{\ell}]a) \leq C_{k,\ell,q} \left( \prod_{i=1}^{\ell} \|h_i\|_{H^{r_{k,\ell}}} \right) p_{r,k+d_P}(a).$$

The infinite regularity reserve therefore absorbs every finite derivative loss. Spatial generators send  $a$  to  $-\mathcal{L}_{\xi}^* a$ ; gauge and Lorentz generators are smooth zeroth-order actions; proper redesciptions pull smooth covectors back to smooth covectors. In each case the displayed estimate applies.

The ordinary product rule preserves the local/separated/smoothing decomposition. Constant configuration differentiation differentiates only finitely many smooth coefficients. Smoothing maps send every distributional rung continuously into every smooth rung. The WDW-5 strong-transpose estimates and projective-tensor Leibniz rule give the same conclusion degreewise on branch  $B$ ; every registered exterior map has a finite degree band.

#### Lemma 2.1 (Non-Hamiltonian smooth-core stability lemma).

Every registered linear constraint action, proper-redescription pullback and inverse, configuration multiplication or differential, local coefficient multiplication, smoothing action, exterior operation, and fixed target map is a total linear endomorphism of  $\Phi_{5b}^{\text{tr}}$ . Its action preserves the exact branch, density, tensor, chirality, exterior-degree, and source labels.

#### Proof.

The estimates above prove preservation of every smooth probe factor. The ordinary and graded Leibniz rules preserve the finite local/separated/smoothing grammar. The imported W5.8–W5.11 estimates give the branch- $B$  statement in each affected completed degree, and the printed finite degree bands preserve finite exterior support.  $\square$

Smooth covectors are dense in every  $H^{-s}$  probe space. Replacing the finitely many probe covectors in one exact WDW-5 expression by smooth approximants gives convergence of every derivative on compact derivative-test sets. Each fixed regulated primitive is continuous in the imported WDW-5 section topology, so induction on the finite syntax word gives a continuous dense inclusion

$$j_b : (\Phi_{5b}^{\text{tr}}, \tau_{5b}|_{\Phi_{5b}^{\text{tr}}}) \longrightarrow (\Phi_{5b}, \tau_{5b}).$$

This argument uses only the already total regulated WDW-5 primitives; it does not assume that the new regulator-free Hamiltonian preserves the core. That remaining analytic step is proved after the capwise endpoint and literal overlap have been constructed in Section 19.3.

## 17. Global local-counterterm assembly

### 17.1 Coordinate-free insertion on the directed quotient

Let

$$\mathcal{J}_{\infty}^{\text{fin}} = \bigcup_{\text{c}} \mathcal{J}_{\text{c}}$$

be the algebraic finite-support union. The actual insertion columns of W6R.8 are natural under the global quotient inclusions. They therefore define a basis-independent filtered map

$$\mathbf{M}_\infty(\epsilon) : \mathcal{J}_\infty^{\text{fin}} \longrightarrow \mathcal{J}_\infty^{\text{fin}}.$$

Write

$$\mathbf{M}_\infty(\epsilon) = (\gamma_3 I + \mathbf{E}_\infty(\epsilon)) \mathbf{D}_\infty(\epsilon).$$

Here  $\mathbf{D}_\infty$  multiplies a heat-weight- $\rho$  source coordinate by  $\epsilon^{-\rho}$ , and  $\mathbf{E}_\infty$  strictly lowers the nonnegative integer heat weight. It preserves every exact branch, density, tensor, exterior, and constraint-role tag.

For  $x \in \mathcal{J}_\infty^{\text{fin}}$ , let  $R(x)$  be the largest heat weight in its finite support. Strict lowering gives

$$\mathbf{E}_\infty(\epsilon)^{R(x)+1} x = 0.$$

Consequently the inverse is an algebraic finite sum on every source element:

$$\mathbf{M}_\infty(\epsilon)^{-1} x = \mathbf{D}_\infty(\epsilon)^{-1} \sum_{k=0}^{R(x)} (-\gamma_3^{-1} \mathbf{E}_\infty(\epsilon))^k \gamma_3^{-1} x.$$

This formula is not an infinite convergence assertion. Its sum terminates because the structural weight is well founded. On every descendant-closed finite cap it agrees with the W6R.9 inverse by uniqueness.

## 17.2 Descendant-closed support

Given a finite cap  $\mathfrak{c}$ , close its finite coefficient support under every nonzero strictly lower block of  $\mathbf{E}_\infty$ . At one step only finitely many new quotient classes appear, and every new class has smaller nonnegative heat weight. The process therefore terminates after at most the largest heat weight in the starting packet. Denote the resulting finite refinement by  $\mathfrak{c}^\dagger$ .

If  $\mathfrak{c} \preceq \mathfrak{c}'$ , a common descendant-closed upper cap contains both  $\mathfrak{c}^\dagger$  and  $(\mathfrak{c}')^\dagger$ . Computation in any such upper cap is the restriction of the same finite global inverse. Thus no later cap can alter a previously computed counterterm coefficient.

## 17.3 Finite normalization and subtraction sign

Let  $\mathbf{h}_{\epsilon,b,\eta}^{\leq 0}[N]$  be the global locally finite operator-coefficient column containing every negative-power and local finite coefficient of the regulated Hamiltonian. It is assembled from the W6R.5 coefficient maps, not from evaluation on a chosen test vector.

Fix the source-attached local finite normalization profile  $\nu_{b,\eta}[N]$  declared in Section 2. Define the global counterterm coefficient

$$\mathbf{b}_{\epsilon,b,\eta}^{\text{ct}}[N] = -\mathbf{M}_\infty(\epsilon)^{-1} (\mathbf{h}_{\epsilon,b,\eta}^{\leq 0}[N] - \nu_{b,\eta}[N]).$$

The actual insertion identity of W6R.8 now gives, before evaluation on any  $F$ ,

$$\text{Coeff}^{\leq 0} [\widehat{\mathcal{H}}_{\epsilon,b,\eta}^0[N] + \Delta \widehat{\mathcal{H}}_{\epsilon,b,\eta}^{\text{ct}}[\mathbf{b}_{\epsilon,b,\eta}^{\text{ct}}[N]]] = \nu_{b,\eta}[N].$$

Every power, logarithmic, mixed, and boundary row has therefore been treated. For the exact closed differential heat class, the logarithmic, mixed, and boundary rows are identically zero, as already proved in W6R.5. The formula nevertheless retains their typed zero records.

The coefficient column carries its exact source locator, first cap, branch, heat and functional order, spatial-jet order, symmetry, constraint role, scale record, finite normalization, and target type. It is an operator-valued source coefficient. No number extracted from the accepted witness enters its definition.

## 17.4 Witness cancellation

For the accepted centered scalar witness  $G_{b,W}$ , W6R.5 identifies the leading operator coefficient and W6.6 supplies the positive detector:

$$p_{\lambda_{b,q_b,*}} (\widehat{\mathcal{H}}_{\epsilon,b}^0[N] G_{b,W}) = C_b \epsilon^{-5} + O(\epsilon^{-4}), \quad C_b > 0.$$

The corresponding class is one coordinate of  $\mathbf{h}_{\epsilon,b,\eta}^{\leq 0}$ . The global insertion identity therefore cancels it together with every lower singular coefficient:

$$p_{\lambda_{b,q_b,*}} (\widehat{\mathcal{H}}_{\epsilon,b}^0[N] G_{b,W} + \Delta \widehat{\mathcal{H}}_{\epsilon,b}^{\text{ct}}[\mathbf{b}_{\epsilon,b}^{\text{ct}}[N]] G_{b,W}) = O(1).$$

The same operator identity applies on branches  $A$  and  $B$ , including the complete affected exterior degree on branch  $B$ . This is not witness-specific tuning.

**Theorem W6R.11 (Complete local counterterm and obstruction-cancellation theorem).**

For every exact WDW-5 branch, candidate, lapse, and source element, the locally finite global inverse above constructs a complete typed local counterterm coefficient. Its actual W5.7-regulated insertion cancels every local negative-power coefficient and realizes the fixed local finite normalization profile. It cancels the accepted scalar  $\epsilon^{-5}$  witness in both branches as one instance of the operator identity. The construction is compatible with descendant-closed cap restriction and introduces no nonlocal subtraction or residual cutoff.

**Proof.**

W6R.4 places every local coefficient and basis operator in one hereditary quotient. W6R.7–W6R.9 identify the actual insertion map, compute its nonzero diagonal, and give its finite-cap inverse. Strict lowering of the global off-diagonal map makes the displayed inverse locally finite and equal to each descendant-closed cap inverse. W6R.8 then converts the coefficient equation into an equality of actual inserted operators. W6R.5 exhausts the allowed coefficient rows, and W6R.6 supplies locality, source attachment, covariance, and type preservation. Applying the operator identity to  $G_{b,W}$  proves the witness statement without using it to select a coefficient.  $\square$

## 18. Finite-cap endpoint

The local subtraction theorem W6R.11 is independent of the ordinary finite-rank endpoint and remains valid on the full regulated source. The regulator-free finite-rank term in this section is narrower: it is proved on

$$\Phi_{5b}^{\epsilon, \text{tr}} = \Phi_{5b}^{\epsilon} \cap \Phi_{5b}^{\text{tr}}.$$

For a general WDW-5  $H^{-s}$  probe, W5.7 supplies the regulated  $\epsilon > 0$  trace, but no ordinary  $L^2$  trace at  $\epsilon = 0$ . The earlier full-cap wording is therefore replaced by the common smooth trace-core statement below; the arbitrary  $L^2$  and rough-probe iterative endpoints remain open rather than being assigned a domain.

### 18.1 Complete finite coefficient

On a descendant-closed cap, split the finite part as

$$\mathcal{F}_{\epsilon}[N] = \mathcal{F}_{\epsilon}^{\text{loc}}[N] + \mathcal{F}_{\epsilon}^{\text{reg}}[N].$$

The first summand is represented in the local quotient and is transported to the prescribed  $\nu_{\epsilon}[N]$  by W6R.11. The second contains the finite-rank and smoothing Hessian traces and every complete typed regular coefficient retained by the Gate-I carrier.

Define

$$\widehat{\mathcal{H}}_{\epsilon}^{\text{ren}}[N] = \nu_{\epsilon}[N] + \mathcal{F}_{\epsilon}^{\text{reg}}[N].$$

The full regulated and subtracted action is

$$\begin{aligned} \widehat{\mathcal{H}}_{\epsilon, \epsilon}^{\text{sub}}[N] &= \widehat{\mathcal{H}}_{\epsilon, \epsilon}^0[N] + \Delta \widehat{\mathcal{H}}_{\epsilon, \epsilon}^{\text{ct}}[\mathbf{b}_{\epsilon, \epsilon}^{\text{ct}}[N]] \\ &= \widehat{\mathcal{H}}_{\epsilon}^{\text{ren}}[N] + \mathcal{R}_{\epsilon, \epsilon}^{\text{full}}[N]. \end{aligned}$$

W6R.10 proves

$$\|\mathcal{R}_{\epsilon, \epsilon}^{\text{full}}[N]\|_{\text{full}, \epsilon} \longrightarrow 0.$$

Continuity of semantic evaluation therefore gives the capwise operator limit.

### 18.2 Uniqueness and typing

Suppose two capwise endpoints use the same local normalization profile and the same retained regular finite coefficient. Their difference is the limit of the difference of two full remainders, hence is zero. This is uniqueness relative to declared normalization data, not scheme-independent uniqueness.

Every term is linear in the lapse. W6R.6 equivariance passes through the finite sum and limit, so proper-redescription covariance holds with the regulator, scale, and normalization data transported as part of the record. Density, branch, tensor, chirality, exterior degree, and fermionic target typing are unchanged. No branch- $B$  conclusion is inferred from degree zero.

**Theorem W6R.S12 (Smooth trace-core capwise renormalized-Hamiltonian theorem).**

For every exact branch, candidate, descendant-closed finite cap, and declared local finite normalization profile, the complete subtracted Hamiltonian has a unique limit

$$\widehat{\mathcal{H}}_{b, \eta; \epsilon}^{\text{ren}}[N] : \Phi_{5b}^{\epsilon, \text{tr}} \longrightarrow \Phi_{5b}^{\epsilon, +}$$

in the declared cap operator topology. The endpoint is continuous on the finite cap, lapse-linear, properly equivariant, and preserves every exact branch, density, exterior, and fermionic target type. No regulator remains in the capwise endpoint. Analytic preservation of the smooth trace core is not inferred from this codomain statement; it is proved in Theorem W6R.S12a below.

**Proof.**

W6R.11 cancels the complete local nonpositive coefficient column and implements the finite profile. W6R.10 makes the complete typed remainder vanish after the already computed inverse. On a smooth trace-core cylinder Hessian, the regular finite-rank channel converges in trace norm by Section 2.1; the local-generator channels retain the Gate-I heat subtraction and finite profile. The displayed limit follows. Finite dimensionality gives capwise continuity. Linearity and type preservation hold before the limit and therefore survive it. The difference argument proves uniqueness relative to the registered profile.  $\square$

## 19. Literal overlap and algebraic union

### 19.1 Restriction identity

Let  $\mathfrak{c} \preceq \mathfrak{c}'$ , and compute both endpoints in one common descendant-closed upper cap. For  $F \in \Phi_{5b}^{\mathfrak{c}, \text{tr}}$ :

- its global heat coefficient column has the same finite support in both calculations;
- the global inverse restricts to the same finite descendant computation;
- the normalization profile is source-wide and restricts literally;
- the finite-rank and smoothing endpoint is the regulator limit of the same W5.7 source action.

Hence

$$\widehat{\mathcal{H}}_{\mathfrak{c}'}^{\text{ren}}[N] \upharpoonright_{\Phi_{5b}^{\mathfrak{c}, \text{tr}}} = \widehat{\mathcal{H}}_{\mathfrak{c}}^{\text{ren}}[N].$$

This is equality of maps, not equivalence after a quotient or redescription.

### 19.2 Algebraic union

The overlap identity defines one linear map

$$\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N] : \Phi_{5b}^{\text{tr}} \longrightarrow \Phi_{5b}$$

by choosing any cap containing the finite syntax of  $F$  and applying its endpoint. The value is independent of that choice. The codomain inclusion at this stage follows because each finite regular coefficient and each finite normalization term is an admitted finite WDW expression. The stronger endomorphism statement is Theorem W6R.S12a.

This algebraic union is not automatically the locally convex inductive limit of the caps. The inherited topology on  $\Phi_{5b}^{\text{tr}}$  is the subspace compact-open Bastiani/direct-sum topology imported from WDW-5. Continuity must first be tested in that topology.

**Lemma 19.1 (Smooth trace-core literal overlap and algebraic-union lemma).**

The W6R.S12 endpoints agree literally on every smooth trace-core cap overlap and define the well-typed algebraic map  $\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N]$  above. Consequently the successor overlap and union clauses are satisfied. The full rough-probe clauses of original AC6R-29–AC6R-35 are not promoted.

**Proof.**

All cap calculations are restrictions of the same source-wide heat coefficient map, locally finite inverse, normalization profile, and regular endpoint. Descendant closure makes every finite inverse calculation independent of the chosen upper cap. Directed exhaustion then defines the union map.  $\square$

### 19.3 Generator-by-generator invariant-domain theorem

It remains to prove that the map just constructed lands back in the analytic class of Section 2.2. Let

$$d^2 F(q) = A_F(q) + K_F(q) + R_F(q)$$

be its structural Hessian decomposition on one finite cap:  $A_F$  is local diagonal and finite order,  $K_F$  is a finite separated sum, and  $R_F$  is smoothing. Every configuration derivative has the same three-way form with smooth coefficients and smooth separated factors.

For the local diagonal part, W6R.5 expresses every heat coefficient as a universal finite polynomial in finite jets of the symbol of  $A_F$ , the kinetic multiplier, the lapse, the reference heat operator, and the registered

smooth coefficients. W6R.11 subtracts the nonpositive local column and replaces its finite local coordinate by the declared smooth normalization profile. Configuration differentiation only differentiates finitely many such smooth jets. The resulting local part therefore remains in the local clause of Section 2.2. The regular tails of the concrete counterterm insertions satisfy the same three-way decomposition by Lemma 8.1, because their fixed representatives are smooth local operator densities; they are included in the separated or smoothing calculations below rather than omitted.

Write the separated part, after the fixed density and bundle identifications, as

$$K_F(q) = \sum_{\alpha=1}^J u_{\alpha}(q) \otimes v_{\alpha}(q), \quad u_{\alpha}(q), v_{\alpha}(q) \in \mathcal{S}_{\infty}^{\vee}.$$

Its ordinary endpoint is the finite sum

$$\mathrm{Tr}(M_{N\mathbb{K}_b(q)} K_F(q)) = \sum_{\alpha=1}^J \langle v_{\alpha}(q), M_{N\mathbb{K}_b(q)} u_{\alpha}(q) \rangle_{L^2}.$$

Every  $\ell$ -th configuration derivative is a finite Leibniz sum of terms

$$\langle d^{\ell_1} v_{\alpha}(q)[\mathbf{h}_1], d^{\ell_0} M_{N\mathbb{K}_b(q)}[\mathbf{h}_0] d^{\ell_2} u_{\alpha}(q)[\mathbf{h}_2] \rangle_{L^2}, \quad \ell_0 + \ell_1 + \ell_2 = \ell.$$

The factors remain in  $\mathcal{S}_{\infty}^{\vee}$  by the estimates in Section 2.3, so these are smooth separated coefficients at every derivative order. This is the analytic step missing from a bare  $L^2$ -seed argument.

For  $R_F$ , composition with the order-zero kinetic multiplier is nuclear on the  $L^2$  pivot. The smoothing-kernel seminorm estimates are stable under every configuration derivative, and the nuclear trace is a continuous linear functional on that kernel class. Hence the smoothing trace and all its derivatives remain in the regular-kernel clause. The first-order ordering correction uses only  $dF$  and registered smooth coefficients, while the potential term is multiplication by a local generator; Lemma 2.1 and the product rule preserve the grammar.

On branch  $B$ , the preceding calculation is applied after an arbitrary continuous seminorm in each of the finitely many affected completed exterior degrees. The imported W5.10–W5.11 strong-transpose estimates preserve the smooth one-particle core; the finite  $d\Gamma$ , Yukawa, exterior, contraction, and ordering maps preserve the printed finite degree band. Thus no branch- $B$  component leaves the common smooth trace domain.

**Theorem W6R.S12a (Common smooth trace-core invariance theorem).**

For every exact branch, candidate label, lapse, ordering, smoothing profile, proper-redescription arrow, and finite normalization profile, the algebraically glued renormalized Hamiltonian and every other registered primitive are total linear endomorphisms

$$\mathcal{G}_b^{\mathrm{ren}} \subset \mathrm{End}_{\mathbb{C}}(\Phi_{5b}^{\mathrm{tr}}).$$

Every type-correct finite word in these maps is therefore an actual total endomorphism of the same analytic trace domain. The conclusion holds degreewise on branch  $B$  with finite exterior support.

**Proof.**

Lemma 2.1 covers every non-Hamiltonian primitive. The local, finite-separated, smoothing, ordering, potential, and branch- $B$  calculations above prove that the renormalized Hamiltonian preserves every clause of the explicit grammar, including all configuration derivatives. Induction on word length is now legitimate because each generator is already a total endomorphism before the word family is formed.  $\square$

## 20. Full-source continuity test

### 20.1 A source-certified cylinder sequence

Fix either branch and the scalar sector used by the accepted WDW-6 witness. Choose a smooth chart center  $q_{\star}$  and a nonnegative lapse  $N$  whose scalar kinetic multiplier is strictly positive on the compact target. The resulting weighted scalar pairing is

$$\langle u, v \rangle_{\kappa} = \int_{\Sigma} N(x) \kappa_b(q_{\star}, x) \overline{u(x)} v(x) d\mu_{\star}(x),$$

with  $\kappa_b(q_{\star}, x) \geq c_b > 0$  on the selected support.

Smooth sections are dense in this weighted scalar  $L^2$  space. Gram–Schmidt applied to one countable smooth total family therefore supplies a smooth  $\langle \cdot, \cdot \rangle_{\kappa}$ -orthonormal sequence  $(u_n)_{n \geq 1}$ . Thus

$$\ell_n(q) = \langle u_n, q - q_{\star} \rangle_{\kappa}$$

is an admitted smooth trace-core WDW-5 cylinder coordinate, and

$$F_n(q) = \ell_n(q)^2$$

belongs to  $\text{Cyl}_b^\infty \cap \Phi_{5b}^{\text{tr}}$ .

## 20.2 Convergence in the exact topology

The functionals  $\ell_n$  are equicontinuous on the fixed Sobolev chart: the weighted  $L^2$  pairing is continuously dominated by the chart norm. They are pointwise null because an orthonormal sequence is weakly null. Equicontinuity and a finite-net argument imply

$$\sup_{q \in K} |\ell_n(q)| \rightarrow 0$$

for every compact  $K$  in the chart. The same statement holds for every compact set of tangent directions.

The derivatives are

$$\begin{aligned} F_n(q) &= \ell_n(q)^2, \\ dF_n(q)[v] &= 2\ell_n(q)\ell_n(v), \\ d^2F_n(q)[v, w] &= 2\ell_n(v)\ell_n(w), \\ d^rF_n &= 0 \quad (r \geq 3). \end{aligned}$$

Uniform compact convergence of the  $\ell_n$  therefore gives

$$\|F_n\|_{r,K,p} \rightarrow 0 \quad \text{for every defining compact-open Bastiani seminorm.}$$

Hence

$$F_n \rightarrow 0 \quad \text{in } (\Phi_{5b}^{\text{tr}}, \tau_{5b}|_{\Phi_{5b}^{\text{tr}}}).$$

## 20.3 The endpoint does not converge

At the chart center,

$$F_n(q_\star) = 0, \quad dF_n(q_\star) = 0, \quad d^2F_n(q_\star) = 2\ell_n \otimes \ell_n.$$

The Hessian is a separated rank-one kernel. Lemma 8.1 and W6R.5 place it in the finite-rank channel, not in the local differential singular quotient. Consequently:

- it has no negative-power heat coefficient;
- its local coefficient column is zero;
- W6R.11 inserts no local counterterm in this channel;
- the potential and first-order ordering terms vanish at  $q_\star$  because  $F_n(q_\star) = dF_n(q_\star) = 0$ .

The W5.7 half-heat trace instead has the finite endpoint

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \text{Tr}(S_\epsilon^\vee M_{N\mathbb{K}_b(q_\star)} d^2F_n(q_\star) S_\epsilon) &= \text{Tr}(M_{N\mathbb{K}_b(q_\star)} d^2F_n(q_\star)) \\ &= 2\langle u_n, u_n \rangle_\kappa = 2. \end{aligned}$$

Including the fixed nonzero kinetic prefactor gives one  $\kappa_b^{\text{WDW}} \neq 0$ , independent of  $n$ , such that

$$(\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N]F_n)(q_\star) = \kappa_b^{\text{WDW}} \quad (n \geq 1).$$

Evaluation at  $q_\star$  is the defining seminorm  $\|\cdot\|_{0,\{q_\star\},|\cdot|}$ . Therefore

$$\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N]F_n \not\rightarrow 0.$$

### Theorem 20.1 (Finite-rank trace continuity obstruction).

The literally glued Gate-II endpoint

$$\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N] : (\Phi_{5b}^{\text{tr}}, \tau_{5b}|_{\Phi_{5b}^{\text{tr}}}) \rightarrow (\Phi_{5b}^{\text{tr}}, \tau_{5b}|_{\Phi_{5b}^{\text{tr}}})$$

is not continuous on either branch. Hence it has no continuous endomorphism extension to the original  $V_b$ . The failure is detected inside the smooth trace-core cylinder algebra by the sequence  $F_n$  above and survives every allowed local singular subtraction.

### Proof.

Section 20.2 proves  $F_n \rightarrow 0$  in the inherited subspace topology. Section 20.3 computes a nonzero constant endpoint under the continuous evaluation seminorm. A continuous linear map must send a convergent zero sequence to a convergent zero sequence. The displayed failure proves discontinuity. A continuous extension to  $V_b$  would restrict to a continuous map on the dense smooth trace-core subspace, which is impossible. The local counterterm is absent because the rank-one Hessian lies in the regular finite-rank channel, so the conclusion is unchanged by the complete W6R.11 subtraction.  $\square$

## 20.4 Disposition of W6R.13–W6R.15

Original W6R.13 requires both literal overlap and continuity of the inductive-union map in  $\tau_{5b}$ . Lemma 19.1 proves the smooth trace-core overlap and algebraic-union clauses, but Theorem 20.1 falsifies the continuity clause even on that dense subspace. Original W6R.13 is therefore **OBSTRUCTED**, not proved.

Original W6R.14 depends on completed W6R.13 and is **NOT-REACHED**. Scale, ordering, smoothing, density, connection, and finite-normalization labels remain standing-bearing data. No equivalence between two such records is asserted.

Original W6R.15 is also **NOT-REACHED**. In particular,

$$\left(\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}\right)^{\prime \times}$$

cannot be formed by the accepted W6.3–W6.5 transpose theorem, because that theorem requires a continuous endomorphism of  $V_b$ . The accepted domain  $\mathcal{D}_{6b} = \mathcal{K}_{6b}$  is not reused.

## 25. Authorized successor boundary

The original Gate-II obstruction is a theorem, not an editorial defect. The present construction therefore leaves

$$V_b = (\Phi_{5b}, \tau_{5b}), \quad \mathcal{K}_{6b} = ((V'_b)^\times, \sigma)$$

unchanged and constructs a new record. The standing-bearing update has two independently visible coordinates:

1. the algebraic source is the dense common smooth trace subspace  $\Phi_{5b}^{\text{tr}}$ , rather than the full arbitrary  $H^{-s}$ -probe source or the arbitrary  $L^2$ -probe class; and
2. the source topology is the graph topology  $\tau_{5b}^{\text{ren}}$ , rather than the inherited compact-open Bastiani topology.

The construction order is fixed. Section 2 first defines the smooth local-separated grammar and proves every non-Hamiltonian primitive stable. W6R.11 constructs the local subtraction algebraically. W6R.S12 then constructs the regulator-free Hamiltonian on the smooth trace-core finite caps, Lemma 19.1 glues it into the ambient source, and W6R.S12a proves that its local, separated, smoothing, ordering, and branch- $B$  outputs return to the same smooth core. Only after all generators are actual total endomorphisms is the word family fixed and the graph topology formed. Thus neither the source definition nor the topology manufactures the operator whose graph it controls.

The old cylinder sequence remains decisive:

$$F_n \longrightarrow 0 \quad \text{in } \tau_{5b}|_{\Phi_{5b}^{\text{tr}}}, \quad \widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N]F_n \not\rightarrow 0.$$

The successor must make this difference visible; it may not quotient the sequence, subtract its regular trace nonlocally, or retain a regulator.

## 26. The minimal renormalization-stable graph topology

### 26.1 Word and seminorm families

By Theorem W6R.S12a, every map below is already a total endomorphism of  $\Phi_{5b}^{\text{tr}}$ . Let  $\mathcal{P}_b^{\text{old}}$  be the complete family of continuous seminorms defining the inherited subspace topology  $\tau_{5b}|_{\Phi_{5b}^{\text{tr}}}$ . The family  $\mathcal{G}_b^{\text{ren}}$  from Section 2.1 contains:

- every algebraically glued  $\widehat{\mathcal{H}}_{\text{ren}, b, \eta}^{\text{alg}}[N]$ ;
- every exact spatial-diffeomorphism, internal-Gauss, and, on branch  $B$ , local-Lorentz source action;
- every proper-redescription pullback and its inverse on the registered arrow;
- every fixed multiplication, differential, exterior, and target map that occurs as a primitive in the finite task registers.

All labels, lapses, arrows, orderings, smoothing profiles, density conventions, and field-space connections remain part of the generator record. No unregistered endpoint map is added later.

Let

$$\mathcal{W}_b^{\text{ren}} = \langle I, \mathcal{G}_b^{\text{ren}} \rangle_{\text{fin}}$$

be the family of all type-correct finite words. It contains the identity and is closed under right composition by every registered primitive. Define

$$\mathcal{P}_b^{\text{ren}} = \{p_W : p \in \mathcal{P}_b^{\text{old}}, W \in \mathcal{W}_b^{\text{ren}}\}, \quad p_W(F) = p(WF),$$

and let  $\tau_{5b}^{\text{ren}}$  be the locally convex topology generated by  $\mathcal{P}_b^{\text{ren}}$ .

Since  $I \in \mathcal{W}_b^{\text{ren}}$ , the new topology is Hausdorff and stronger than the inherited topology. It remains a genuine locally convex topology: every  $p_W$  is a seminorm because every word is linear.

## 26.2 Continuity and minimality

For  $A \in \mathcal{G}_b^{\text{ren}}$ ,

$$p_W(AF) = p((W \circ A)F) = p_{W \circ A}(F).$$

Because  $W \circ A \in \mathcal{W}_b^{\text{ren}}$ , every registered primitive is continuous on  $(\Phi_{5b}^{\text{tr}}, \tau_{5b}^{\text{ren}})$ . The same identity proves continuity of every finite word.

Now let  $v$  be any locally convex topology on the same algebraic source that is stronger than the inherited topology and makes every member of  $\mathcal{G}_b^{\text{ren}}$  continuous. Every finite word is then  $v$ -continuous, and every  $p_W$  is  $v$ -continuous. Hence

$$\tau_{5b}^{\text{ren}} \subseteq v.$$

The successor topology is therefore the coarsest topology with the declared continuity property. It is not an arbitrarily chosen maximal or discrete topology.

## 26.3 Exact exclusion of the old null sequence

Let

$$p_*(G) = |G(q_*)|$$

in the scalar output channel used in Theorem 20.1. This is an inherited continuous seminorm. Therefore

$$q_{*,N}(F) = p_*(\widehat{\mathcal{H}}_{\text{ren}}^{\text{alg}}[N]F)$$

belongs to  $\mathcal{P}_b^{\text{ren}}$ . For the smooth trace-core sequence,

$$q_{*,N}(F_n) = |\kappa_b^{\text{WDW}}| > 0.$$

Thus  $F_n \not\rightarrow 0$  in  $\tau_{5b}^{\text{ren}}$ . The topology records the exact operation that the old topology failed to control.

### Theorem W6R.S13 (Successor overlap, graph-continuity, and minimality theorem).

For each exact branch and registered successor candidate, the compatible smooth trace-core cap endpoints define a linear endomorphism of  $\Phi_{5b}^{\text{tr}}$ . Every renormalized Hamiltonian, linear constraint action, proper-redescription action, and registered primitive word is continuous in  $\tau_{5b}^{\text{ren}}$ . This topology is the coarsest locally convex topology stronger than the inherited subspace topology with that property. Its identity comparison to the inherited topology is continuous, while the reverse identity is not continuous.

#### Proof.

Lemma 19.1 gives literal overlap, and Theorem W6R.S12a proves analytic endomorphism totality before any word is formed. The  $p_W(AF) = p_{W \circ A}(F)$  identity proves continuity. The preceding universal-property argument proves minimality. The identity word proves continuity toward the inherited topology. The sequence  $F_n$  converges to zero in the inherited topology and fails to do so in the graph topology, proving that the reverse identity is not continuous.  $\square$

## 27. Completion and comparison map

### 27.1 Branch A

Let

$$V_{A,0}^{\text{ren}} = (\Phi_{5A}^{\text{tr}}, \tau_{5A}^{\text{ren}})$$

and define  $V_A^{\text{ren}}$  to be its Hausdorff completion. Continuous linear maps are uniformly continuous for the canonical uniformity of a locally convex space. Every registered primitive therefore extends uniquely to a continuous endomorphism of  $V_A^{\text{ren}}$ . Algebraic identities among finite words persist after completion because the algebraic source is dense and the completed codomain is Hausdorff.

### 27.2 Branch B and finite exterior degree

No Fock completion is introduced. Write the algebraic source as the finite-particle direct sum

$$\Phi_{5B}^{\text{tr}} = \bigoplus_{n \geq 0}^{\text{alg}} \Phi_{5B,n}^{\text{tr}}.$$

For each input degree  $n$ , give  $\Phi_{5B,n}^{\text{tr}}$  the graph topology generated by all component seminorms

$$F \mapsto p_m(P_m W F),$$

where  $P_m$  is the output-degree projection and  $W$  ranges over the registered finite words. W5.8–W5.11 and the Gate-I typing register imply that every primitive is finite-band in exterior degree, so only finitely many  $m$  occur for one  $W$  and one input degree.

Complete each degree and set

$$V_B^{\text{ren}} = \bigoplus_{n \geq 0} \widehat{\Phi}_{5B,n}^{\text{tr,ren}}$$

with the locally convex direct-sum topology. Its elements retain finite exterior support. The finite-band component maps extend degreewise and assemble to continuous endomorphisms. This is the completed finite-particle successor carrier, not a hidden Hilbert or infinite-wedge space.

### 27.3 Comparison with the old source

On the algebraic core, the identity gives

$$j_b : (\Phi_{5b}^{\text{tr}}, \tau_{5b}^{\text{ren}}) \longrightarrow (\Phi_{5b}, \tau_{5b}).$$

It is continuous because the identity word retains every old seminorm, injective because no quotient is taken, and dense because smooth  $L^2$ -representable covectors are dense in  $H^{-s}$  and finite-probe cylinder derivatives converge uniformly on compact derivative-test sets.

No inverse continuous comparison exists: its restriction to  $\Phi_{5b}^{\text{tr}}$  would send the old-null sequence  $F_n$  to the same sequence in the graph topology, contradicting Section 26.3. The completion is therefore a new standing-bearing source record. No continuous extension of  $j_b$  into the uncompleted old  $V_b$  is asserted.

## 28. Scale, scheme, ordering, and smoothing transport

Let a complete Gate-II candidate record be

$$r = (b, \eta, \mu, \nu, \mathfrak{D}, \mathcal{K}, \delta, \nabla^{\text{fld}})$$

with the already fixed branch, source, couplings, target types, and counterterm basis understood. Its smooth trace-core endpoint is

$$\widehat{\mathcal{H}}_r^{\text{ren}}[N] = \nu_r[N] + \mathcal{F}_r^{\text{reg}}[N].$$

For two declared records  $r, r'$ , the complete output difference is

$$\Delta_{r' \leftarrow r}[N] = \widehat{\mathcal{H}}_{r'}^{\text{ren}}[N] - \widehat{\mathcal{H}}_r^{\text{ren}}[N] = (\nu_{r'} - \nu_r)[N] + (\mathcal{F}_{r'}^{\text{reg}} - \mathcal{F}_r^{\text{reg}})[N].$$

This formula compares the complete local finite, finite-rank, smoothing, ordering, density, connection, and branch- $B$  target-valued outputs; it does not compare principal parts alone.

For the exact closed differential heat class used here, every logarithmic coefficient vanishes. Holding the finite normalization profile fixed therefore gives

$$\widehat{\mathcal{H}}_{\mu'}^{\text{ren}}[N] = \widehat{\mathcal{H}}_{\mu}^{\text{ren}}[N].$$

If the declared finite profile itself depends on scale, the complete transport is instead

$$\widehat{\mathcal{H}}_{\mu'}^{\text{ren}}[N] - \widehat{\mathcal{H}}_{\mu}^{\text{ren}}[N] = \nu(\mu')[N] - \nu(\mu)[N].$$

No beta function is manufactured from the identically zero logarithmic rows.

For pairwise comparison, define the joint graph topology from the composition closure of  $\mathcal{G}_r^{\text{ren}} \cup \mathcal{G}_{r'}^{\text{ren}}$ . Both endpoints and  $\Delta_{r' \leftarrow r}$  are continuous on that joint comparison source, and the identity maps from the joint topology to each individual graph topology are continuous.

This comparison is not a physical-equivalence certificate. A complete equivalence would require a bicontinuous intertwiner transporting the operator domains, full constraint and anomaly records, state carrier, physical form, observables, and continuation interfaces. Those downstream objects have not yet been constructed. Only literally identical complete records receive the identity certificate at Gate II; every nonidentical scale, finite-profile, ordering, smoothing, density, or connection record remains standing-bearing and unmerged.

**Theorem W6R.S14 (Successor transport and classification-guard theorem).**

The displayed formulas give the complete Gate-II transport for renormalization scale, finite normalization, ordering, smoothing, density, and field-space connection changes. Every transport difference is continuous on the corresponding joint successor source. Scale transport is the identity when the finite profile is scale independent. No two distinct records are declared physically equivalent without the complete downstream intertwiner required by the fixed standard.

**Proof.**

The endpoint decomposition gives the difference identity term by term. The zero logarithmic rows give the scale formula. The joint word family contains both endpoints, so Theorem W6R.S13 gives continuity of both and their difference. The final statement is the fixed physical-equivalence predicate applied without suppressing its unconstructed coordinates.  $\square$

## 29. Complete continuous-dual realization

### 29.1 Successor carrier and evaluation embedding

For either completed branch define

$$\mathcal{K}_{6b}^{\text{ren}} = \left( ((V_b^{\text{ren}})')^\times, \sigma(((V_b^{\text{ren}})')^\times, (V_b^{\text{ren}})') \right).$$

The evaluation map

$$\iota_{6b}^{\text{ren}} : V_b^{\text{ren}} \longrightarrow \mathcal{K}_{6b}^{\text{ren}}, \quad \iota_{6b}^{\text{ren}}(F)(\lambda) = \lambda(F),$$

is continuous and injective because the continuous dual separates points of a Hausdorff locally convex space.

Its image is weakly dense. Indeed, a basic weak neighborhood tests only finitely many linearly independent  $\lambda_1, \dots, \lambda_m \in (V_b^{\text{ren}})'$ . The map

$$F \mapsto (\lambda_1(F), \dots, \lambda_m(F))$$

is onto  $\mathbb{C}^m$ ; otherwise a nonzero linear combination of the  $\lambda_j$  would vanish identically. Hence every finite weak coordinate prescription is attained by an evaluation vector.

The weak algebraic dual is complete: it is the closed linearity subspace of the complete product  $\mathbb{C}^{(V_b^{\text{ren}})'}$ .

### 29.2 Algebraic-transpose extension

Let  $A : V_b^{\text{ren}} \rightarrow V_b^{\text{ren}}$  be any registered continuous source endomorphism. Its continuous transpose

$$A' : (V_b^{\text{ren}})' \longrightarrow (V_b^{\text{ren}})'$$

defines

$$A'^\times : \mathcal{K}_{6b}^{\text{ren}} \longrightarrow \mathcal{K}_{6b}^{\text{ren}}, \quad (A'^\times \Lambda)(\lambda) = \Lambda(A'\lambda).$$

This map is weakly continuous and satisfies

$$A'^\times \iota_{6b}^{\text{ren}}(F) = \iota_{6b}^{\text{ren}}(AF).$$

It is the unique weakly continuous extension because the evaluation image is weakly dense and the carrier is Hausdorff.

Every registered primitive, product, commutator, and finite word is a continuous endomorphism of  $V_b^{\text{ren}}$ . Therefore

$$\mathcal{D}_{6b}^{\text{ren}} = \mathcal{K}_{6b}^{\text{ren}}$$

is one literal common domain for all declared transpose actions and their finite compositions. This domain statement does not assert quantum constraint closure; Gate III must compute the commutators and anomalies.

**Theorem W6R.S15 (Successor completion, continuous-dual, and common-domain theorem).**

The smooth trace-core graph source has the branchwise complete realization  $V_b^{\text{ren}}$  above. Every registered source primitive extends uniquely and continuously to it. The carrier  $\mathcal{K}_{6b}^{\text{ren}}$  is complete in its declared weak topology, contains the continuous dense evaluation image, and supports the unique full-domain algebraic-transpose extension of every registered primitive, product, commutator, and finite word. Branch  $B$  retains finite exterior support degree-wise. No Hilbert carrier or old  $\mathcal{K}_{6b}$  domain is reused.

**Proof.**

Sections 27.1–27.2 give the completions and continuous extensions. Section 29.1 proves separation, density, and completeness. Section 29.2 proves the transpose identity, uniqueness, and full common domain. The branch- $B$  finite-band construction preserves finite exterior support.  $\square$

## Part IV

# Renormalized constraint and anomaly closure

## 4. Proper-redescription orbit carrier

### 4.1 Orbit and fibre family

Fix one branch  $b$ , one accepted complete successor record  $r$ , and its proper-redescription orbit

$$\mathcal{O}_r = \{gr : g \in \mathcal{G}_{\text{prop}}, s(g) = r\}.$$

The WDW-2 action groupoid restricts to  $\mathcal{G}_r \rightrightarrows \mathcal{O}_r$ . Every individual orbit has constant isotropy type and carries its smooth tame homogeneous-orbit structure. No global slice or smooth orbit space is assumed. Isotropy acts in the fibre, and disconnected large transformations remain outside  $\mathcal{G}_r$ .

Over  $\mathcal{O}_r$ , form the locally convex groupoid bundle

$$\mathbb{E}_r^{\text{ren}} = \coprod_{r' \in \mathcal{O}_r} V_{r'}^{\text{ren}} \longrightarrow \mathcal{O}_r.$$

For an arrow  $g : r' \rightarrow r''$ , the fibre action is the accepted bicontinuous map

$$U_g : V_{r'}^{\text{ren}} \longrightarrow V_{r''}^{\text{ren}}.$$

The groupoid laws hold literally:

$$U_{1_{r'}} = 1, \quad U_{g^{-1}} = U_g^{-1}, \quad U_{g_2 g_1} = U_{g_2} U_{g_1}.$$

This construction does not identify  $r'$  and  $r''$ . They remain different base objects connected by a typed arrow.

### 4.2 Local section grammar

Let  $\Phi_{r, \text{orb}}^{\text{tr}}$  be the space of smooth orbitwise sections  $\Psi$  such that, in every WDW-2 orbit chart and every groupoid trivialization by  $U_g$ :

1.  $\Psi(r')$  lies in the accepted smooth trace core  $\Phi_{5b, r'}^{\text{tr}}$ ;
2. every finite orbit derivative lies in that same trace core;
3. every type-correct finite fibre word applied before or after any finite orbit derivative again has the accepted local, finite-separated, smoothing, ordering, potential, and branch- $B$  grammar; and
4. only finitely many exterior degrees are occupied on branch  $B$ .

These conditions are invariant under a change of orbit chart. Indeed, if  $g : r' \rightarrow r''$ , complete-record covariance maps every registered word  $W_{r'}$  to

$$W_{r''}^g = U_g W_{r'} U_g^{-1}.$$

The Gate-II smooth-core theorem applies in every fibre, while WDW-2 smoothness supplies the finite orbit derivatives of  $U_g$ .

### 4.3 Mixed graph topology

Choose an orbit chart, a compact chart set  $K$ , an integer  $m \geq 0$ , a continuous fibre seminorm  $p$ , and a type-correct registered fibre word  $W$ . Define

$$q_{K, m, p, W}(\Psi) = \max_{0 \leq j \leq m} \sup_{r' \in K} p_{r'}(W_{r'} \nabla_{\text{orb}}^j \Psi(r')).$$

The connection symbol here denotes orbit-chart differentiation with the transition correction induced by  $U_g$ ; it introduces no independent physical connection coordinate. On chart overlaps, the chain rule and the bicontinuity of  $U_g$  make the two seminorm families equivalent.

Let

$$\tau_{r, \text{orb}}^{\text{ren}} = \text{lc}\{q_{K, m, p, W}\}$$

and write

$$\mathbb{V}_r^{\text{ren}} = (\widehat{\Phi_{r, \text{orb}}^{\text{tr}}, \tau_{r, \text{orb}}^{\text{ren}}}).$$

The hat is the Hausdorff locally convex completion. This topology is the coarsest locally convex section topology that:

1. retains smooth orbit differentiation;
2. makes evaluation into every fibre continuous;

3. makes every registered fibre word continuous; and
4. makes every proper groupoid transport bicontinuous.

The continuous-dual carrier is reconstructed rather than inherited:

$$\mathbb{K}_r^{\text{ren}} = ((\mathbb{V}_r^{\text{ren}})')^\times$$

with the weak algebraic-dual topology used at the accepted successor. As before, it is complete, fibre evaluations have weakly dense span, and every continuous source endomorphism has a literal full-domain transpose action.

## 5. Reissued orbit Gate II

### 5.1 Fibrewise primitive actions

For a descriptor  $\sigma$  transported over the orbit, define

$$\begin{aligned} (\mathbb{H}[N]\Psi)(r') &= H_{r'}[N]\Psi(r'), \\ (\mathbb{C}_{\text{lin}}[\sigma]\Psi)(r') &= C_{\text{lin},r'}[\sigma]\Psi(r'). \end{aligned}$$

The Hamiltonian in this display differentiates only the physical configuration variables inside its fibre. Orbit differentiation belongs to the diagonal proper-redescription generator constructed below.

Every fibre map preserves the local section grammar. Mixed orbit derivatives are finite Leibniz sums of:

- orbit derivatives of the complete coefficients;
- orbit derivatives of the section; and
- the accepted finite fibre words.

Complete-record smooth covariance and W6R.S12a control every term. Therefore every primitive and finite word is continuous in  $\tau_{r,\text{orb}}^{\text{ren}}$ , extends uniquely to  $\mathbb{V}_r^{\text{ren}}$ , and has a full-domain transpose on  $\mathbb{K}_r^{\text{ren}}$ .

### 5.2 Inter-fibre transport

For an arrow  $g$ , define the induced action on sections by

$$(\mathbb{U}_g\Psi)(r') = U_g\Psi(g^{-1}r').$$

The groupoid and fibre laws give

$$\mathbb{U}_{g_2g_1} = \mathbb{U}_{g_2}\mathbb{U}_{g_1}, \quad \mathbb{U}_{g^{-1}} = \mathbb{U}_g^{-1}.$$

The mixed graph seminorms prove bicontinuity. Complete covariance gives

$$\mathbb{U}_g\mathbb{H}[N]\mathbb{U}_g^{-1} = \mathbb{H}[gN]$$

and the corresponding identities for all linear constraints and registered primitives.

### 5.3 Orbit-carrier reissue theorem

**Theorem W6R.O12–W6R.O15 (Orbit Gate-II reissue).**

For every accepted proper-redescription orbit, the local section grammar, mixed graph topology, completion, continuous-dual carrier, fibrewise renormalized Hamiltonians, linear constraints, and complete groupoid transports above are well defined. Every primitive is a continuous endomorphism, every finite word is total, every proper arrow is bicontinuous, and every transpose acts on the literal full dual domain. The family is compatible with cap restriction, scheme labels, ordering, smoothing, density, connection, and finite-profile transport exactly when those coordinates belong to the same accepted complete-record orbit.

**Proof.**

W6R.S12a supplies analytic endomorphism invariance in each fibre. W6R.S13 supplies the fibre graph universal property. W6R.S14 supplies the complete covariance and classification guard, and W6R.S15 supplies completion and transpose reconstruction. WDW-2 supplies the groupoid laws and smooth orbit charts. The mixed seminorm calculation in Sections 4.2–4.3 proves continuity of orbit derivatives and fibre words simultaneously. Completion and the weak algebraic-dual argument then repeat without reusing either the original compact-open carrier or one selected fibre as the orbit carrier.  $\square$

### Classification

The identity from a fibre into the section carrier is an evaluation insertion, not an equivalence of records. Orbit position, isotropy, and the complete transported renormalization record remain standing-bearing. The construction supplies inter-fibre comparison without quotienting the base objects.

## 6. W6R.O16 linear closure

The fibrewise linear constraints obey the already proved complete semidirect algebra. The induced section action preserves brackets because it is a groupoid representation. Thus, with

$$\mathbb{D}_\xi, \quad \mathbb{G}_\lambda, \quad \mathbb{J}_\vartheta$$

denoting the spatial, internal, and Lorentz generators,

$$\begin{aligned} [\mathbb{D}_\xi, \mathbb{D}_\eta] &= -i\hbar \mathbb{D}_{[\xi, \eta]}, \\ [\mathbb{D}_\xi, \mathbb{G}_\lambda] &= -i\hbar \mathbb{G}_{\mathcal{L}_\xi \lambda}, \\ [\mathbb{D}_\xi, \mathbb{J}_\vartheta] &= -i\hbar \mathbb{J}_{\mathcal{L}_\xi \vartheta}, \\ [\mathbb{G}_\lambda, \mathbb{G}_\mu] &= -i\hbar \mathbb{G}_{[\lambda, \mu]}, \\ [\mathbb{J}_\vartheta, \mathbb{J}_\chi] &= -i\hbar \mathbb{J}_{[\vartheta, \chi]}, \\ [\mathbb{G}_\lambda, \mathbb{J}_\vartheta] &= 0. \end{aligned}$$

The identities hold on the smooth orbit core, extend to  $\mathbb{V}_r^{\text{ren}}$ , and transpose to the full domain  $\mathbb{K}_r^{\text{ren}}$ . Branch  $B$  retains every matter sector, every completed finite exterior degree, and all six Lorentz directions.

### Theorem W6R.O16 (Orbit linear-closure theorem).

The complete linear quantum constraint family closes on the orbit source, completion, and transpose carrier with the displayed graded semidirect algebra.

#### Proof.

W6R.S16 proves the identities in every fibre. The section action is the differential of the exact groupoid representation  $\mathbb{U}$ , so bracket preservation is literal. Continuity and transpose extension follow from W6R.O12–W6R.O15.  $\square$

## 7. W6R.O17 Hamiltonian-linear closure

### 7.1 Diagonal infinitesimal action

Let  $g_t$  be a one-parameter proper-redescription family generated by  $\alpha$ . Define

$$\mathbb{L}_\alpha = \left. \frac{d}{dt} \right|_{t=0} \mathbb{U}_{g_t}, \quad \mathbb{C}_\alpha = -i\hbar \mathbb{L}_\alpha.$$

In a local orbit trivialization,

$$\mathbb{L}_\alpha = L_\alpha^{\text{field}} + L_\alpha^{\text{record}}.$$

The first term differentiates the fibre action  $U_{g_t}$ ; the second differentiates the base argument  $g_t^{-1}r'$ . The split depends on a local trivialization, but their sum is the intrinsic infinitesimal groupoid action.

### 7.2 Exact cancellation of the predecessor defect

The orbit Hamiltonian is fibrewise:

$$(\mathbb{H}[N]\Psi)(r') = H_{r'}[N]\Psi(r').$$

The exact finite covariance identity is

$$\mathbb{U}_{g_t} \mathbb{H}[N] \mathbb{U}_{g_t}^{-1} = \mathbb{H}[g_t N].$$

Differentiating on the common smooth orbit core gives

$$[\mathbb{L}_\alpha, \mathbb{H}[N]] = \mathbb{H}[\dot{N}_\alpha].$$

With  $\mathbb{C}_\alpha = -i\hbar \mathbb{L}_\alpha$ , the convention used in the predecessor is

$$[\mathbb{H}[N], \mathbb{C}_\alpha] = i\hbar \mathbb{H}[\dot{N}_\alpha].$$

To compare with the predecessor formula, evaluate the local split at one record  $r$ . The field part alone gives

$$[H_r[N], C_\alpha^{\text{field}}] = i\hbar H_r[\dot{N}_\alpha] + i\hbar \mathfrak{V}_r[\alpha; N].$$

The base derivative acts on the multiplication field  $r' \mapsto H_{r'}[N]$  and gives

$$[H[N], C_\alpha^{\text{record}}] \Big|_r = -i\hbar \mathfrak{V}_r[\alpha; N].$$

Their sum cancels the vertical term exactly. No equality  $r = g_t r$  is asserted.

### 7.3 Domain and branch closure

Both sides are continuous endomorphisms of  $\mathbb{V}_r^{\text{ren}}$ . The equality therefore extends from the smooth orbit core to the completion and, by the transpose identity, to the full dual carrier. Internal and Lorentz descriptors have  $\dot{N}_\alpha = 0$ . Spatial descriptors give the fixed Lie derivative of the lapse. Every regular, finite-rank, smoothing, ordering, finite-profile, and branch- $B$  channel is included because the finite covariance identity transports the complete Hamiltonian.

**Theorem W6R.O17 (Orbit Hamiltonian-linear closure theorem).**

On the standing-bearing proper-redescription orbit carrier,

$$[\mathbb{H}[N], \mathbb{C}_\alpha] = i\hbar \mathbb{H}[\dot{N}_\alpha]$$

on the literal common source, completion, and transpose domains. The predecessor defect  $\mathfrak{V}_r$  is the local expression of the record-coordinate part of the diagonal action and cancels without collapsing distinct complete records.

**Proof.**

W6R.O12–W6R.O15 type all products and derivatives. Differentiate the exact induced covariance identity. The local two-term calculation shows that the derivative formerly left outside a fixed fibre is now an explicit part of the generator. Density of the smooth orbit core, continuity, and Hausdorff separation extend the identity; the full-domain transpose follows functorially.  $\square$

#### First dependency closed

W6R.S17 remains a correct boundary theorem for a single fixed record. W6R.O17 is a new theorem on a new carrier and discharges the predecessor's authorized option 3. W6R.O18 may therefore be computed, but it is not thereby predetermined.

## 8. Retraction of the compressed-cap obstruction

### 8.1 Restriction is not composition

Let  $I : \mathcal{E}_{\text{cap}} \rightarrow \Phi_{5b}^{\text{tr}}$  insert a finite-cylinder cap, let  $R$  read its selected coordinates, and put  $P = IR$ . The compressed Hamiltonian is

$$h_N = RH_N I.$$

For two lapses,

$$\begin{aligned} [h_N, h_M] - R[H_N, H_M]I &= -RH_N(1-P)H_M I \\ &\quad + RH_M(1-P)H_N I. \end{aligned}$$

Consequently

$$[h_N, h_M] = R[H_N, H_M]I$$

holds only when the complementary excursions vanish. In particular, it is not a consequence of cap compatibility or of the fact that both operators are individually defined on the cap.

The predecessor stage conformal cap does not satisfy this invariance condition. At the chart centre its selected covector is

$$\ell^{ij} = u \bar{h}^{ij} \rho,$$

while the lapse- $u$  inverse-DeWitt action has the spatial profile

$$\mathbb{K}_u \ell \propto u^2 \bar{h}_{ij}.$$

For smooth positive nonconstant  $u$ , this is not in the one-dimensional span of  $u \bar{h}_{ij}$ . Thus  $(1-P)H_u I \neq 0$ , and the two complementary terms in the displayed identity were omitted in predecessor stage.

The moment determinant

$$I_2 I_4 - I_3^2 > 0$$

therefore remains a correct calculation for the compressed pair  $(h_1, h_u)$ . It proves that this compression is not a representation of the source operator algebra. It does not prove a nonzero cubic symbol for  $[H_1, H_u]$ .

**Proposition 8.1 (Compression noninterchange).**

The predecessor stage one-mode calculation cannot be promoted to W6R.O18 because the selected conformal cap is not invariant under the principal Hamiltonian tensors. The compressed-cap obstruction does not apply to the complete orbit-carrier theorems W6R.O12–W6R.O17 are unaffected.

**Proof.**

The first identity is direct multiplication using  $P = IR$ . The explicit inverse-DeWitt action proves noninvariance. Since the missing terms contribute precisely when the second Hamiltonian differentiates the coefficient created by the first, the compressed Wronskian is not the restriction of the source commutator symbol.  $\square$

## 9. Source-wide functional principal symbol

### 9.1 Regulator-free principal tensor

On the smooth trace core the accepted successor has already removed the outer heat regulator. Its even second-order functional tensor is

$$\mathbb{K}_N(q) = M_N \mathbb{K}(q),$$

with the exact DeWitt, scalar, Yang–Mills, and Higgs blocks of W5.10. The multiplication operator  $M_N$  acts in the spatial variable and is independent of the configuration  $q$ . Every block of  $\mathbb{K}(q)$  is ultralocal in the configuration variables.

The half-heat factors of W5.7 are not present in this endpoint. The retained finite-separated and smoothing Hessian channels belong to the source grammar and to the regular value of the functional trace; they do not replace  $\mathbb{K}_N$  by a nonlocal principal tensor.

For a smooth cotangent field  $p$ ,

$$d\mathbb{K}_M(q)[\mathbb{K}_N(q)p] = MN d\mathbb{K}(q)[\mathbb{K}(q)p].$$

Interchanging  $M, N$  gives the identical expression. Hence the source-wide cubic commutator symbol vanishes:

$$\text{Sym}(d\mathbb{K}_M[\mathbb{K}_N p] - d\mathbb{K}_N[\mathbb{K}_M p]) = 0.$$

This is an identity before cap restriction and therefore remains true on every cylinder descendant after the complete source commutator has been formed.

### 9.2 Pure kinetic rows

Fix the baseline Gate-II ordering record

$$r_0 = (b, \eta, \mu, \nu^0, \mathfrak{D}^0, \mathcal{K}, \delta, \nabla^{\text{fld},0}),$$

where:

1. bosonic coefficients stand to the left and the additional first-order ordering vector is zero;
2. the field-dependent Gauss and Lorentz smearings in the raw spatial generator use the same left-coefficient convention;
3. the local affine configuration connection is the fixed flat chart connection;
4. branch  $B$  uses the number-preserving holomorphic CAR ordering; and
5.  $\nu^0$  retains the inherited classical potential and density terms and sets the additional local heat finite part to zero, exactly as declared by the accepted Gate-II successor.

Write  $\mathbb{T}_N^0$  for the renormalized kinetic action at this record. Source-wide functional differentiation gives only diagonal contact contributions to  $[\mathbb{T}_N^0, \mathbb{T}_M^0]$ . Every such coefficient is the same finite jet of  $\mathbb{K}$ , multiplied by  $NM$ . Antisymmetrization therefore kills the cubic, quadratic, first-order, and zero-order pure-kinetic rows:

$$[\mathbb{T}_N^0, \mathbb{T}_M^0] = 0.$$

Equivalently, the complete cap notation of Section 34.2 of the predecessor satisfies

$$\mathfrak{T}_{N,M} = 0, \quad \mathfrak{S}_{N,M}^{\text{kin}} = 0, \quad \mathfrak{R}_{N,M}^{\text{kin}} = 0, \quad \mathfrak{Z}_{N,M}^{\text{kin}} = 0.$$

The argument is sectorwise and survives direct sums and the symmetric  $(H, H^\dagger)$  cross-polarization. No positivity of the DeWitt block is used.

#### **Theorem W6R.O18a (Source-wide kinetic-symbol theorem).**

On the baseline orbit record, the complete source-wide pure-kinetic commutator vanishes. In particular, both the cubic and quadratic differential rows vanish identically on the common smooth trace core, its completion, and the full transpose carrier.

**Proof.**

Every functional coefficient jet is supported on the spatial diagonal. After the functional delta distributions are composed, both ordered products carry the identical factor  $N(x)M(x)$  multiplying the identical typed jet of  $\mathbb{K}(q; x)$ . Antisymmetrization gives zero at every differential order. W6R.O12–W6R.O15 then extend the identity from the smooth core to the completed source and transpose carrier.  $\square$

## 10. Composite finite-part resolution

### 10.1 The complete common-regulator product

Write the even baseline endpoint as

$$H_N^{\text{ev}} = \mathbb{T}_N^0 + V_N, \quad V_N(q) = \int_{\Sigma} N \mathcal{V}_b(q),$$

where  $\mathbb{T}_N^0$  is the regulator-free renormalized Hessian action and  $V_N$  is multiplication by the inherited classical potential. The disputed term is not evaluated by assigning a value to its endpoint. It is evaluated at the common regulated task before the finite part is taken.

For every smooth trace-core word  $F$ ,

$$d^2(V_M F) = V_M d^2 F + 2 \text{Sym}(dV_M \otimes dF) + F d^2 V_M.$$

The three summands have different Gate-II types:

| Hessian summand                 | Gate-II channel                                              | Fate in the commutator                                                               |
|---------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $V_M d^2 F$                     | the same<br>local/separated/smoothing<br>channels as $d^2 F$ | cancels literally against $V_M \mathbb{T}_N^0 F$                                     |
| $2 \text{Sym}(dV_M \otimes dF)$ | finite separated channel                                     | gives the first-order configuration<br>vector                                        |
| $F d^2 V_M$                     | finite-order spatial-diagonal local<br>channel               | its complete nonpositive heat<br>column is replaced by the declared<br>local profile |

The last classification is exact.  $V_M$  is one integral of a local finite-jet density, so its second variation is a finite-order differential kernel supported on the spatial diagonal. It has no finite-rank or smoothing component in Lemma 8.1. In the common-parameter scaled-diagonal expansion, every order-zero coefficient is therefore a class in the local quotient of W6R.4. W6R.11 replaces that entire local column, including its finite coefficient, by the prescribed profile. The off-diagonal part is  $O(\epsilon^K)$  for every  $K$ , and the positive-order local remainder tends to zero by W6R.10. Thus no inherited finite-rank, smoothing, or regular scalar term survives from  $F d^2 V_M$ .

The baseline profile has zero additional composite local coordinate. The inherited classical potential, density conversion, and left-coefficient terms are all lapse-linear multiplications; their two-lapse product is proportional to  $NM$  and is symmetric. The baseline first-order ordering vector is zero. Consequently the complete local scalar antisymmetrization is

$$\text{FP}_{\epsilon \downarrow 0} [\mathcal{L}_{N,\epsilon}^{\text{loc}}(V_M) - \mathcal{L}_{M,\epsilon}^{\text{loc}}(V_N)] = 0.$$

The separated term has the ordinary trace endpoint

$$2 dF[\mathbb{K}_N dV_M].$$

It is not sent through the local quotient. Hence

$$[\mathbb{T}_N^0, V_M]F - [\mathbb{T}_M^0, V_N]F = -2\hbar^2 dF[\mathbb{K}_N dV_M - \mathbb{K}_M dV_N].$$

#### Theorem W6R.O18b (Composite finite-part theorem).

For the zero-additional-local-finite, left-coefficient baseline record, the complete common-regulator finite part of the even kinetic–potential commutator has no scalar remainder. Its prescribed local, inherited local, separated, and smoothing rows are respectively

$$0, \quad 0, \quad -2\hbar^2 dF[\mathbb{K}_N dV_M - \mathbb{K}_M dV_N], \quad 0.$$

In the predecessor notation the composite contribution to  $\mathfrak{Z}_{N,M}$  is zero.

#### Proof.

The displayed product Hessian is an identity before renormalization. W6R.4, Lemma 8.1, W6R.10, and W6R.11 give the mutually exclusive channel classification above. Linearity of the common-regulator trace and of the counterterm insertion cancels the factorized first summand before the limit. Trace-norm continuity evaluates the separated summand. The third is wholly local and is assigned the zero additional baseline coordinate; the complete scaled-diagonal remainder vanishes. The inherited lapse-linear multiplications commute.  $\square$

## 10.2 Branch-A lower row

The accepted gravity and clock variation identities are used only after Theorem W6R.O18b has completed the quantum product. With

$$s^i = h^{ij}(N\partial_j M - M\partial_j N),$$

they give

$$-2(\mathbb{K}_N dV_M - \mathbb{K}_M dV_N) = X_{\text{raw},A}[s].$$

Since the source-wide pure-kinetic rows vanish by W6R.O18a, it follows that

$$[H_N^A, H_M^A] = i\hbar D_A[s].$$

**Corollary W6R.O18<sup>A</sup> (Baseline branch-A closure).**

W6R.O18 is proved for the metric-clock branch of the declared baseline orbit. The identity holds on the common smooth source and extends by W6R.O12–W6R.O15 to the completed source and full transpose carrier.  $\square$

## 11. Exact branch-B coefficient dictionary

### 11.1 Even and odd pieces

On the baseline degree-wise exterior-section carrier the candidate is

$$H_N^B = \mathbb{T}_N^0 + V_N + Q_N.$$

The even entries are exactly

$$\mathbb{T}_N^0 = -\hbar^2 \text{FP Contr}_{\mathbb{K}_B, M_N} d^2, \quad V_N = \int_{\Sigma} N \mathcal{V}_B, \quad \mathbb{K}_N = M_N \mathbb{K}_B,$$

with the DeWitt, clock, three electric Yang–Mills, and independent  $(H, H^\dagger)$  momentum blocks obtained by polarizing W4B.6 and printed in W5.10. Coefficients stand to the left, the configuration connection is the flat affine-chart connection, and there is no additional first-order ordering vector.

Let  $\mathcal{D}_i(e, \mathcal{A})$  be the W4B torsion-free spin, internal gauge, and half-density connection and let  $\alpha^i(e)$  be Clifford multiplication by the spatial coframe. The one-particle Dirac map is fixed by the actual W4B symmetric density:

$$\langle u, a_{D,N}(q)v \rangle = -\frac{i}{2} \sum_{\Psi, r} \int_{\Sigma} N [u^\dagger \alpha^i \mathcal{D}_i v - (\mathcal{D}_i u)^\dagger \alpha^i v].$$

Because  $\partial\Sigma = \emptyset$ , this is equivalently

$$a_{D,N} = \frac{1}{2} (M_N D_q + D_q M_N),$$

where  $D_q$  includes all species, generation, chirality, color, weak, hypercharge, spin, and half-density entries. The Yukawa map is

$$a_{Y,N} = M_N Y_H(q),$$

where  $Y_H$  is the off-diagonal chiral multiplication determined by the fixed matrices  $Y_u, Y_d, Y_e$ , the accepted Higgs doublet, and its Hermitian-conjugate blocks. Put

$$a_N = a_{D,N} + a_{Y,N}.$$

### 11.2 The forced $\hbar$ normalization

W5.8 fixes

$$\widehat{\Psi}[\eta] = \eta \wedge, \quad \widehat{\Psi}^\dagger[u] = \hbar \iota_u.$$

Normal ordering of the classical bilinear  $\Psi^\dagger a_N \Psi$  therefore gives

$$Q_N = d\Gamma_\hbar(a'_N) := \hbar d\Gamma(a'_N).$$

Equivalently, if the W5.10 notation  $d\Gamma(A'_N)$  is retained, its one-particle coefficient is

$$A_N = \hbar a_N.$$

This is the only reading compatible with the W5.8 CAR normalization. On every finite exterior degree,

$$[d\Gamma_\hbar(A), d\Gamma_\hbar(B)] = \hbar d\Gamma_\hbar([A, B]),$$

and no scalar central block occurs.

### 11.3 The rows produced by bosonic differentiation

The renormalized product rule for a target-valued coefficient gives, before any use of W4B.7,

$$\begin{aligned} [\mathbb{T}_N^0, Q_M]F &= -2\hbar^3 \text{Contr}_{\mathbb{K}_N} \text{Sym}(d\Gamma(da'_M) \otimes dF) \\ &\quad - \hbar^3 d\Gamma(\text{FP Contr}_{\mathbb{K}_N} d^2 a'_M) F. \end{aligned}$$

Antisymmetrizing defines the target-valued mixed first-order row

$$\boxed{\begin{aligned} \mathfrak{S}_{N,M}^{\text{mix}} F &= -2\hbar^3 \left( \text{Contr}_{\mathbb{K}_N} \text{Sym}(d\Gamma(da'_M) \otimes dF) \right. \\ &\quad \left. - \text{Contr}_{\mathbb{K}_M} \text{Sym}(d\Gamma(da'_N) \otimes dF) \right). \end{aligned}}$$

The accompanying target-valued zero-order row is

$$\mathfrak{Z}_{N,M}^{\text{mix}} = -\hbar^3 d\Gamma(\text{FP} [\text{Contr}_{\mathbb{K}_N} d^2 a'_M - \text{Contr}_{\mathbb{K}_M} d^2 a'_N]).$$

The CAR–CAR term is

$$[Q_N, Q_M] = \hbar d\Gamma_{\hbar}([a'_N, a'_M]),$$

which is zero order in the configuration derivative. It cannot cancel a nonzero  $\mathfrak{S}^{\text{mix}}$ . The bosonic potential multiplications commute with  $Q_N$  at fixed  $q$ .

These formulas are the missing W5.10 substitution. In particular, W4B.7 can identify the resulting vector only after  $\mathfrak{S}^{\text{mix}}$  and  $\mathfrak{Z}^{\text{mix}}$  have been evaluated.

## 12. Branch-B coefficient calculation

### 12.1 Half-density Dirac variation

The W4B variable is not an ordinary spinor. It is the half-density  $\Psi = h^{1/4}\psi$ . Let  $D_{1/2}(q)$  denote the spatial Dirac operator conjugated to the fixed half-density carrier. For a symmetric metric variation  $k$ , its exact first variation in a local orthonormal frame is

$$\delta_k D_{1/2} = -\frac{1}{2} k^{ij} c_i \mathcal{D}_j - \frac{1}{4} c(\text{div } k).$$

The trace-gradient term present for ordinary spinors is cancelled by differentiating the half-density conjugation. Therefore, for every smooth scalar  $f$ ,

$$\begin{aligned} \delta_{fk} D_{1/2} &= -\frac{1}{2} f k^{ij} c_i \mathcal{D}_j - \frac{1}{4} c(f \text{ div } k + k^\sharp df) \\ &= \frac{1}{2} (f \delta_k D_{1/2} + \delta_k D_{1/2} f). \end{aligned}$$

Thus

$$\boxed{\delta_{fk} D_{1/2} = \frac{1}{2} \{f, \delta_k D_{1/2}\}.$$

The identity includes the derived Levi–Civita spin-connection variation; no independent connection variable has been suppressed.

Gauge-connection variation is zero order and obeys

$$\delta_{f\mathcal{A}} D_{1/2} = f \delta_{\mathcal{A}} D_{1/2}.$$

The Yukawa map is ultralocal in the coframe, gauge, and Higgs variables, so its variation has the same scalar-linearity. Clock and unrelated even blocks do not vary  $a_N$ .

Now let  $v = \mathbb{K}_{BP}$  be any smooth even tangent generated from a cotangent test direction. Since  $\mathbb{K}_N p = Nv$  and  $a_{D,M} = \frac{1}{2} \{M, D_{1/2}\}$ , the double anticommutator identity gives

$$\begin{aligned} da_{D,M}[Nv] &= \frac{1}{4} \{M, \{N, \delta_v D_{1/2}\}\}, \\ da_{D,N}[Mv] &= \frac{1}{4} \{N, \{M, \delta_v D_{1/2}\}\}. \end{aligned}$$

Multiplication by  $M$  and  $N$  commutes, so the two expressions are identical. The gauge and Yukawa identities give the same conclusion:

$$\boxed{da_M[\mathbb{K}_N p] = da_N[\mathbb{K}_M p] \quad \text{for every smooth } p.}$$

After transposition and finite exterior lifting this proves

$$\mathfrak{S}_{N,M}^{\text{mix}} = 0$$

on every exterior degree. This cancellation would fail if the W4B half-density were replaced by an ordinary-spinor coefficient; the half-density convention is load bearing.

## 12.2 Target-valued contact row

The kernel of  $d^2 a_M$  is a finite-order spatial-diagonal, target-valued local kernel:  $D_{1/2}$  depends on the coframe and its first spatial jet, on the gauge connections without their momenta, and on the Higgs field through the ultralocal Yukawa multiplication. It has no finite-rank or smoothing component in the Gate-II structural Hessian decomposition.

Consequently the complete common-regulator contraction

$$\text{Contr}_{\mathbb{K}_N} d^2 a'_M$$

belongs to the branch-B target-valued local quotient. W6R.11 replaces its entire nonpositive column, including the finite contact coefficient, by the declared branch-B local profile. The baseline assigns zero to this additional target-valued contact coordinate. The scaled-diagonal off-diagonal and positive-order remainders vanish by W6R.10. Therefore

$$\boxed{\mathfrak{Z}_{N,M}^{\text{mix}} = 0.}$$

This is not inferred from the scalar composite calculation: it is the same complete local-profile theorem applied in the printed target-valued branch-B type.

## 12.3 Dirac–Dirac, curvature, and Yukawa rows

Write

$$a_N = \frac{1}{2}\{N, D_{1/2}\} + NY_H.$$

Expand the one-particle commutator before applying  $d\Gamma$ . The  $NMD_{1/2}^2$  terms cancel between the two ordered products. The symmetric Clifford contraction gives

$$s^i \mathcal{D}_i + \frac{1}{2} \mathcal{D}_i s^i, \quad s^i = h^{ij}(N \partial_j M - M \partial_j N),$$

the covariant Lie derivative on spatial spinor half-densities. The antisymmetric Clifford contraction is

$$\frac{1}{4}(\mathcal{D}_i s_j - \mathcal{D}_j s_i) \Sigma^{ij},$$

which is the induced Lorentz rotation. The connection part of  $\mathcal{D}_s$  supplies the internal and spin covariant terms. Hence the complete Dirac coefficient is the raw one-particle generator

$$t_{\text{raw},\Psi}[s] = t_{D,\Psi}[s] + t_{G,\Psi}[\iota_s \mathcal{A}] + t_{J,\Psi}[\iota_s \omega_{LC}].$$

Terms containing a derivative of  $Y_H$  occur with the common scalar factor  $NM$  and cancel. Terms containing a derivative of a lapse cancel pairwise because the chiral Yukawa multiplication anticommutes with spatial Clifford multiplication. The Higgs/Yukawa Ward terms are already included in the same internal representation action. Thus

$$\boxed{[a_N, a_M] = t_{\text{raw},\Psi}[s].}$$

This is the coefficient-level content behind the fermionic row of W4B.7; it has now been obtained from the actual W5.10 one-particle operator rather than imported from the classical bracket.

The exterior lift gives

$$\begin{aligned} [Q_N, Q_M] &= \hbar^2 d\Gamma([a'_N, a'_M]) \\ &= i\hbar C_{\text{raw},\Psi}[s]. \end{aligned}$$

Here the convention is

$$C_{\text{raw},\Psi}[s] = -i\hbar d\Gamma(t_{\text{raw},\Psi}[s]'),$$

so the last identity fixes rather than suppresses the factor of  $i$  and both powers of  $\hbar$ . No vacuum scalar occurs because  $d\Gamma$  is an exact representation on the algebraic finite-particle exterior carrier.

## 12.4 Complete branch-B row

Theorem W6R.O18a removes every pure-kinetic row. Theorem W6R.O18b gives the even gravity, clock, Yang–Mills, and Higgs raw spatial vector and zero scalar contact term. Sections 12.1–12.2 give

$$\mathfrak{S}_{N,M}^{\text{mix}} = 0, \quad \mathfrak{Z}_{N,M}^{\text{mix}} = 0.$$

Section 12.3 gives the fermionic raw spatial coefficient. Their sum is

$$[H_N^B, H_M^B] = i\hbar (D[s] + G[\iota_s \mathcal{A}] + J[\iota_s \omega_{LC}]).$$

The pure improved  $D[s]$  formula is only the joint Gauss/Lorentz-surface restriction.

### Theorem W6R.O18<sup>B</sup> (Baseline branch-B closure).

For the W5.8-normalized, W4B-symmetrized half-density Dirac/Yukawa coefficient and the zero-additional-local-finite, left-coefficient, flat-chart baseline record,

$$\mathfrak{T}_{N,M} = 0, \quad \mathfrak{S}_{N,M} = 0, \quad \mathfrak{R}_{N,M} = -i\hbar C_{\text{lin}}[\beta_{r_0}(N, M)], \quad \mathfrak{Z}_{N,M} = 0,$$

with  $\beta_{r_0}(N, M) = -s$  in the predecessor convention. W6R.O18 is proved on branch  $B$  on the common smooth source, its completion, and the full transpose domain.

### Proof.

Every row has been computed in Sections 9–12 before W4B.7 is used to name the resulting raw generator. W6R.O12–W6R.O15 extend the common-core identity.  $\square$

## 13. Baseline W6R.O18 closure and scheme guard

Combining W6R.O18<sup>A</sup> and W6R.O18<sup>B</sup> proves the complete Hamiltonian–Hamiltonian theorem for the declared baseline record  $r_0$ . This is the record used by the anomaly and physical-realization stages below.

For a different complete renormalization record  $r$ , define the exact operator difference

$$\Delta_r[N] = H_r[N] - H_{r_0}[N]$$

and the corresponding commutator defect

$$\begin{aligned} \mathfrak{A}_r[N, M] &= [H_{r_0}[N], \Delta_r[M]] + [\Delta_r[N], H_{r_0}[M]] \\ &\quad + [\Delta_r[N], \Delta_r[M]]. \end{aligned}$$

No identity  $\mathfrak{A}_r[N, M] = 0$  is inferred from the baseline calculation. A change of ordering, finite profile, smoothing, density, field-space connection, polarization, or one-particle normalization is a change of standing-bearing record and requires its own closure proof. Thus the proved W6R.O18 scope is exact: both branches close at  $r_0$ , while other complete records are neither identified with nor promoted by the baseline theorem.

## 2. Exact W6R.O19 target

### 2.1 Spatial and canonical coordinates

The spatial target is:

- a compact, connected, boundaryless hyperbolic three-manifold  $\Sigma$ ;
- the accepted orientation;
- one accepted ordinary spin structure  $\sigma$ ;
- the regular Sobolev branch with  $s > 7/2$ ;
- the oriented tetrad carrier with derived torsion-free spin connection;
- the accepted local scalar clock and its already registered failure boundary.

Only orientation-preserving transformations stabilizing  $\sigma$  act within this target. An orientation-reversing transformation or a mapping class that changes  $\sigma$  is a sector/target map, not an anomaly loop of this record.

### 2.2 Internal global form and representation

Let

$$G_6 = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\langle (\omega_3, -\mathbf{1}_2, e^{2\pi i/6}) \rangle}$$

with the hypercharge normalization

| left-handed field    | $Q$                        | $u^c$                            | $d^c$                            | $L$                        | $e^c$                      |
|----------------------|----------------------------|----------------------------------|----------------------------------|----------------------------|----------------------------|
| $SU(3) \times SU(2)$ | $(\mathbf{3}, \mathbf{2})$ | $(\bar{\mathbf{3}}, \mathbf{1})$ | $(\bar{\mathbf{3}}, \mathbf{1})$ | $(\mathbf{1}, \mathbf{2})$ | $(\mathbf{1}, \mathbf{1})$ |
| $Y$                  | $1/6$                      | $-2/3$                           | $1/3$                            | $-1/2$                     | $1$                        |
| $q = 6Y$             | $1$                        | $-4$                             | $2$                              | $-3$                       | $6$                        |

The displayed  $U(1)$  factor is represented by the honest character

$$\rho_q(z) = z^q, \quad q \in \mathbb{Z},$$

not by a fractional power. Write  $2j \in \mathbb{Z}/2$  for weak isospin parity and  $t \in \mathbb{Z}/3$  for color triality, with  $t(\mathbf{3}) = 1$ ,  $t(\bar{\mathbf{3}}) = 2$ , and  $t(\mathbf{1}) = 0$ . The five fields obey

$$q \equiv 2j \pmod{2}, \quad q \equiv t \pmod{3}.$$

Equivalently, the quotient generator acts as follows:

|                                       | $Q$         | $u^c$        | $d^c$       | $L$          | $e^c$       |
|---------------------------------------|-------------|--------------|-------------|--------------|-------------|
| $(t, 2j, q)$                          | $(1, 1, 1)$ | $(2, 0, -4)$ | $(2, 0, 2)$ | $(0, 1, -3)$ | $(0, 0, 6)$ |
| $\omega_3^t (-1)^{2j} e^{2\pi i q/6}$ | $1$         | $1$          | $1$         | $1$          | $1$         |

There are three copies of this set, no right-handed neutrino, one Higgs doublet, the accepted trivial spatial principal bundle, and no line insertion. Thus every displayed fermion representation is a bona fide representation of  $G_6$ . The Higgs is nonchiral and contributes no fermionic anomaly line.

## Closed anomaly base and exact obligations

Let  $\mathfrak{B}_{\text{WDW}}^{0,(4,5)}$  be the symmetric-monoidal category whose objects are closed four-dimensional spin- $G_6$  backgrounds with the fixed orientation, matter representation, and zero-theta record, and whose arrows are collared relative five-dimensional spin- $G_6$  bordisms. For every fixed manifold, bundle, spin structure, and collar type, its metrics, connections, tetrads, and Higgs/Yukawa backgrounds carry their ordinary smooth parameter spaces. Different topological types remain different components.

The admitted groupoid arrows are induced spin- $G_6$  bundle isomorphisms and orientation- and spin-preserving diffeomorphisms, including the retained large transformations. A loop  $\gamma$  has a closed spin- $G_6$  mapping-torus representative  $Y_\gamma$  and a classifying map

$$f_\gamma : Y_\gamma \longrightarrow BG_6.$$

This includes pure gauge, pure diffeomorphism, and combined loops. A map that reverses orientation or changes the fixed spin structure is a sector map, not a loop of this record.

There is a forgetful functor

$$\mathcal{F}_{45} : \mathfrak{B}_{\text{WDW}}^{0,(4,5)} \longrightarrow \text{Bord}_{4,5}^{\text{Spin-}G_6}$$

that retains exactly the metric Clifford module, spin- $G_6$  twisting, and product-collar data required by the determinant-line and Dai–Freed theorems. The closed anomaly theory is constructed on the full target and then pulled back; no field equation or preferred Euclidean time is assumed.

W6R.O19 requires one compatible record containing:

| Coordinate              | Required object                                                       |
|-------------------------|-----------------------------------------------------------------------|
| local and mixed anomaly | total six-form polynomial and its vanishing                           |
| determinant line        | product Fredholm determinant line with connection                     |
| Pfaffian row            | exact applicability and mod-two orientation check                     |
| bordism                 | $\Omega_5^{\text{Spin}}(BG_6)$ evaluated                              |
| large transformations   | mapping-torus holonomy for every admitted loop                        |
| quantum integrand       | the declared character of $\Omega_4^{\text{Spin}}(BG_6)$              |
| fermionic measure       | a coherent closed determinant-line section                            |
| regularization          | compatibility with the fixed heat/finite-profile record               |
| replacements            | explicit non-use of BRST, BFV, master constraint, and group averaging |

No row substitutes for another. In particular, local polynomial cancellation does not by itself trivialize global holonomy, and  $\Omega_5^{\text{Spin}}(BG_6) = 0$  does not select the declared four-dimensional theta character.

## 4. The total chiral Fredholm family

### 4.1 Historywise chiral operator

For a smooth four-dimensional spin history  $X$  represented in  $\mathfrak{B}_{\text{WDW}}^0$ , let

$$D_{X,R}^+ : \Gamma(S_X^+ \otimes E_R) \longrightarrow \Gamma(S_X^- \otimes E_R)$$

be the Euclidean chiral Dirac operator in one left-handed  $G_6$ -representation  $R$ . The coframe determines the metric and torsion-free spin connection; the accepted gauge fields determine the connection on  $E_R$ . Compactness and ellipticity make the Sobolev extension Fredholm. Its family is natural under every admitted WDW-2 history arrow.

The accepted Higgs/Yukawa coefficients are equivariant zero-order perturbations of the direct-sum chiral kinetic family. They may move zero modes and the numerical determinant section, but they do not change the Fredholm principal symbol, representation anomaly polynomial, or stable determinant-line anomaly class. Thus they are retained without being falsely counted as new chiral representations.

For a Fredholm operator  $D$ , define

$$\text{Det}(D) = \det \ker D \otimes (\det \text{coker } D)^*.$$

This definition remains a one-dimensional line when zero modes occur. It must not be replaced by the numerical determinant of an invertible operator.

For one generation set

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_Q \otimes \mathcal{L}_{u^c} \otimes \mathcal{L}_{d^c} \otimes \mathcal{L}_L \otimes \mathcal{L}_{e^c},$$

and define the three-generation total line

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{gen}}^{\otimes 3} \longrightarrow \mathfrak{B}_{\text{WDW}}^0.$$

It carries the product Quillen metric and product Bismut–Freed connection  $\nabla^{\text{tot}}$ .

#### Theorem W6R.O19a (Target determinant-line construction).

$\mathcal{L}_{\text{tot}}$  is a smooth Hermitian line over the exact WDW history/groupoid base, including zero-mode strata, and every admitted history arrow acts by a connection-preserving line isomorphism.

#### Proof.

The chiral operators are a smooth equivariant Fredholm family on compact spin histories. The determinant-line functor glues the local kernel/cokernel models [10, 11] across rank-changing strata and is natural under conjugating bundle isomorphisms. Tensoring the five species and three generations preserves smoothness, Hermitian structure, and equivariance. No invertibility or numerical determinant is assumed.  $\square$

## 5. Exact local anomaly polynomial

### 5.1 Coefficient register

All right-handed physical fermions are written as left-handed charge-conjugate fields. Normalize  $T(\mathbf{3}) = T(\mathbf{2}) = 1/2$  and the cubic  $SU(3)$  coefficient of  $\mathbf{3}$  as  $+1$ . For one generation:

$$\begin{aligned} \mathcal{A}_{SU(3)^3} &= 2 - 1 - 1 = 0, \\ \mathcal{A}_{SU(3)^2 Y} &= 2 \left( \frac{1}{6} \right) \left( \frac{1}{2} \right) + \left( -\frac{2}{3} \right) \left( \frac{1}{2} \right) + \left( \frac{1}{3} \right) \left( \frac{1}{2} \right) = 0, \\ \mathcal{A}_{SU(2)^2 Y} &= 3 \left( \frac{1}{6} \right) \left( \frac{1}{2} \right) + \left( -\frac{1}{2} \right) \left( \frac{1}{2} \right) = 0, \\ \mathcal{A}_{Y^3} &= 6 \left( \frac{1}{6} \right)^3 + 3 \left( -\frac{2}{3} \right)^3 + 3 \left( \frac{1}{3} \right)^3 + 2 \left( -\frac{1}{2} \right)^3 + 1 = 0, \\ \mathcal{A}_{\text{grav}^2 Y} &= 6 \left( \frac{1}{6} \right) + 3 \left( -\frac{2}{3} \right) + 3 \left( \frac{1}{3} \right) + 2 \left( -\frac{1}{2} \right) + 1 = 0. \end{aligned}$$

The perturbative  $SU(2)^3$  coefficient is zero because the weak representations are pseudoreal. Mixed coefficients with one traceless nonabelian generator vanish by tracelessness. There is no pure local gravitational anomaly for a four-dimensional Weyl theory: the relevant six-form has no tangent-only term. The only gravitational contribution in this target is the displayed mixed  $U(1)$ -gravitational coefficient, which vanishes.

Multiplication by three generations preserves every zero. The Higgs, gauge, clock, and gravitational bosonic sectors contribute no chiral fermion anomaly polynomial.

## 5.2 Curvature theorem

Let

$$I_6^{\text{tot}} = [\widehat{A}(TX) \text{ch}_{\rho_{\text{tot}}}(F)]_6$$

in the accepted normalization. The preceding coefficient register is the complete decomposition of  $I_6^{\text{tot}}$  into independent invariant polynomials for the declared representation.

**Theorem W6R.O19b (Local polynomial and curvature vanishing).**

On the exact baseline branch,

$$I_6^{\text{tot}} = 0, \quad F_{\nabla^{\text{tot}}} = 2\pi i \left[ \int_X I_6^{\text{tot}} \right]_{(2)} = 0.$$

Thus  $\nabla^{\text{tot}}$  is flat.

**Proof.**

Each independent gauge and mixed coefficient vanishes by the displayed rational identities. The four-dimensional tangent-only term is absent. The family-index curvature formula is linear in the representation and therefore in the five species and three generations. Substitution gives zero curvature.  $\square$

### Flat is not yet trivial

Theorem W6R.O19b removes local curvature only. A flat line can retain nontrivial holonomy around large gauge, diffeomorphism, or combined loops. No global conclusion is taken until the exact bordism row is computed.

## 6. Pfaffian applicability and weak mod-two row

The complete  $G_6$  Standard Model representation is complex, so its primary global object is  $\mathcal{L}_{\text{tot}}$ , not one universal real Pfaffian line. Restriction to the weak pseudoreal diagnostic exposes the possible real orientation sign responsible for the ordinary Witten anomaly.

One generation has

$$N_2 = 3 \text{ colored quark doublets} + 1 \text{ lepton doublet} = 4.$$

The fixed three-generation target has

$$N_2^{\text{tot}} = 12 \equiv 0 \pmod{2}.$$

If  $\nu(\gamma) \in \mathbb{Z}_2$  is the mod-two spectral flow of a weak loop, the restricted product Pfaffian changes by

$$(-1)^{12\nu(\gamma)} = 1.$$

No isospin-3/2 multiplet is present, and the target uses an ordinary spin structure rather than a spin- $SU(2)$  structure. The newer  $\int w_2 w_3$  anomaly therefore belongs to a different tangential and representation target.

**Theorem W6R.O19c (Pfaffian row).**

The full target requires the complex determinant line of Theorem W6R.O19a. On the only target-relevant real weak restriction, its Pfaffian orientation holonomy is trivial. No independent Pfaffian obstruction remains.

**Proof.**

The full  $G_6$  hypercharges and color representations make the chiral sum complex. The restricted weak mod-two sign is the parity of the doublet spectral flow. Its exponent is multiplied by twelve and is therefore even for every loop. The newer spin- $SU(2)$  invariant has neither the required tangential structure nor representation.  $\square$

## 7. Compatibility with the baseline renormalization record

The baseline renormalized Hamiltonian uses its common covariant spatial heat family and zero-additional-local finite profile on the accepted orbit. The four-dimensional determinant line uses the canonical Quillen/heat regularization of the chiral Fredholm family. These are separately typed regulators. Compatibility requires their groupoid actions and local anomaly class to agree; it does not require their kernels to be identical. The determinant variation is precisely the local anomaly polynomial already shown zero. The empty spatial boundary supports no edge central extension or physical boundary inflow sector. A temporal cut is nevertheless an analytic three-dimensional boundary of each cut-open history. the zero-theta baseline treats it by the direct Cauchy-data/Clifford-bimodule construction of Section 11. The product collar and finite-dimensional harmonic correction implement composition; they are not new edge fields, inflow degrees of freedom, or a changed physical boundary package.

**Theorem W6R.O19d (Regulator and local-trivialization compatibility).**

The product determinant connection and the baseline renormalized operator record are simultaneously equivariant under the connected proper groupoid. The determinant construction introduces no new local counterterm, coupling, central block, or finite-profile coordinate.

**Proof.**

lower-order Gate-III stage proves exact local constraint closure with its covariant common spatial regulator. The Quillen construction is natural under the same spin- $G_6$  arrows, and the only infinitesimal obstruction to its equivariance is the curvature/local polynomial of Theorem W6R.O19b, which is zero. The determinant line is attached downstream and changes no operator coefficient or finite profile. The empty spatial boundary removes edge central and inflow rows, while the non-use of inflow prevents a target-changing bulk term.  $\square$

**8. Exact  $\mathbb{Z}_6$  spin-bordism evaluation****8.1 The target group**

The faithful global group is exactly the  $\Gamma_6$  case of the registered Standard Model family:

$$G_6 = \frac{SU(3) \times SU(2) \times U(1)}{\Gamma_6}.$$

It admits the fibration

$$\mathbb{Z}_3 \longrightarrow U(2) \times SU(3) \longrightarrow G_6$$

and hence the Puppe-induced fibration

$$B(U(2) \times SU(3)) \longrightarrow BG_6 \longrightarrow K(\mathbb{Z}_3, 2).$$

This is the target bridge. A result for the unquotiented product,  $\Gamma_2$ , or  $\Gamma_3$  is not silently substituted.

**8.2 Total-degree-five page**

For total degree at most five, the fiber bordism groups used by the published computation are

$$\Omega_q^{\text{Spin}}(B(U(2) \times SU(3))) = \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}_2, & q = 1, \\ \mathbb{Z} \oplus \mathbb{Z}_2, & q = 2, \\ 0, & q = 3, \\ \mathbb{Z}^4, & q = 4, \\ 0, & q = 5. \end{cases}$$

The required homology of  $K(\mathbb{Z}_3, 2)$  is

|                        |                |   |                |   |                |   |
|------------------------|----------------|---|----------------|---|----------------|---|
| $p$                    | 0              | 1 | 2              | 3 | 4              | 5 |
| $H_p(-; \mathbb{Z})$   | $\mathbb{Z}$   | 0 | $\mathbb{Z}_3$ | 0 | $\mathbb{Z}_3$ | 0 |
| $H_p(-; \mathbb{Z}_2)$ | $\mathbb{Z}_2$ | 0 | 0              | 0 | 0              | 0 |

The generalized spectral-sequence entries on  $p + q = 5$  are therefore:

| $(p, q)$ | Coefficient group                | Entry                                                |
|----------|----------------------------------|------------------------------------------------------|
| (0, 5)   | 0                                | 0                                                    |
| (1, 4)   | $\mathbb{Z}^4$                   | $H_1(K; \mathbb{Z}^4) = 0$                           |
| (2, 3)   | 0                                | 0                                                    |
| (3, 2)   | $\mathbb{Z} \oplus \mathbb{Z}_2$ | $H_3(K; \mathbb{Z}) \oplus H_3(K; \mathbb{Z}_2) = 0$ |
| (4, 1)   | $\mathbb{Z}_2$                   | $H_4(K; \mathbb{Z}_2) = 0$                           |
| (5, 0)   | $\mathbb{Z}$                     | $H_5(K; \mathbb{Z}) = 0$                             |

Every total-degree-five entry is already zero on the second page. No unknown differential or extension can create a later nonzero group.

**Theorem W6R.O19e (Exact target bordism theorem).**

$$\Omega_5^{\text{Spin}}(BG_6) = 0.$$

**Proof.**

The displayed fibration, fiber groups, and base homology are the exact  $\Gamma_6$  computation. The six possible total-degree-five entries all vanish. The spectral sequence therefore has no associated-graded piece in degree five, and the bordism group is zero. This reproduces Eq. (4.45) of the primary calculation [12] at the interface needed for W6R.O19.  $\square$

**What the zero group does and does not say**

The zero group is stronger than a calculation on the accepted trivial spatial bundle: it covers every closed spin five-dimensional mapping torus carrying a  $G_6$ -bundle. It does not identify different global forms, spin structures, line sectors, or four-dimensional physical histories. It classifies the anomaly phase on the fixed symmetry and tangential target.

**8.4 Exact four-dimensional component group**

The same registered target calculation resolves the degree-four extension and gives

$$\Omega_4^{\text{Spin}}(BG_6) \cong \mathbb{Z}^4.$$

This is Eq. (4.50) of the primary source [12]. It is not an anomaly obstruction: anomalies are classified by the degree-five phase after the local polynomial vanishes. It is instead the component group on which the anomaly-free theory still requires a coherent quantum integrand.

Choose an ordered integral basis  $\beta = (\beta_1, \dots, \beta_4)$  only to display the character space. Every class has a unique coordinate vector  $\mathbf{n} = (n_1, \dots, n_4) \in \mathbb{Z}^4$ , and every unitary character is

$$\chi_\theta \left( \sum_{a=1}^4 n_a \beta_a \right) = \exp \left( i \sum_{a=1}^4 n_a \theta_a \right), \quad \theta \in (\mathbb{R}/2\pi\mathbb{Z})^4.$$

The character space is therefore

$$\text{Hom}(\Omega_4^{\text{Spin}}(BG_6), U(1)) \cong U(1)^4.$$

Changing the basis reparametrizes the four angles. The trivial character

$$\chi_0([X, f]) = 1$$

is basis-independent.

**Definition 8.3 (the zero-theta baseline retained zero-theta record).**

The baseline quantum-integrand datum is the declared character  $\chi_0$ . It is an additional standing-bearing phase record on non-null-bordant four-dimensional sectors. It is not inferred from W6R.O19b, W6R.O19e, or any kernel necessity claim.

**Anomaly freedom does not choose theta**

The vanishing of  $\Omega_5^{\text{Spin}}(BG_6)$  removes every closed five-dimensional anomalous phase. It neither collapses  $\Omega_4^{\text{Spin}}(BG_6)$  nor selects a character on it. the zero-theta baseline retains the predecessor stage declaration of  $\chi_0$ ; another  $\chi_\theta$  is a distinct, anomaly-free quantum-integrand record.

**9. Eta invariant and global line holonomy**

For a loop  $\gamma$  in the full admitted background/history groupoid, let  $(Y_\gamma, f_\gamma)$  be its five-dimensional spin  $G_6$  mapping torus. In the convention fixed by the determinant-line theorem [10, 13], write

$$\text{Hol}_\gamma(\nabla^{\text{tot}}) = \exp(2\pi i \bar{\eta}_{\rho_{\text{tot}}}(Y_\gamma, f_\gamma)).$$

Only the normalized phase matters below; changing the sign convention for  $\bar{\eta}$  changes both sides of the imported holonomy dictionary and not the conclusion.

Because  $I_6^{\text{tot}} = 0$ , the exponentiated eta invariant depends only on

$$[Y_\gamma, f_\gamma] \in \Omega_5^{\text{Spin}}(BG_6).$$

Theorem W6R.O19e makes this class zero for every loop. The normalized phase of the null class is one.

**Theorem W6R.O19f (Global holonomy theorem).**

For every connected, large-gauge, large-diffeomorphism, or combined admitted loop  $\gamma$ ,

$$\text{Hol}_\gamma(\nabla^{\text{tot}}) = 1.$$

**Proof.**

Local cancellation makes the eta phase a spin- $G_6$  bordism homomorphism. Its domain group is zero by Theorem W6R.O19e. The unique homomorphism from the zero group sends its sole class to the identity phase.  $\square$

## 10. Large-diffeomorphism row

Let

$$\text{MCG}^+(\Sigma, \sigma) = \pi_0(\text{Diff}^+(\Sigma)_\sigma)$$

be the orientation- and chosen-spin-structure-preserving mapping-class stabilizer. WDW-2 retains this group outside the connected proper groupoid. the zero-theta baseline does not quotient or erase its action.

For  $[\varphi] \in \text{MCG}^+(\Sigma, \sigma)$ , the background history groupoid of Section 2.3 supplies the corresponding cobordism and composition loops. The five-dimensional anomaly theory transgresses to the possible projective multiplier of the canonical fermionic implementation. That multiplier is the restriction of the determinant-line holonomy computed in Theorem W6R.O19f.

**Theorem W6R.O19g (Large-diffeomorphism anomaly theorem).**

Every admitted mapping class has trivial anomalous phase. Equivalently, the determinant contribution to the large-action projective cocycle is a coboundary and is made identically one by the coherent  $\chi_0$ -trivialization constructed in Section 11.

This theorem does **not** assert that the honest mapping-class action on states or observables is the identity. It proves only that the chiral fermion measure supplies no projective/sign obstruction to representing that retained action.

**Proof.**

Each admitted mapping class defines a loop  $\gamma_\varphi$  in the history/groupoid base. Theorem W6R.O19f gives  $\text{Hol}_{\gamma_\varphi} = 1$ . A mapping class that reverses orientation or changes  $\sigma$  is not a loop of this target and is correctly excluded rather than called anomaly-free.  $\square$

## 11. Coherent closed quantum integrand

### 11.1 Construct first on the full symmetry bordism target

Let

$$\mathcal{C}_{4,5}^{G_6, \text{full}} = \text{Bord}_{4,5}^{\text{Spin-}G_6}$$

be the full bordism category of closed spin- $G_6$  four-backgrounds and collared relative five-bordisms. Its monoidal product is disjoint union, its dual is orientation reversal, and its composition is collar gluing.

The phase part of the Quillen–Dai–Freed determinant theory [10, 11] is a symmetric-monoidal functor

$$\mathcal{A}_{\rho_{\text{tot}}} : \mathcal{C}_{4,5}^{G_6, \text{full}} \longrightarrow \text{Pic}_{\mathbb{C}}.$$

It assigns the total determinant line  $\mathcal{L}_X$  to  $X$ , and a unitary line arrow

$$T_W : \mathcal{L}_{X_0} \longrightarrow \mathcal{L}_{X_1}$$

to  $W : X_0 \rightarrow X_1$ . Its structural maps give

$$\mathcal{L}_{X \sqcup X'} \cong \mathcal{L}_X \otimes \mathcal{L}_{X'}, \quad \mathcal{L}_{-X} \cong \mathcal{L}_X^\vee, \quad T_{W_2 \circ W_1} = T_{W_2} T_{W_1}.$$

The monoidal unit is the empty background with line  $\mathbb{C}$ .

The WDW closed-background category has the forgetful functor

$$\mathcal{F}_{45} : \mathcal{C}_{4,5}^{G_6, \text{WDW}} \longrightarrow \mathcal{C}_{4,5}^{G_6, \text{full}}$$

which retains the spin- $G_6$  principal-symbol data. Higgs/Yukawa coefficients and the extra WDW history labels are zero-order or downstream data and do not change the anomaly functor. The accepted predecessor stage construction builds the trivialization on the full category and uses  $\mathcal{F}_{45}^* \mathcal{A}_{\rho_{\text{tot}}}$  on the WDW carrier. It therefore never assumes that an arbitrary full bordism satisfies extra WDW history conditions.

On every fixed smooth background component the pullback is the Fredholm determinant-line family of W6R.O19a. Different manifold and bundle types remain different components; the bordism arrows compare them without inventing a cross-topology smooth path.

## 11.2 Evaluation-normalized $\Omega_4$ basis model

Fix smooth representatives

$$B_a = (X_a, f_a), \quad [B_a] = \beta_a, \quad 1 \leq a \leq 4,$$

of the ordered integral basis in Section 8.4. Orientation reversal is the bordism dual. The product cylinder on  $B_a$ , regarded with both boundary components incoming or both outgoing, supplies evaluation and coevaluation bordisms

$$E_a : B_a \sqcup (-B_a) \longrightarrow \emptyset, \quad \overline{E}_a : \emptyset \longrightarrow B_a \sqcup (-B_a).$$

These are arrows, not object isomorphisms.

Declare one unit vector  $\varepsilon_a \in \mathcal{L}_{B_a}$ . Define the dual vector  $\varepsilon_{-a} \in \mathcal{L}_{-B_a}$  by the exact normalization

$$T_{E_a}(\varepsilon_a \otimes \varepsilon_{-a}) = 1.$$

The evaluation/coevaluation triangle identities and W6R.O19f then give

$$T_{\overline{E}_a}(1) = \varepsilon_a \otimes \varepsilon_{-a}.$$

Thus a cancellation pair has a specified determinant transport; it is not deleted by notation.

For  $\mathbf{n} = (n_1, \dots, n_4) \in \mathbb{Z}^4$ , let

$$R_{\mathbf{n}} = \bigsqcup_{a=1}^4 \left( B_a^{\sqcup n_a^+} \sqcup (-B_a)^{\sqcup n_a^-} \right), \quad n_a^{\pm} = \max(\pm n_a, 0).$$

Let  $\varepsilon_{\mathbf{n}} \in \mathcal{L}_{R_{\mathbf{n}}}$  be the ordered tensor of the declared positive and evaluation-dual negative vectors, and set  $\varepsilon_{\mathbf{0}} = 1$ .

For  $\mathbf{n}, \mathbf{m}$ , define

$$\kappa_a(\mathbf{n}, \mathbf{m}) = \min(n_a^+, m_a^-) + \min(n_a^-, m_a^+).$$

Starting from  $R_{\mathbf{n}+\mathbf{m}}$ , insert exactly  $\kappa_a(\mathbf{n}, \mathbf{m})$  coevaluation pairs, then apply the canonical symmetry maps to order the factors as  $R_{\mathbf{n}} \sqcup R_{\mathbf{m}}$ . This defines a comparison bordism

$$C_{\mathbf{n}, \mathbf{m}} : R_{\mathbf{n}+\mathbf{m}} \longrightarrow R_{\mathbf{n}} \sqcup R_{\mathbf{m}}.$$

Its reverse evaluates the same pairs. By the chosen normalization,

$$\boxed{T_{C_{\mathbf{n}, \mathbf{m}}} \varepsilon_{\mathbf{n}+\mathbf{m}} = \varepsilon_{\mathbf{n}} \otimes \varepsilon_{\mathbf{m}}.}$$

The comparison arrows obey:

$$\begin{aligned} C_{\mathbf{n}, \mathbf{0}} &= C_{\mathbf{0}, \mathbf{n}} = \text{id}, \\ \sigma C_{\mathbf{n}, \mathbf{m}} &= C_{\mathbf{m}, \mathbf{n}}, \end{aligned}$$

and

$$(C_{\mathbf{n}, \mathbf{m}} \sqcup \text{id}) C_{\mathbf{n}+\mathbf{m}, \ell} = (\text{id} \sqcup C_{\mathbf{m}, \ell}) C_{\mathbf{n}, \mathbf{m}+\ell}.$$

At the bordism level these are the symmetric-monoidal coherence and the cylinder snake identities. At the determinant level they are ordinary associativity and commutativity of tensor products and one-dimensional evaluation pairings. If two collar presentations are used, closing their difference gives a five-bordism of phase one by W6R.O19f. Orientation reversal exchanges evaluation with coevaluation and gives the dual coherence law.

### Exact categorical boundary

No statement of the form  $R_{\mathbf{n}} \sqcup R_{\mathbf{m}} \cong R_{\mathbf{n+m}}$  is used when cancellation occurs. The objects are connected only by the displayed bordism  $C_{\mathbf{n,m}}$ , and its determinant arrow is present in every monoidal comparison.

## 11.3 Relative-bordism section on the full category

Let  $X$  be any closed spin- $G_6$  four-background with  $[X] = \mathbf{n}$ . Equality in  $\Omega_4^{\text{Spin}}(BG_6)$  gives a full symmetry bordism

$$W_X : R_{\mathbf{n}} \longrightarrow X.$$

Define

$$\tau_0(X) = T_{W_X} \varepsilon_{\mathbf{n}} \in \mathcal{L}_X.$$

If  $W'_X$  is another such bordism, then  $\overline{W'_X} \circ W_X$  is closed and W6R.O19f assigns it phase one. Hence

$$T_{W_X} \varepsilon_{\mathbf{n}} = T_{W'_X} \varepsilon_{\mathbf{n}}.$$

The construction is therefore independent of the selected relative bordism.

For  $X$  of class  $\mathbf{n}$  and  $X'$  of class  $\mathbf{m}$ , use the actual composite

$$R_{\mathbf{n+m}} \xrightarrow{C_{\mathbf{n,m}}} R_{\mathbf{n}} \sqcup R_{\mathbf{m}} \xrightarrow{W_X \sqcup W_{X'}} X \sqcup X'.$$

The boxed transport identity in Section 11.2 gives

$$\tau_0(X \sqcup X') = \tau_0(X) \otimes \tau_0(X').$$

The empty and dual formulas follow from the unit, evaluation, and coevaluation bordisms:

$$\tau_0(\emptyset) = 1, \quad \tau_0(-X) = \tau_0(X)^\vee.$$

The comparison-arrow coherence proves the unit, symmetry, associativity, and dual diagrams rather than assuming them.

On a smooth path inside one fixed component,  $W_X$  can be changed by a cylinder. The section is consequently smooth and parallel there. For an admitted groupoid arrow  $g : X \rightarrow X'$ ,

$$T_g \tau_0(X) = \tau_0(X').$$

Finally, pullback along  $\mathcal{F}_{45}$  gives the same section on the WDW background groupoid. No basis representative or relative bordism is required to be a WDW-admitted history.

## Original-contract Gate-III compatibility theorem

The anomaly stage is evaluated against the declared Gate-III contract. That contract requires one compatible record containing regulator and renormalization anomalies; local gauge, gravitational, and mixed anomalies; determinant and applicable Pfaffian data; global gauge and large-diffeomorphism phases; bordism obstructions; and the equivalence of any replacement formalism that is actually used. The last qualifier is essential: the construction closes the original constraint family directly and invokes no BRST, BFV, master-constraint, or group-averaging replacement.

The fixed baseline uses the same (G\_6) global form, spin structure, orientation, branch-(B) matter representation, regulator data, and local finite profile throughout. It also declares the trivial character

$$\chi_0 : \Omega_4^{\text{Spin}}(BG_6) \longrightarrow U(1), \quad \chi_0 = 1,$$

as standing-bearing quantum-integrand data. This is a selected baseline coordinate, not a consequence of anomaly cancellation and not a claim that other theta characters are inconsistent.

**Theorem W6R.19 (compatible total-anomaly record at the declared zero-theta baseline).** The results W6R.O16–W6R.O18 and W6R.O19a–g, together with the evaluation-normalized closed determinant section for  $\chi_0$ , define one compatible anomaly record. Explicitly,

$$I_6^{\text{tot}} = 0, \quad F_{\nabla^{\text{tot}}} = 0, \quad \Omega_5^{\text{Spin}}(BG_6) = 0, \quad \text{Hol}_\gamma(\nabla^{\text{tot}}) = 1$$

for every admitted loop  $\gamma$ , the weak Pfaffian orientation row is trivial, the regularization record is the same record used by the renormalized operators, and the closed determinant section is coherent under disjoint union, orientation reversal, and closed bordism gluing. Therefore the renormalized linear, Hamiltonian-linear, and Hamiltonian-Hamiltonian constraint closures coexist with the complete original-contract anomaly certificate.

**Proof.** W6R.O16 proves closure of the complete linear constraint family on the common successor domain. W6R.O17 proves the Hamiltonian-linear commutators, including the proper-redescription transport terms. The source-wide functional principal-symbol calculation, the common-regulator composite finite part, and the branch-specific lower-order computations prove W6R.O18 on both branches without importing the classical closure theorem as a quantum identity.

For the anomaly coordinate, W6R.O19a constructs the total determinant line including zero-mode strata. W6R.O19b computes every independent local and mixed coefficient and makes its Bismut–Freed curvature zero. W6R.O19c discharges the only target-relevant real Pfaffian diagnostic. W6R.O19d places this flat determinant line on the same groupoid and local finite-profile record as the renormalized constraints. W6R.O19e computes the exact fifth spin bordism group of the faithful Standard Model quotient to be zero; after local cancellation, the Dai–Freed phase is therefore the unique homomorphism from the zero group. W6R.O19f gives unit holonomy, and W6R.O19g restricts that conclusion to every admitted large-diffeomorphism loop without erasing the honest mapping-class action.

The evaluation/coevaluation construction for the declared character  $\chi_0$  gives a closed symmetric-monoidal determinant section. Its data are defined on the same  $(G_6)$ , spin, orientation, regulator, matter, and operator record as the preceding rows, so the rows coexist rather than merely vanishing separately. No replacement formalism is invoked; consequently no replacement-state, form, or observable equivalence is assumed. This proves the original Gate-III compatibility claim.  $\square$

#### Exact Gate-III boundary

The theorem does not claim a natural transformation from the trivial  $(3,4)$  theory to a direct anomaly twist, algebraic dualizability of all filling modules, canonical one-dimensional history intertwiner spaces, or a smooth equivariant polarization-gerbe trivialization. Those are stronger temporal extensions and remain open. They are not premises of W6R.16–19 and do not enter the physical-state construction below.

## Part V

# Physical realization and certified endpoint

## Terminal-map criterion and the noncircular W6R.20 burden

For an accepted record  $r$  and branch  $b$ , let

$$\mathcal{J}_{r,b} = \text{span}_{\mathbb{C}}\{(\widetilde{C}_i^{\text{ren}})'e : i \in \mathcal{L}_{\text{full}}, e \in \mathcal{E}_{r,b}^{\text{ren}}\}, \quad \mathcal{Q}_{r,b} = \mathcal{E}_{r,b}^{\text{ren}} / \overline{\mathcal{J}_{r,b}}^{\tau_E}.$$

This is the topological coinvariant of the complete transposed constraint images. Its nontriviality is not implied merely by constraint closure or by independent occupancy of a persistence target.

**Definition 0.1** (Pointed terminal-map witness). A pointed terminal-map witness for  $(r,b)$  is a quadruple

$$(R_{r,b}, \pi_{0,r,b}, e_{r,b}^*, v_{r,b}^*)$$

such that  $R_{r,b}$  is Hausdorff locally convex,  $\pi_{0,r,b} : (\mathcal{E}_{r,b}^{\text{ren}}, \tau_E) \rightarrow R_{r,b}$  is continuous linear,

$$\pi_{0,r,b} \circ (\widetilde{C}_i^{\text{ren}})' = 0 \quad (i \in \mathcal{L}_{\text{full}}),$$

and  $v_{r,b}^* = \pi_{0,r,b}(e_{r,b}^*) \neq 0$ . When  $R_{r,b}$  linearizes the R1 persistence interior, the same-target clause requires  $v_{r,b}^*$  to represent the fixed non-null class  $\delta_{r,b}$ .

**Lemma 0.2** (Closed-image annihilation). *Every pointed terminal-map witness satisfies*

$$\overline{\mathcal{J}_{r,b}}^{\tau_E} \subseteq \ker \pi_{0,r,b}.$$

*Proof.* The defining annihilation identities kill every generator and therefore its finite linear span. The kernel of a continuous linear map into a Hausdorff space is closed, so it contains the closure of that span.  $\square$

**Theorem 0.3** (Terminal-map equivalence). *For every accepted  $(r, b)$ , the following are equivalent:*

1.  $\mathcal{Q}_{r,b} \neq 0$ ;
2. *there is a nonzero  $\Psi_{r,b} \in (\mathcal{E}_{r,b}^{\text{ren}}, \tau_E)'$  satisfying  $\Psi_{r,b} \circ (\widetilde{C}_i^{\text{ren}})' = 0$  for every  $i \in \mathcal{L}_{\text{full}}$ ;*
3. *a pointed terminal-map witness exists.*

*Proof.* (1  $\Rightarrow$  2). The quotient is nonzero, Hausdorff, and locally convex. Choose  $q \neq 0$ . Continuous Hahn–Banach separation [7, Ch. 18] supplies  $\lambda \in (\mathcal{Q}_{r,b})'$  with  $\lambda(q) = 1$ . If  $p_{r,b} : \mathcal{E}_{r,b}^{\text{ren}} \rightarrow \mathcal{Q}_{r,b}$  is the quotient map, then  $\Psi_{r,b} = \lambda \circ p_{r,b}$  is continuous, nonzero, and annihilates every complete constraint image.

(2  $\Rightarrow$  3). Take  $R_{r,b} = \mathbb{C}$ ,  $\pi_{0,r,b} = \Psi_{r,b}$ , and choose  $e_{r,b}^*$  outside the kernel of  $\Psi_{r,b}$ .

(3  $\Rightarrow$  1). Lemma 0.2 and  $\pi_{0,r,b}(e_{r,b}^*) \neq 0$  imply that  $e_{r,b}^* \notin \overline{\mathcal{T}_{r,b}}^{\tau_E}$ . Its quotient class is therefore nonzero.  $\square$

**Corollary 0.4** (Pointed rather than global quotient identification). *W6R.20 does not require an isomorphism between the entire R1 persistence quotient and  $\mathcal{Q}_{r,b}$ . One continuous same-target witness carrying the declared non-null class to a nonzero coinvariant is sufficient. Conversely, calling a desired state a terminal map does not lower the burden: complete annihilation and one nonzero value already imply  $\mathcal{Q}_{r,b} \neq 0$ .*

## Independence of target occupancy and module coinvariants

The logical independence is witnessed by a simple exact system. Give  $E = \mathbb{C}[x]$  its finest locally convex topology, equivalently the locally convex direct limit of its finite-dimensional degree truncations. The derivative  $D = d/dx : E \rightarrow E$  is continuous and surjective, and its one-generator commutator algebra closes:  $[D, D] = 0$ . Nevertheless,

$$\overline{D(E)} = E, \quad E/\overline{D(E)} = 0, \quad \{\Psi \in E' : \Psi \circ D = 0\} = \{0\}.$$

An independently occupied non-null target class can coexist with this zero coinvariant because no attachment map has been supplied. The example is not the WDW operator family; it proves only that target occupancy and closed constraints do not entail the missing morphism.

The construction below meets the exact criterion rather than inferring it. Its functional  $\Psi_{A,r}(u) = u(F_{A,r}^{(0)})$  is defined on the declared strong-dual carrier, is continuous by bounded-set evaluation, annihilates every complete transposed constraint because  $F_{A,r}^{(0)}$  is a genuine joint solution, and is nonzero by the fixed normalization.

## Direct Gate-IV completion

### Fixed-profile scale cylinder and normalized joint solution

The finite-anticipation, exact coefficient, scale-cylinder, phase-jet, and constant-normalization inputs used at this boundary are retained as the proof-bearing technical supplements T1–T5 in the ARC package [14, 15, 16, 17, 18]. Their role is limited: they authenticate the fixed-profile source/operator packet and the precise target boundary. They do not themselves supply the physical form, observable algebra, complete matching, or continuation interfaces proved below.

For a fixed accepted record  $r_0$ , the scale-cylinder theorem holds all non-scale coordinates fixed and varies only  $\mu > 0$  under a scale-independent finite normalization profile. Gate-II scale transport then has vanishing logarithmic coefficients, so the complete registered Branch-A operator family, graph topology, and joint kernel are literally common across the cylinder:

$$\mathcal{Z}_{A,r(\mu)} = \mathcal{Z}_{A,r(\nu)} =: \mathcal{Z}_A \quad (\mu, \nu > 0).$$

The source-bound packet theorem supplies a common normalized solution

$$F_{A,r}^{(0)} \in \mathcal{Z}_A \setminus \{0\}, \quad F_{A,r}^{(0)}(q^*) = 1,$$

at every retained address. This is operator/source identity, not yet physical equivalence of complete records.

**Proposition (source-bound endpoint packet).** At every retained fixed-profile scale address, the endpoint-attached record has the unchanged finite-cap descriptor, exact operator syntax, finite-extension authority, complete constraint-label family, and normalized joint solution  $F_{A,r}^{(0)}$ . The identity source lift preserves  $F_{A,r}^{(0)}$ , and every complete constraint satisfies

$$C_{A,r,i} F_{A,r}^{(0)} = 0.$$

Changing the endpoint coordinate at this stage is not a complete-record equivalence and does not identify distinct scale addresses.

*Proof.* The sparse-jet cancellation and deterministic finite-anticipation results fix the exact cap restriction and coefficient oracle [14, 15]. The scale-cylinder result then proves literal equality of the source, complete operator family, graph topology, normalized joint solution, and value-one anchor at every retained address [16]. The endpoint attachment changes none of those source-side fields, so its source lift is the identity and retains  $F_{A,r}^{(0)}$  in the complete joint kernel. The regular-history phase-jet theorem and constant-normalization obstruction delimit, respectively, the strongest inherited history datum and the quotient that may not be used [17, 18]. Thus the packet is source-bound and complete at the stated fixed-profile scope, while record equivalence remains unavailable until the state, form, algebra, and matching coordinates are populated.  $\square$

## W6R.20: pointed physical-state attachment

Let  $\mathbf{e}_{\delta_{A,r}}$  denote the basis vector labelled by the independently occupied non-null R1 class and let  $R_{A,r}^{\text{pt}} = \mathbb{C}\mathbf{e}_{\delta_{A,r}}$ . Define

$$\Psi_{A,r}(u) = u(F_{A,r}^{(0)}), \quad \pi_{A,r}^{\text{pt}}(u) = \Psi_{A,r}(u)\mathbf{e}_{\delta_{A,r}}.$$

**Theorem W6R.20 (pointed physical-state attachment).** For every  $r \in \mathfrak{R}_A^{\text{pt},\mu}$ ,  $\pi_{A,r}^{\text{pt}}$  is continuous and nonzero, annihilates every complete transposed constraint, and takes the normalized profile  $u_{A,r}^*$  to  $\mathbf{e}_{\delta_{A,r}}$ . It therefore satisfies pointed recordwise W6R.20[A].

*Proof.* The singleton  $\{F_{A,r}^{(0)}\}$  is bounded in the smooth graph source, so evaluation at  $F_{A,r}^{(0)}$  is a continuous strong-dual coordinate. For every complete label  $i$ ,

$$\pi_{A,r}^{\text{pt}}(C'_{A,r,i}u) = u(C_{A,r,i}F_{A,r}^{(0)})\mathbf{e}_{\delta_{A,r}} = 0.$$

Normalization gives  $\pi_{A,r}^{\text{pt}}(u_{A,r}^*) = \mathbf{e}_{\delta_{A,r}} \neq 0$ . The endpoint-packet theorem retains the source, operator AST, joint solution, and anchor while attaching precisely this target line at each scale address.  $\square$

The proof uses a genuine joint solution. Thus a constraint transpose that does not annihilate the constant source 1 is irrelevant: the terminal map evaluates at  $F_{A,r}^{(0)}$ , not at 1. No refuted normalized-history quotient is resurrected.

## W6R.21: positive physical form and completion

Define

$$\eta_r(u) = \overline{\Psi_{A,r}(u)}\Psi_{A,r}, \quad B_{A,r}(u, v) = \overline{\Psi_{A,r}(u)}\Psi_{A,r}(v).$$

**Theorem W6R.21 (pointed positive quotient).** The form  $B_{A,r}$  is continuous, Hermitian, positive semidefinite, nonzero, and null on every complete constraint image. Moreover

$$\text{Rad } B_{A,r} = \ker \Psi_{A,r}, \quad \mathcal{E}_{A,r}^{\text{ren}} / \widehat{\text{Rad } B_{A,r}} \cong \mathbb{C}.$$

Hence minimal recordwise W6R.21[A] holds throughout  $\mathfrak{R}_A^{\text{pt},\mu}$ .

*Proof.* Continuity follows from continuity of  $\Psi_{A,r}$ . Hermiticity is immediate, and  $B_{A,r}(u, u) = |u(F_{A,r}^{(0)})|^2 \geq 0$ , with value one on  $u_{A,r}^*$ . Complete annihilation puts every constraint image in both null slots. The square vanishes exactly when  $\Psi_{A,r}(u) = 0$ , giving the radical identity. The map  $[u] \mapsto \Psi_{A,r}(u)$  is an isometric surjection onto the complete space  $\mathbb{C}$ .  $\square$

## W6R.22: represented observable algebra and separation

For  $z \in \mathbb{C}$ , set  $M_r(z)u = zu$ . Scalar multiplication preserves the complete constraint images and  $\ker \Psi_{A,r}$ , and therefore descends to

$$\rho_r(z)[u] = [zu] = z[u] \quad \text{on} \quad \mathcal{H}_{A,r}^{\text{pt}} \cong \mathbb{C}.$$

**Theorem W6R.22 (exact pointed observable closure).** The represented physical star-algebra is

$$\mathfrak{A}_{A,r}^{\text{pt}} = \mathbb{C}, \quad \rho_r(z) = zI, \quad \rho_r(z)^* = \rho_r(\bar{z}).$$

It is faithful, unital, and defined on the full physical carrier. The projective ray space and normalized positive state space are both singletons; therefore the algebra separates every distinct admitted ray and every distinct admitted normalized state. Minimal recordwise W6R.22[A] holds.

The separation is exact rather than evidentiary: the complete admitted spaces contain no distinct pair. It does not assert a hidden non-scalar spectrum.

### W6R.23: complete matching and one live class

For  $r_\mu, r_\nu \in \mathfrak{R}_A^{\text{pt},\mu}$ , the complete matching contains eleven coordinates: the WDW-5 object, carrier, domain, constraint-label map, regulator/counterterm endpoint, anomaly record, normalized solution, physical carrier, represented algebra, and the two continuation-interface types. At this fixed-profile scope every coordinate is transported by the identity on common data, with only the retained source and target address tags changed.

**Theorem W6R.23 (complete pointed scale matching).** The matchings contain identities, are closed under componentwise inverse, and satisfy the composition law

$$\mathfrak{M}_{\lambda \leftarrow \nu}^{\text{pt}} \circ \mathfrak{M}_{\nu \leftarrow \mu}^{\text{pt}} = \mathfrak{M}_{\lambda \leftarrow \mu}^{\text{pt}}.$$

Every required intertwining, isometry, represented-action, and transport square commutes. Hence every two members of  $\mathfrak{R}_A^{\text{pt},\mu}$  are physically equivalent. If  $\text{Live}_{18}$  denotes the records satisfying Q6R-01–18, then

$$\text{Live}_{18} = \mathfrak{R}_A^{\text{pt},\mu}, \quad |\text{Live}_{18}/\sim_{6R}| = 1.$$

This is the first point at which scale addresses are merged. Earlier operator identity alone did not prove complete physical equivalence; the state, form, quotient, and observable coordinates are now present.

### W6R.24: positive continuation interfaces

The WDW-7 clock handoff attaches the inherited nonempty WDW-4A clock domain, its ten-part failure boundary, the pointed joint state, the positive form, and the represented algebra. It licenses a local clock class and preserves every failure condition; it does not assert conditional quantum evolution.

The WDW-8 recovery handoff attaches the inherited source-bound covariant Einstein–Klein–Gordon history, the pointed state and algebra, a nonempty recovery-seed anchor, the exact source-comparison coordinate, and the guarded register

$$(E_{\text{HJ}}, E_{\text{Ein}}, E_{\text{mat}}, E_{\text{obs}}, E_{\text{back}}; \text{UNESTIMATED-WDW8}).$$

The explicit status prevents absent estimates from being read as zero error.

**Theorem W6R.24 (twenty-clause certification).** Both interface objects are occupied at every  $r \in \mathfrak{R}_A^{\text{pt},\mu}$ , and the complete matching transports their payloads by identities. Thus

$$\text{Cert}_{20} = \mathfrak{R}_A^{\text{pt},\mu}, \quad \text{Cert}_{20}/\sim_{6R} = \{[r_1]_{6R}\}, \quad |\text{Cert}_{20}/\sim_{6R}| = 1.$$

Consequently W6R.24[A] holds, the pointed Branch-A WDW-6 capstone is complete, and WDW-7A is authorized.

## Complete consolidated theorem

### Pointed Branch-A WDW-6 capstone theorem

For the fixed target and certified family  $\mathfrak{R}_A^{\text{pt},\mu}$ :

1. Every exact unrenormalized WDW-5 lift fails continuum regulator removal on the typed scalar heat-trace witness and remains excluded.
2. The distinct counterterm-enriched route constructs the finite-cap renormalization calculus, renormalization-stable successor carrier, and the declared zero-theta constraint/anomaly closure.
3. The normalized joint solution defines a continuous nonzero pointed terminal map that annihilates every complete constraint transpose. Hence W6R.20[A] holds.
4. Its rank-one Hermitian form has radical exactly  $\ker \Psi_{A,r}$  and positive completed quotient  $\mathbb{C}$ . Hence W6R.21[A] holds.
5. Scalar multipliers descend to the faithful physical star-algebra  $\mathbb{C}I$ , which separates the fully classified singleton ray and state spaces. Hence W6R.22[A] holds.
6. The complete eleven-coordinate matchings identify all positive scale addresses and form a groupoid.

Hence W6R.23[A] holds and the live quotient has one class.

7. The positive clock and guarded recovery handoffs occupy Q6R-19 and Q6R-20 on every live record.  
Hence W6R.24[A] holds and the twenty-clause certified quotient has one class.

Therefore the pointed Branch-A WDW-6 endpoint is proved at the stated scope, and WDW-7A is authorized from its unique certified class.

## Adversarial and scope closure

| Proposed objection or overreading                                    | Exact disposition                                                                                                                                                     |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Later counterterms revive the killed exact lifts.                    | No. The counterterm- enriched records are a distinct typed comparison class; the exact lifts remain killed.                                                           |
| The successor carrier merely renames the discontinuous old endpoint. | No. The old topology remains obstructed; the smooth trace source and graph completion are separately declared and constructed.                                        |
| Open temporal or gerbe claims invalidate Gate III.                   | No. They are stronger extensions, not clauses of the declared Gate-III contract.                                                                                      |
| R1 occupancy was treated as a WDW state.                             | No. Occupancy only labels the target contrast. The source-side continuous map $u \mapsto u(F_{A,r}^{(0)})\mathbf{e}_\delta$ is constructed independently.             |
| The constant-source obstruction defeats the pointed map.             | No. The map evaluates at a complete joint solution $F_{A,r}^{(0)}$ , so every complete constraint transpose is annihilated even when the constant source is not.      |
| History-complete JSS is required for W6R.20.                         | No. JSS asks separation of all reconstruction classes. Pointed W6R.20 asks for the named nonzero attachment constructed here. JSS remains a stronger open enrichment. |
| Every possible record coordinate must be populated first.            | No. W6R.20–24 are checked recordwise on the populated survivor family. Seven optional non-scale classes retain their own open status.                                 |
| The scalar algebra hides an unproved observable spectrum.            | No. The claimed algebra is exactly $\mathbb{C}I$ ; separation follows from complete classification of the admitted singleton spaces.                                  |
| Scale records were identified before full matching.                  | No. Their addresses remain distinct through W6R.22. They merge only under the eleven-coordinate W6R.23 certificate.                                                   |
| The WDW-8 interface proves recovery.                                 | No. Its error coordinates are explicitly marked unestimated. The handoff is positive; recovery remains a downstream theorem.                                          |
| The result closes Branch B.                                          | No. The theorem and certified family are explicitly Branch A.                                                                                                         |

## Exact current boundary

| Object                                | Status           | Exact reading                                       |
|---------------------------------------|------------------|-----------------------------------------------------|
| Exact unrenormalized lifts            | KILLED-AT-Q6R-07 | No revival.                                         |
| Counterterm-enriched Gate I–III chain | PROVED           | Successor carrier and zero-theta original contract. |
| Pointed Branch-A W6R.20–24            | PROVED           | Every populated scale record.                       |
| Certified quotient                    | CARDINALITY-1    | One physical-equivalence class.                     |
| Pointed Branch-A WDW-6                | COMPLETE         | Exact scope of this capstone.                       |
| WDW-7A                                | AUTHORIZED       | Dynamics not yet claimed.                           |
| WDW-8 recovery                        | UNAUTHORIZED     | Positive guarded interface only.                    |
| History-complete JSS                  | OPEN-ENRICHMENT  | Not load bearing here.                              |
| Seven optional non-scale classes      | OPEN             | Neither killed nor certified.                       |
| Branch B / R9                         | UNAUTHORIZED     | Separate finite-pullback boundary.                  |

### No downstream overpromotion

Authorization of WDW-7A means only that a nonempty certified WDW-6 handoff exists. This manuscript does not prove conditional Schrödinger evolution, clock-time unitarity, a global clock, WKB/Hamilton–Jacobi recovery, the Einstein limit, the matter Schrödinger limit, quantitative recovery errors, or backreaction control.

## Publication handoff

This unified capstone is the citation surface for the WDW-6 endpoint theorem, its exact Branch-A scope, and the positive WDW-7/WDW-8 interfaces. The stagewise statements above retain the dependency order needed to locate every load-bearing object without importing editorial chronology into the theorem text.

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